SINGLE CHOICE CORRECT

1. The organic compound contains mass % of carbon, hydrogen, deuterium as 24%, 3%, 2% respectively and rest being chlorine. If 5 gm of same organic compound occupies 2 litre at 1 atm, 500K.

(R = 0.08 atm-L/mole-K)

Select the correct statement.

(A) In one molecule number of H-atoms are 6

(B) In empirical formula, ratio of C to Cl atoms is 2

(C) vapour density of organic compound is 100.

(D) All are incorrect.

2. Human lungs can absorb 8 gm O_2 per hour by respiration. If all oxygen atoms are converted to carbohydrates ($C_6H_{12}O_6$) how long will it take to produce 180 gm $C_6H_{12}O_6$

(A) 8 hour (B) 12 hour (C) 10 hour (D) 6 hour

3. If 0.5 mol of $BaCl_2$ is mixed with 0.1 mole of Na_3PO_4, the maximum number of mole of $Ba_3(PO_4)_2$ that can be formed is:-

(A) 0.7 (B) 0.05 (C) 0.30 (D) 0.10

4. A compound contains 10^{-2} % of phosphorous. If atomic mass of phosphorus is 31, the molecular mass of the compound having one phosphorus atom per molecule is :–

(A) 31 (B) 31×10^2 (C) 31×10^4 (D) 31×10^3

5. When 10 ml of ethanol of density 0.7893 gm/ml is mixed with 20 ml of water of density 0.9971 gm/ml at 25°C final solution has a density of 0.9571 gm/ml. Calculate percentage change in total volume on mixing:

(A) 2.167% (B) 3.067% (C) 0.067% (D) 1.027%

6. 120 ml of gaseous mixture consisting of equal proportions by volume of CO and H_2 are reacted with 400 ml of air containing 20% by volume of O_2. Assuming that all gas volumes are measured at room temperature and pressure, calculate the percentage by volume of N_2 remaining at the end of the reaction. (The gaseous mixture of CO and H_2 is known as water gas).

(A) 20% (B) 40% (C) 80% (D) 60%

7. What weight of inorganic rubber phosphoric dichloride [$(PNCl_2)_n$] can be obtained if the synthesis begins with 1.4 g N_2, 5.6 lit of H_2 at STP, 137.5 g of PCl_3 and large excess of S_2Cl_2 and HCl.

[At. wt. = N = 14, P = 31, Cl = 35.5]

$$N_2 + 3H_2 \longrightarrow 2NH_3$$

$$NH_3 + HCl \longrightarrow NH_4Cl$$

$$3PCl_3 + S_2Cl_2 \longrightarrow 2PSCl_3 + PCl_5$$

$$nPCl_5 + nNH_4Cl \xrightarrow[120^\circ C]{\Delta} (PNCl_2)_n + nHCl$$

(A) 38.6 g (B) 11.6 g (C) 11.6 ng (D) 38.6 ng

8. In the extraction of iron from haematite ore, first the coke is made to react with air to produce CO gas according to the equation :

$$\underset{(coke)}{2C(s)} + \underset{(from\ air)}{O_2(g)} \longrightarrow 2CO(g)$$

This CO gas causes the reduction of haematite ore (Fe_2O_3) according to following reactions :

$$3Fe_2O_3 + CO \longrightarrow 2Fe_3O_4 + CO_2$$

$$Fe_3O_4 + CO \longrightarrow 3FeO + CO_2$$

$$FeO + CO \longrightarrow \underset{\text{(Molten iron)}}{Fe} + CO_2$$

The raw materials available are : Haematite ore (Fe_2O_3) is 240 kg, coke is present in excess and volume of air (containing 20% by volume of oxygen0 is 224 m^3 at STP. Find the amount of molten iron (in kg) obtained finally. (Take molar volume of 1 mole of an ideal gas at STP = 22.4 litre).

(A) 140 kg (B) 120 kg (C) 70 kg (D) 50 kg

9. H_3PO_4 undergoes dissociation as

$$H_3PO_4 \longrightarrow 3H^+ + PO_4^{3-}$$

If $\dfrac{n_{cation} + n_{anion}}{(n_{H_3PO_4})_{undissociated}} = \dfrac{8}{3}$

Calculate % dissociation of H_3PO_4

(A) 40 (B) 20 (C) 80 (D) 60

10. The action of bacteria on meat and fish produces a poisonous compound called cadaverine. As its name and origin imply, it stinks. It is observed that 510 gm. of it contains 300 gm. carbon, 70 gm. hydrogen & rest nitrogen. Given that its molar mass is 102g/mol. Determine the molecular formula of cadaverine.

(A) $C_5H_{14}N_2$ (B) $C_5H_{12}N_2$ (C) $C_4H_{14}N_2$ (D) $C_5H_{10}N_2$

11. A mixture of hydrocarbon C_2H_2, C_2H_4 and CH_4 in mole ratio of 2 : 1 : 2 is burnt completely in the presence of air containing 80% N_2 and 20% O_2 by volume. The mass of air required for the complete combustion of one gm of mixture is :

(A) $\dfrac{1728}{112}$ (B) $\dfrac{1528}{73}$ (C) $\dfrac{1920}{120}$ (D) $\dfrac{112}{1728}$

MORE THAN ONE CORRECT

12. 10 mole of A_2B_3 contains 100gm A atom & 60 gm B atoms. Choose the correct statements -

(A) Molecular weight of A_2B_3 is equal to 16

(B) Atomic weight of A is equal to 5

(C) Weight of one atom of B is equal to 2

(D) Atomic weight of B is equal to 6

13. For reaction :

$$2Fe_2S_3 + 6H_2O + 3O_2 \longrightarrow 4\,Fe(OH)_3 + 6S$$

If 4 moles of Fe_2S_3 react completely with 2 mole of H_2O and 3 moles of O_2, then which of the following statement (s) is/are correct-

(A) H_2O is limiting reagent

(B) Moles of $Fe(OH)_3$ formed is 4/3

(C) Moles of Fe_2S_3 remaining is 10/3

(D) Mass of O_2 remaining is 32 gm

14. Choose the incorrect statement(s)

(A) 1 gm-molecule always contains same number of atoms

(B) weight of one molecule in gm is equal to its molar mass

(C) Number of atoms in 2 gm of hydrogen is greater than 11.35 litre hydrogen at STP

(D) Volume of 16 gm oxygen gas at 2 atm, 300 K is greater than volume of 2 gm hydrogen gas at 1 atm, 300K.

15. Select the incorrect statement(s)

(A) During a reaction moles and mass of atoms remain constant

(B) For reaction $2A + 3B \longrightarrow C + 3D$, for maximum product formation per gram of reactant mixture, mass ratio of A & B must be 2 : 3

(C) Both molarity and mole fraction are temperature dependent

(D) 22.7 litre of water at S.T.P. conditions contains 6×10^{24} protons.

16. For reaction

$$MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2(g)$$

261 gm MnO_2 mixed with 448 litre of HCl gas at 273°C & 1 atm pressure to produce product.

[$N_A = 6 \times 10^{23}$; Atomic mass of Mn = 55, Cl = 35.5]

Select correct statement(s).

(A) MnO_2 is limiting reagent

(B) Chlorine gas produced contains 15×10^{23} molecules.

(C) Moles of excess reactant left is 0.5 moles

(D) If % yield of reaction is 50%, then mass of $MnCl_2$ obtained will be 315.

17. A chloride mixture is prepared by grinding together pure $BaCl_2.2H_2O$, KCl and NaCl. What is the smallest and largest volume of 0.15 M $AgNO_3$ solution that may be used for complete precipitation of chloride from a 0.3 g sample of the mixture which may contain any one or all of the constituents ?

(A) Smallest volume is 16.38 mL
(B) Largest volume is 34.18 mL
(C) Largest volume is 16.38 mL
(D) Smallest volume is 8.18 mL

18. When 1 mol of CO is formed from 1 mol of carbon and $\frac{1}{2}$ mole of O_2, 25 Kcal heat is evolved. When 1 mol of CO_2 is formed from 1 mol of carbon and 1 mol of O_2, 75 Kcal heat is evolved.

[It is known that the formation of CO or CO_2 depends upon the relative amount of O_2 and carbon. When O_2 is limited it gives CO, when in excess gives CO_2. CO can be converted to CO_2 when reacted with O_2]

Select the option(s) in which heat evolved is more than 100 Kcal.

(A) 10 mol of carbon and 1.5 mol of O_2
(B) 36 gm of carbon and 96 gm of O_2
(C) 6 mol of carbon and 3 mol of O_2
(D) 72 gm of carbon and 32 gm of O_2

COMPREHENSION TYPE

Paragraph for Q. 19 to Q.21.

In parallel reaction common reactants react together to form different product. In such an experiment N_2 & H_2 react together to form $N_2H_2(\ell)$ & $NH_3(g)$ simultaneously, such that none of the reactants are left behind. If 5 gm-molecules of nitrogen is taken with 20 gm atoms of hydrogen in a container.

19. Total volume of gases collected (in litre) in container at 300K, 1520 mm Hg pressure

(R = 0.08 atm-L/mole-K)

(A) 60 (B) 30 (C) 120 (D) 75

20. Mass of $N_2H_2(\ell)$ (in gm) obtained is

(A) 60 (B) 90 (C) 150 (D) 75

21. If ammonia gas produced is separated and dissolved to form 250 ml solution. Molarity of solution is

(A) 2 M (B) 20 M (C) 10 M (D) 60 M

Paragraph for Q.22 to Q. 24

81gm mixture of $MgCO_3(s)$ and $NH_2COONH_4(s)$ is heated to constant mass. If vapour density of gaseous mixture evolved was found to be 61/4 then

Given that :

$$MgCO_3(s) \longrightarrow MgO(s) + CO_2(g)$$

$$NH_2COONH_4(s) \longrightarrow 2NH_3(g) + CO_2(g)$$

22. Mole % of $MgCO_3$ in original sample -

(A) 50% (B) 60% (C) 75% (D) None of these

23. Volume (in litre) of the total gases produced at 1 atm & 273K -

(A) 22.4 (B) 44.8 (C) 179.2 (D) 89.6

24. If above gaseous mixture at 273K is passed through KOH solution, contraction in volume (in litre) will be -

(A) 11.2 (B) 22.4 (C) 44.8 (D) 89.6

ANSWER KEY (Quiz # 1)

1. (D)	**2. (B)**	**3. (B)**	**4. (C)**	**5. (B)**	**6. (C)**	**7. (B)**	**8. (A)**	**9. (A)**	**10. (A)**
11. (A)	**12.(A,B)**	**13. (A,B,C)**		**14. (A,B,D)**		**15. (B,C,D)**		**16. (B,C)**	
17. (A,B)		**18. (B, C)**		**19. (A)**	**20. (D)**	**21. (B)**	**22. (A)**	**23. (B)**	**24. (B)**

QUIZ # 2

SINGLE CHOICE CORRECT

1. 300 gm, 30% (w/w) NaOH solution is mixed with 500 gm 40% (w/w) NaOH solution. What is % (w/v) NaOH if density of final solution is 2 gm /mL :

(A) 72.5 (B) 65 (C) 62.5 (D) None of these

2. If mole fraction of $NaCl_{(aq)}$ is same as that of water then molality of NaCl is

(A) 5.55 (B) 55.55 (C) 0.18 (D) 58.5

3. On heating 60 ml mixture containing equal volume of chlorine gas and it's gaseous oxide, volume becomes 75 ml due complete decomposition of oxide. On treatment with KOH volume becomes 15 ml. What is the formula of oxide of chlorine

(A) ClO_2 (B) Cl_2O_3 (C) Cl_2O (D) Cl_2O_7

4. A gaseous mixture of C_3H_8 and CH_4 exerts a pressure of 320 mm Hg at temperature TK in a V lt. flask. On complete combustion of mixture, flask containing only CO_2 exerts a pressure of 448 mm Hg under identical condition. The mole fraction of C_3H_8 in the given mixture is

(A) 0.35 (B) 0.5 (C) 0.2 (D) 0.4

5. 50 ml mixture of NH_3 and H_2 was completely decomposed by sparking at room temperature into nitrogen and hydrogen. 40 ml of oxygen was added and the mixture was sparked again. After cooling to the room temperature, the mixture was shaken with alkaline pyrogallol and the contraction of 6ml was observed. The mole % of NH_3 is the original mixture is

(A) 12% (B) 34% (C) 68/3% (D) 72%

6. 50 ml Toluene [$C_6H_5CH_3(g)$] undergoes combustion with excess of oxygen. Calculate the volume contraction.

(A) 50 ml (B) 150 ml (C) 75 ml (D) 100 ml

7. 200 ml of an aqueous solution of glucose ($C_6H_{12}O_6$) has molarity of 0.01M. Which of the following operations can be done to this solution so as to increase molarity to 0.015 M ?

(A) Evaporate 50 ml water from this solution

(B) Add 0.18 g glucose to solution without changing its volume

(C) Add 50 ml water to this solution

(D) None of these

COMPREHENSION TYPE

Paragraph for Q. 8 to Q.10

Air sample from an industrial town, heavily polluted by CO_2 was collected and analyzed. In one analysis, 56 L of air measured at NTP was passed through a 250 mL of 0.025M NaOH solution, where $CO_2(g)$ was absorbed completely. 25mL of the above solution was then treated with excess of $BaCl_2$ solution where all the carbonate was precipitated as $BaCO_3(s)$. The solution was filtered off and the filtrate required 25mL of a 0.005M HCl solution for neutralization. Answer the following four questions based on the given observations.

8. ppm strength of $CO_2(g)$, volume by volume i.e., mL of CO_2 per 10^6 mL of air was -

(A) 560 (B) 5600 (C) 100 (D) 1000

9. Weight (in milligrams) of precipitate $BaCO_3(s)$ obtained from the 25 ml of test solution was

Molar masses : Ba = 137, C = 12, O = 16

(A) 27.58 (B) 275.8 (C) 4.925 (D) 49.25

10. Fraction of original NaOH (by mole) that reacted with CO_2 was -

(A) 0.2 (B) 0.4 (C) 0.6 (D) 0.8

ONE OR MORE THAN ONE CHOICE CORRECT

11. For 7 molal NaOH solution. Select the correct statement -

(A) $\%\left(\frac{w}{w}\right)=21.7$ (B) $\%\left(\frac{w}{w}\right)=72$ (C) $X_{H_2O}=\frac{7}{47}$ (D) $X_{NaOH}=\frac{7}{47}$

12. Aqueous solution containing 30gm CH_3COOH are -

(A) 250 ml of 2M CH_3COOH solution

(B) 600 gm of 5% (by wt.) CH_3COOH solution

(C) 111gm of solution in which mole fraction of CH_3COOH is 0.1

(D) 500 gm of 7m CH_3COOH solution

13. Molarity of solute A (mol.wt. = 60) and B (mol.wt. = 40) in an aqueous solution are 2M each. (Density of aqueous solution is 1.2 gm/mℓ.)

(A) Molality of A in the solution is 2m (B) Molality of B in the solution is 2m

(C) wt% of B in the solution is 10% (D) wt % of A in the solution is 10 %

MATCH THE COLUMN

14. Match the column-

Column-I (Concentration of aqueous solution)	**Column-II (Density of given solutions is 1.2 g/ml)**
(A) 2M NaOH solution	(P) 16gm solute in 240gm solution
(B) $8\%\left(\frac{w}{V}\right)$ KOH solution	(Q) 60gm solute in 240 gm solution
(C) $25\%\left(\frac{w}{W}\right)$ $CaCO_3$ solution	(R) 8gm solute in 100 ml solution
(D) $X_{C_3H_7OH}=\frac{1}{11}$	(S) 30 gm solute in 100 ml solution
	(T) 1 mole solute in 400 gm solution

INTEGER TYPE

15. Find the sum of molarity of all the ions present in an aqueous solution of 5M $NaNO_3$ and 3m $BeCl_2$. The specific gravity of the given solution is 1.665. Assume 100 % dissociation of each salt.

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.(19)

16. 1000 ml solution of $CaBr_2$ contain 200 gm $CaBr_2$ if density of solution is 1200 kg/m^3 . Calculate molarity of solution

17. 25 ml of a solution containing HCl and H_2SO_4 required 10 ml of 1 M NaOH solution for complete neutralisation. 20 ml of the same acid mixture on being treated with excess of $AgNO_3$ gives 0.1435 gm of AgCl. If molarity of HCl is 'x' and that of H_2SO_4 is 'y'. Find the value of x & y. **[Give your answer as (x + y) × 1000]**

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

18. 2 m solution is formed when 10 gm solute is present in 100 ml solvent. Calculate the molality of solution which is formed due to dissolution of 25 gm solute in 500 ml solvent.

19. A mixture of butane (C_4H_{10}) with a gaseous hydrocarbon of alkene series occupied 20mm. To burn the mixture completely 125 mm pressure of O_2 were required and after combustion 80 mm of CO_2 were formed. All the pressure were measured at same temperature and volume. Find the mole % of alkene in the given mixture.

20. A solvent 'X' (mol mass 50) contains solute A (mol. mass 125) & solute 'B' (mol mass 100) If solution is 4M 'A' & 6M 'B' find ratio of moles of A : B : X. [Given : $d_{solution}$ = 1.3 gm/ml].

If your answer is A : B : C, then fill your answer is A + B + C.

ANSWER KEY (Quiz # 2)

1. (A)	2. (B)	3.(C)	4. (C)	5. (D)	6. (B)	7.(B)	8. (D)	9. (D)	10. (D)

11. (A) **12. (A,B,C)** **13. (A,B,D)** **14. (A) - P,R ; (B) - P,R ; (C) - Q,S,T ; (D) - S, Q**

15. (1) **16. (1)** **17. (9)** **18. (1m)** **19. (050)** **20. (017)**

QUIZ # 1 EQUIVALENT CONCEPT PHYSICAL CHEMISTRY

1. Find the **n** factor in following non-redox interaction.

(a) **Of base**

(i) $Ba(OH)_2 + HCl \rightarrow BaCl_2 + H_2O$

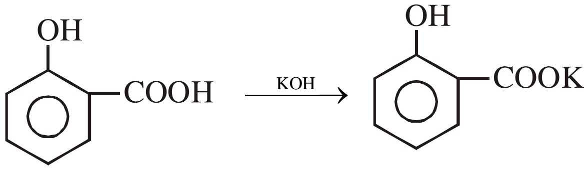
(ii) $Al(OH)_3 + H_2SO_4 \rightarrow Al(OH)\,(HSO_4)_2 + H_2O$

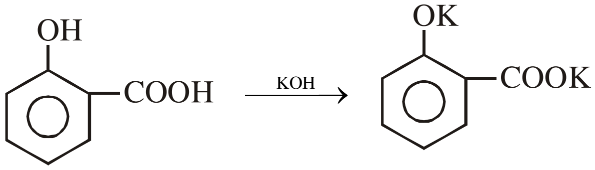
(iii) $\begin{array}{l} CH_2-NH_2 \\ | \\ CH_2-NH_2 \end{array} + HCl \rightarrow \begin{array}{l} CH_2-NH_3^{\oplus}Cl^- \\ | \\ CH_2-NH_3^{\oplus}Cl^- \end{array}$

(b) **Of base**

(i) $H_3SbO_4 \xrightarrow{KOH} KH_2SbO_4 + H_2O$

(ii) $H_3SbO_4 + KOH \rightarrow K_2HSbO_4 + H_2O$

(iii) 

(iv) 

2. Find the **n** factor of underlined compound in following interaction

(i) $\underline{Pb(NO_3)_2} + \underline{Cr_2(SO_4)_3} \longrightarrow \underline{PbSO_4} + \underline{Cr(NO_3)_3}$

(ii) $\underline{KMnO_4} + \underline{MnSO_4} \longrightarrow \underline{MnO_2}$

(iii) $\underline{P_4} \longrightarrow \underline{H_2PO_2^-} + \underline{PH_3}$

3. 5 ml of N–HCl, 20 ml of $\frac{N}{2}$ H_2SO_4 and 30 ml of $\frac{N}{3}$ HNO_3 are mixed together and the volume made of 1 litre. What is the weight of pure NaOH required to neutralize the solution?

(A) 1 gm (B) 0.5 gm (C) 0.1 gm (D) 2 gm

4. In basic medium CrO_4^{2-} oxidises $S_2O_3^{2-}$ to form $Cr(OH)_4^-$ & SO_4^{2-}. How many ml of 0.25 M Na_2CrO_4 are required to react with 40 ml of 0.3 M $Na_2S_2O_3$.

(A) 16 ml (B) 32 ml (C) 128 ml (D) 42 ml

5. What is the molarity of H_2O_2 solution whose 100 ml produce the 0.5 mole of I_2 when reacted with excess KI solution.

(A) 0.5 M (B) 1 M (C) 2.5 M (D) 5 M

ANSWER KEY (QUIZ # 1)

1. **(a) (i) 2, (ii) n = 2, (iii) 2 ; (b) (i) 1, (ii) 2, (iii) 1,(iv) 2 ,**

2. **(i) 2, 6, 2, 3 (ii) 3, 2, 6/5 (iii) n_f = 3, n_f = 1, n_f = 3**

3. (A) **4.** (C) **5.** (D)

QUIZ # 2

1. Moles of $K_2Cr_2O_7$ used to oxidise 1 mol $Fe_{0.92}O$ to Fe^{+3} are

(A) $\frac{0.92}{6}$ (B) $\frac{70}{92}\times\frac{1}{6}$ (C) $\frac{0.76}{6}$ (D) $\frac{70}{92}\times\frac{1}{3}$

2. Mass of Cl_2 produced by the complete reaction of 230 gm As_2O_5 with 182.5 gm HCl according to reaction is

$As_2O_5 + HCl \longrightarrow AsCl_3 + Cl_2 + H_2O$

(A) 71 gm (B) 142 gm (C) 177.5 gm (D) 35.5 gm

3. What are the value of **p, q, r** and **s** for the following reaction

$$\boldsymbol{p}\ O_3 + \boldsymbol{q}\ HI \longrightarrow \boldsymbol{r}\ I_2 + \boldsymbol{s}\ H_2O$$

(A)1, 6, 3, 1 (B) 1, 6, 3, 3 (C) 1, 6, 6, 3 (D) 1, 6, 3, 6

4. Which of the following is not a redox reaction.

(A) $CO + NO_2 \longrightarrow CO_2 + NO$

(B) $3SnCl_2 + 6HCl + 2NO \longrightarrow 3SnCl_4 + 2NH_2OH$

(C) $PCl_3 + Cl_2 \longrightarrow PCl_5$

(D) $SiO_2 + 4HF \longrightarrow SiF_4 + 2H_2O$

Paragraph for Q.5 to Q.7

Potassium dichromate ($K_2Cr_2O_7$) is an orange coloured compound, very frequently used in laboratory as an oxidising agent as well as in a redox titration. It is generally prepared from chromite ($FeCr_2O_4$) ore according to the following reactions:

(1) Fusion of chromite ore with sodium carbonate in excess of air.

$$FeCr_2O_4 + Na_2CO_3 + O_2 \longrightarrow Na_2CrO_4 + Fe_2O_3 + CO_2$$

(2) Acidifying filtered sodium chromate solution with sulphuric acid.

$$Na_2CrO_4 + H_2SO_4 \longrightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$$

(3) Treating sodium dichromate with potassium chloride.

$$Na_2Cr_2O_7 + KCl \longrightarrow K_2Cr_2O_7 + NaCl$$

Answer the following questions using above information.

5. If you are initially provided with 224 gm of pure chromite ore & 169.6 gm of sodium carbonate, the minimum volume of air required at 1 atm and 273 K to consume at least one of the reactant completely, if air contains 20% by volume of oxygen gas is

(A) 156.8 L (B) 196 L (C) 28 L (D) 152.4 L

6. If the number of moles of reactants available for reaction are: {$FeCr_2O_4$ = 0.25 moles, O_2 = 0.35 moles, Na_2CO_3 = 0.60 moles, H_2SO_4 = 0.2 moles, KCl = 0.1 moles}, then the maximum number of moles of $K_2Cr_2O_7$, that can be produced is :

(A) 0.05 moles (B) 0.1 moles (C) 0.2 moles (D) 0.5 moles

7. If whole of the chromite ore given in the previous question gets consumed & sufficient amount of rest of the reactants are given, then the mass of $K_2Cr_2O_7$ obtained is :

(A) 14.7 gm (B) 7.35 gm (C) 73.5 gm (D) 147 gm

ANSWER KEY (QUIZ # 2)

1. (C) **2.** (A) **3.** (B) **4.** (D) **5.** (A) **6.** (A) **7.** (C)

QUESTION BANK

MOLE & EQUIVALENT CONCEPT

SINGLE CHOICE QUESTIONS

1. H_2O_2 acts as both oxidising as well as reducing agent, its product is H_2O, when act as oxidising agent and its product is O_2 when act as of reducing agent. The strength of '10V' means one liter of H_2O_2 solution on decomposition at ST.P. condintion liberate 10 liters of oxygen gas $(H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2)$ 15 gm $Ba(MnO_4)_2$ sample containing inert impurity is completely reacting with 100 ml of '11.2 V H_2O_2, then what will be the % purity of Ba $(MnO_4)_2$ in the sample ? (Atomic mass Ba = 137, Mn = 55)

(A) 5 % (B) 10 % (C) 50 % (D) none

2. 1.2 gm of carbon is burnt completely in oxygen (limited supply) to produce CO and CO_2. This mixture of gases is treated with solid I_2O_5 (to know the amount of CO produced), the liberated iodine required 120 ml of 0.1 M hypo solution for complete titration. The % of carbon converted into CO is :

(A) 60 % (B) 100 % (C) 50 % (D) 30 %

3. Nitrogen (N), phosphorus (P) and Potassium (K) are the main nutrients in plant fertilizers. According to an industry convention, the numbers on the label refer to the mass % of N, P_2O_5 and K_2O in that order calculate the N : P : K ratio of a 30 : 10 : 10, fertilizer in terms of moles of each element and express it as x : y: 1.

(A) 10:0.67 : 1 (B) 20:0.37:1 (C 0.37:10:1 (D 5: 2:1

4. In what ratio should a 15% solution of acetic acid be mixed with a 3% solution of the acid to prepare a 10% solutionfall percentages are mass/mass percentages):

(A) 7:3 (B) 5 : 7 (C) 7 : 5 (D) 7:10

5. 105 ml of pure water at 4°C saturated with NH_3 gas yielded a solution of density 0.9 g/ml and containing 30% NH_3 by mass. Find the valume of resulting NH_3 solution.

(A) 66.67 ml (B) 166.67 ml (C) 133.33 ml (D) 266.67 ml

6. The equivalent mass of H_3BO_3 (M = Molar mass of H_3BO_3) in its reaction with NaOH to form $Na_2B_4O_7$ is equal to-

(A) M (B) $\frac{M}{2}$ (C) $\frac{M}{4}$ (D) $\frac{M}{6}$

7. x gram of pure As_2S_3 is completely oxidised to respective highest oxidation states by 50 ml of 0.1 M hot acidified $KMnO_4$ then x mass of As_2S_3 taken is : (Molar mass of As_2S_3 = 246)

(A) 22.4 g (B) 43.92 g (C) 64.23 g (D) None

8. The number of moles of ferrous oxalate oxidised by one mole of $KMnO_4$ is

(A) $\frac{5}{2}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{5}{3}$

9. How many moles of $KMnO_4$ are needed to oxidised a mixture of 1 mole of each $FeSO_4$ & FeC_2O_4 in acidic medium

(A) $\frac{4}{5}$ (B) $\frac{5}{4}$ (C) $\frac{3}{4}$ (D) $\frac{5}{3}$

10. In the reaction $Na_2S_2O_3 + 4Cl_2 + 5H_2O \longrightarrow Na_2SO_4 + H_2SO_4 + 8HCl$

the equivalent weight of $Na_2\,S_2\,O_3$ will be

(A) M/4 (B) M/8 (C) M/1 (D) M/2

(M = molecular weight of $Na_2S_2O_3$)

11. Which of the following equations is a balanced one-

(A) $5BiO_3^- + 22H^+ + Mn^{2+} \rightarrow 5Bi^{3+} + 7H_2O + MnO_4^-$

(B) $5BiO_3^- + 14H^+ + 2Mn^{2+} \rightarrow 5Bi^{3+} + 7H_2O + 2MnO_4^-$

(C) $2BiO_3^- + 4H^+ + Mn^{2+} \rightarrow 2Bi^{3+} + 2H_2O + MnO_4^-$

(D) $6BiO_3^- + 12H^+ + 3Mn^{2+} \rightarrow 6Bi^{3+} + 6H_2O + 3MnO_4^-$

12. The following equations are balanced atomwise and chargewise.

(i) $Cr_2O_7^{2-} + 8H^+ + 3H_2O_2 \longrightarrow 2Cr^{3+} + 7H_2O + 3O_2$

(ii) $Cr_2O_7^{2-} + 8H^+ + 5H_2O_2 \longrightarrow 2Cr^{3+} + 9H_2O + 4O_2$

(iii) $Cr_2O_7^{2-} + 8H^+ + 7H_2O_2 \longrightarrow 2Cr^{3+} + 11H_2O + 5O_2$

The precise equation/equations representing the oxidation of H_2O_2 is/are

(A) (i) only (B) (ii) only (C) (iii) only (D) all the three

13. An excess of NaOH was added to 100 ml of a ferric chloride solution. This caused the precipitation of 1.425 g of $Fe(OH)_3$. Calculate the normality of the ferric chloride solution

(A) 0.20 N (B) 0.50N (C) 0.25 N (D) 0.40 N

14. In the reaction $CrO_5 + H_2SO_4 \rightarrow Cr_2(SO_4)_3 + H_2O + O_2$ one mole of CrO_5 will liberate how many moles of O_2

(A) 5/2 (B) 5/4 (C) 9/2 (D) none of these

15. 0.4g of a polybasic acid H_nA (all the hydrogens are acidic) requires 0.5g of NaOH for complete neutralization The number of replaceable hydrogen atoms in the acid and the molecular weight of 'A' would be : (Molecular weight of the acid is 96 gms.)

(A) 1, 95 (B) 2, 94 (C) 3, 93 (D) 4, 92

16. Number of moles of $K_2Cr_2O_7$ required in acidic medium to oxidise 4 mole of a mixture containing 1 : 1 molar ratio of $Na_2C_2O_4$ and FeC_2O_4 will be :

(A) 2.5 (B) 5 (C) 5/3 (D) 3/4

17. 25.0 g of $FeSO_4 . 7H_2O$ was dissolved in water containing dilute H_2SO_4, and the volume was made up to 1.0 L. 25.0 ml of this solution required 20 mL of an N/10 $KMnO_4$ solution for complete oxidation. The percentage of $FeSO_4$ $7H_2O$ in the acid solution is

(A) 78 % (B) 98 % (C) 89 % (D) 79 %

18. l mol of Fe reacts completely with 0.65 molof O_2 to give a mixture of only FeO and Fe_2O_3. The mole ratio of ferrous oxide to ferric oxide is

(A) 2:3 (B) 4 : 3 (C) 1: 2 (D) 2 : 7

19. 25 mL of a solution containing $HC\ell$ and H_2SO_4 required 10mLofa 1 N NaOH solution for neutralization. 20 mL of the same acid mixture on being treated with an excess of $AgNO_3$ gives 0.1435 g of $AgC\ell$. The normality of the $HC\ell$. and the normality of the H_2SO_4 are respectively

(A) 0.40 N and 0.05 N (B) 0.05 N and 0.35 N

(C) 0.50 N and 0.25 N (D) 0.40 N and 0.5 N

20. If 10 gm of V_2O_5 is dissolved in acid and is reduced to V^{2+} by zinc metal, how many mole of I_2 could be reduced by the resulting solution if it is further oxidised to VO^{2+} ions ? [Assume no change in state of Zn^{2+}ions] (V = 51, O = 16,1= 127) :

(A) 0.11 mole of I_2 (B) 0.22 mole of I_2 (C) 0.055 mole of I_2 (D) 0.44 mole of I_2

21. 0.70 g of mixture $(NH_4)_2$ SO_4 was boiled with 100 ml of 0.2 N NaOH solution till all the $NH_3(g)$ evolved and get dissolved in solution itself. The remaining solution was diluted to 250 mL 25 mL of this solution was neutralized using 10 ml of a 0.1 N H_2SO_4 solution. The percentage purity of the $(NH_4)_2SO_4$ sample is

(A) 94.3 (B) 50.8 (C) 47.4 (D) 79.8

22. A mixed solution of potassium hydroxide and sodium carbonate required 15 ml of an N/20 HCI solution when titrated with phenolphthalein as an indicator. But the same amount of the solution when titrated with methyl orange as an indicator required 25 ml of the same acid. The amount of KOH present in the solution is

(A) 0.014 g (B) 0.14g (C) 0.028 g (D) 1.4g

23. The percentage of copper in a copper(ll) salt can be determined by using a thiosulphate titration. 0.305 gm of a copper(ll) salt was dissolved in water and added to an excess of potassium iodide solution liberating iodine according to the following equation

$2Cu^{2+}(aq) + 4I^-(aq) \rightleftharpoons 2CuI(s) + I_2(aq)$

The iodine liberated required 24.5 cm^3 of a 0.100 mole dm^{-3} solution of sodium thiosulphate

$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$

the percentage of copper, by mass in the copper (II) salt is. [Atomic mass of copper = 63.5]

(A) 64.2 (B) 51.0 (C) 48.4 (D) 25.5

MULTIPLE CHOICE QUESTIONS

24. Which of the following samples of reducing agents is /are chemically equivalent to 25 ml of 0.2 N $KMnO_4$ to be reduced to Mn^{2+} and water.

(A) 25 ml of 0.2 M $FeSO_4$ to be oxidized to Fe^{3+}

(B) 50 ml of 0.1 M H_3AsO_3 to be oxidized to H_3AsO_4

(C) 25 ml of 0.1 M H_2O_2 to be oxidized to H^+ and O_2

(D) 25 ml of 0.1 M $SnCI_2$ to be oxidized to Sn^{4+}

25. There are two sample of HCI having molarity 1M and 0.25 M. Find volume of these sample taken in order to prepare 0.75 M HCI solution. (Assume no water is used)

(A) 20 ml, 10ml (B) 100ml, 50 ml (C) 40 ml, 20ml (D) 50 ml, 25 ml

26. Chloride of an element is given by the formula MCI_x and it is 100% ionised in 0.01 M aqueous solution. Then

(A) if $[Cl^-]$ = 0.03 M then the value of x is 3.

(B) if $[Cl^-]$ = 0.05 M then the value of x is 5

(C) $[M^{x+}]$ = 0.01 M, irrespective of $[Cl^-]$

(D) $[M^{x+}]$ depends on $[Cl^-]$

27. If 100 ml of 1M H_2SO_4 solution is mixed with 100 ml of 98 % (w/w) H_2SO_4 solution (d = 0.1 gm/ml) then :

(A) concentration of solution remains same

(B) volume of solution become 200 ml

(C) mass of H_2SO_4 in the solution is 98 gm

(D) mass of H_2SO_4 in the solution is 19.6 gm

28. 3 moles of the gas C_2H_6 is mixed with 60 gm of this gas and 2.4×10^{24} molecules of the gas is removed. The left over gas is combusted in the presence of excess oxygen then :

($N_A = 6 \times 10^{23}$) (Density of water = 1 gm/ml)

(A) 2 Moles of C_2H_6 left for combustion

(B) Volume of CO_2 at S.T.P. produced after combustion 44.8 litre.

(C) Volume of water produced is 54 ml

(D) None

29. Which of the following contains the same number of molecules ?

(A) 1g of O_2, 2g of SO_2

(B) 1g of CO_2,1g of N_2O

(C) 112 ml of O_2 at STP, 224 ml of He at 0.5 atm and 273 K

(D) 1g of oxygen, 1g of ozone

30. A100 ml mixture of CO and CO_2 is passed through a tube containing red hot charcoal. The volume becomes 160 ml. The volumes are measured under the same conditions of temperature and pressure. Amongst the following, select the correct statement(s).

(A) Mole percent of CO_2 in the mixture is 60.

(B) Mole fraction of CO in the mixture is 0.40

(C) The mixture contains 40 ml of CO_2

(D) The mixture contains 40 ml of CO

31. Choose the correct statement: (Cu_2S oxidised into Cu^{+2} and SO_2)

(A) 1 mole of MnO_4^- ion can oxidised 5 moles of Fe^{2+} ion in acidic medium

(B) 1 mole of $Cr_2O_7^{2-}$ ion can oxidised 6 moles of Fe^{2+} ion in acidic medium

(C) 1 mole of Cu_2S can be oxidised by 1.6 moles of MnO_4^- ion in acidic medium

(D) 1 mole of Cu_2S can be oxidised by 1.33 moles of $Cr_2O_7^{2-}$ ion in acidic medium

32. 0.1 M solution of KI reacts with excess of H_2SO_4 and and KIO_3 solutions, according to equation

$5I^- + IO_3^- + 6H^+ \longrightarrow 3I_2 + 3H_2O$; which of the following statement is correct

(A) 200 ml of the KI soluition react with 0.004 mole KIO_3

(B) 0.5 litre of the KI solution produced 0.005 mole of I_2

(C) Equivalent weight of KIO_3 is equal to $\left(\frac{\text{MolecularWeight}}{5}\right)$

(D) None of these

MATRIX MATCH QUESTIONS

33.

Column -1	Column - II
(A) $CaCO_3 + H_3PO_4 \rightarrow Ca_3(PO_4)_2 + CO_2 + H_2O$	(p) 1 mole of oxidising agent gains more electrons than lost by 1 mole of reducing agent.
(B) $MnO_2 + HCl \rightarrow MnCl_2 + Cl_2 + H_2O$	(q) No change in oxidation number of any element takes place.
(C) $Na_2CrO_4 + H_2SO_4 \rightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$	(r) All elements (except oxygen) are in their maximum oxidation states on the reactant side.
(D) $H_2O + F_2 \rightarrow HF + O_2$	(s) Number of moles of oxidising agent consumed is less than number of moles of reducing agent.

34. **[H = 1, C = 12, O = 16, Na = 23, P = 31, S = 32]**

Column -1	Column - II
(A) 4.1 g H_2SO_3	(p) 200 ml of 0.5 N NaOH is used for complete neutralization
(B) 4.9 g H_3PO_4	(q) 200 millimoles of oxygen atoms
(C) 4.5 g oxalic acid ($H_2C_2O_4$)	(r) Central atom has its highest oxidation number
(D) 5.3 g Na_2CO_3	(s) May react with an oxidising agent
	(t) Shape around central atom is regular.

35. Match the following :

	Column I	**Column II**
(A)	4.5 m solution of $CaCO_3$ density 1.45 gm/ml	(p) mole fraction of solute is 0.2
(B)	3 M 100 ml H_2SO_4 mixed with 1 M 300 ml H_2SO_4 solution	(q) mass of the solute is 360 gm
(C)	14.5 m solution of Ca	(r) molarity = 4.5
(D)	in 4 M 2 litre solution of NaOH, 40 gm NaOH is added.	(s) molarity 1.5
(E)	5m (molal) NaOH solution	(t) 16.66 % (w/w) of NaOH in solution.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

632 g of sodium thiosulphate ($Na_2S_2O_3$) reacts with copper sulphate to form ciipric thiosulphate which is reduced by sodium thiosulphate to give cuprous compound which is dissolved in excess of sodium thiosulphate to form a complex compound sodium cuprothiosulphate
($Na_4[Cu_6(S_2O_3)_5]$).

$$CuSO_4 + Na_2S_2O_3 \longrightarrow CuS_2O_3 + Na_2SO_4$$
$$2CuS_2O_3 + Na_2S_2O_3 \longrightarrow Cu_2S_2O_3 + Na_2S_4O_8$$
$$3Cu_2S_2O_3 + 2Na_2S_2O_3 \longrightarrow \underset{\text{Sodium cuprothiosulphate}}{Na_4[Cu_6(S_2O_3)_5]}$$

In this process, 0.2 mole of sodium cuprothiosulphate is formed. (O = 16, Na = 23, S = 32)

36. The average oxidation states of sulphur in $Na_2S_2O_3$ and $Na_2S_4O_6$ are respectively.

(A) + 5 & + 2 (B) + 2 & + 2.5 (C) + 5 & 2.5 (D) + 2 & + 4

37. Moles of sodium thiosulphate reacted and unreacted after the reaction are respecticvely.

(A) 3 & 2 (B) 2 & 3 (C) 2.2 & 1.8 (D) 1.8 & 2.2

38. If instead of given amount of sodium thiosulphate, 2 moles of sodium thiosulphate along with 3 moles of $CuSO_4$ were taken initially. Then moles of sodium cuprothiosulphate formed is

(A) 0 (B) 1 (C 1.5 (D) 2

Comprehension # 2

We know that balancing of a chemical equation is entirely based on law of conservation of mass. However the concept of Principle of Atom Conservation (POAC) can also be related to law of conservation of mass in a chemical reaction. So, POAC can also act as a technique for balancing a chemical equation. For example, for a reaction:

$$ABC_3 \longrightarrow AB + C_2$$

On applying POAC for A, B & C and relating the 3 equations, we get : $\frac{n_{ABC_3}}{2} = \frac{n_{AB}}{2} = \frac{n_{C_2}}{3}$

(n_x : number of moles of X)

Thus, the coefficients of ABC_3, AB & C_2 in the balanced chemical equation will be 2, 2 & 3 respectively and the balanced chemical equation can be represented as :

$$2ABC_3 \longrightarrow 2AB + 3C_2$$

Now answer the following questions :

39. Which of the following relation is correct regarding the numerical coefficients p,q,r in the balanced chemical equation :

$$pA + qB_2 \longrightarrow rA_2B_5$$

(A) $2p = r$ (B) $q = 1.25\,p$ (C) $r = 2q$ (D) $q = 0.8\,p$

40. If the weight ratio of C and O_2 present is 1 :2 and both of reactants completely consume and form CO and CO_2 and we will obtain a gasous mixture of CO and CO_2. What would be the weight ratio of CO and CO_2 in mixture.

(A) 11 : 7 (B) 7 :11 (C) 1 : 1 (D) 1 : 2

41. If the atomic masses of X and Y are 10 & 30 respectively, then the mass of XY_3 formed when 120 g of Y_2 reacts completely with X is : Reaction $X + Y_2 \longrightarrow XY_3$

(A) 133.3g (B) 200g (C) 266.6 g (D) 400g

Comprehension # 3

Oleum is considered as a solution of SO_3 in H_2SO_4, which is obtained by passing SO_3 in solution of H_2SO_4. When 100 g sample of oleum is diluted with desired weight of H_2O then the total mass of H_2SO_4 obtained after dilution is known as % labelling in oleum.

For example, a oleum bottle labelled as '109% H_2SO_4' means the 109 g total mass of pure H_2SO_4 will be formed when 100 g of oleum is diluted by 9 g of H_2O which combines with all the free SO_3 to form

H_2SO_4 as $SO_3 + H_2O \longrightarrow H_2SO_4$.

42. What is the % of free SO_3 in an oleum that is labelled as '104.5% H_2SO_4' ?

(A) 10 (B) 20 (C) 40 (D) None of these

43. If excess water is added into a 100 g bottle sample labelled as "112% H_2SO_4" and is reacted with 5.3 g Na_2CO_3, then find the volume of CO_2 evolved at 1 atm pressure and 300 K temperature after the completion of the reaction : [R = 0.0821 L atm moM K^{-1}]

$$H_2SO_4 + Na_2CO_3 \longrightarrow Na_2SO_4 + H_2O + CO_2$$

(A) 2.46 L (B) 24.6L (C) 1.23L (D) 12.3

44. 1 g of oleum sample is diluted with water.The solution required 54 ml of 0.4 N NaOH for complete neutralization. The % of free SO_3 in the sample is :

(A) 74 (B) 26 (C) 20 (D) None of these

Comprehension # 4

Volumetric analysis is based on the principle of equivalence in that substances react in the ratio of their equivalents. At the equivalence point of the reaction, involving the reactahtsAand B: Number of gram equivalents of A = Number of gram equivalents of B. If V_A ml of solution A having normality N_A react just completely with V_B ml of solution B having normality N_B then

$$\frac{N_AV_A}{1000} = \frac{N_BV_B}{1000};\ N_AV_A = N_BV_B \quad .... (1)$$

Equation(1), called normality equation is very useful in numerical calculations of volumetric analysis. Equivalent masses of different substances :

Basically the equivalent mass of a substance is defined as the parts by mass of it which combine with or displace 1.0078 parts ($\approx$ 1 part) by mass of hydrogen, 8 parts by mass of oxygen and 35.5 parts by mass of chlorine.

Mass of a substance expressed in gram equal to its equivalent mass is called gram equivalent mass. Equivalent mass of a substance is not constant but depends upon the reaction in which the substance participates.

Equivalent mass of an acid in acid-base reaction is its mass in gram which contains 1 mole of replaceable H^+ ions (= 1.0078 g « 1 g). On the other hand, equivalent mass of a base is its mass which contains 1 mole of replaceable OH^- ions. 1 g equivalent mass each of an acid and base on reaction gives salt and 1 mole of water (= 18 g).

Equivalent mass of an oxidising agent is its mass which gains 1 mole of electrons. It can be obtained by dividing the molecular mass or formula mass by the total decrease in oxidation number of one or more elements per molecule.

On the other hand, equivalent mass of **a reducing agent** is the mass of the substance which loses 1 mole of electorns. It can be calculated by dividing the molecular or formula mass of the substance by the total increase in oxidation number of one or more elements per molecular or formula mass.

45. When 5.00 g of a metal is strongly heated, 9.44 g of its oxide is obtained. The equivalent mass of the metal is

(A) 2.22 g (B) 5.00g (C) 9.00g (D) 4.44g

46. How many gram of phosphoric acid (H_3PO_4) would be required to neutralize 58 g of magnesium hydroxide?

(A) 65.3 g (B) 70.2g (C) 85.0g (D) 112.0 g

47. Which of the following solutions, when mixed with 100 ml of 0.05 M NaOH, will give a neutral solution?

(A) 100 ml of 0.1 M H_2SO_4 (B) 50 ml of 0.05 M H_2SO_4

(C) 100ml of 0.05MCH_3COOH (D) None of these

48. 100 ml of 0.1 M H_3PO_2 is titrated with 0.1 M NaOH. The volume of NaOH needed will be

(A) 300 ml (B) 200 ml (C) 100 ml (D) None of these

49. 9.8 g of an acid on treatment with excess of an alkali forms salt along with 3.6 g of water. What is the equivalent mass of the acid?

(A) 98.0 g (B) 49.0g (C) 24.5g (D) 73.5g

Comprehension # 5

Molality : It is defined as the moles of the solute present in 1 kg of the solvent. It is denoted by 'm'.

$$\text{Molality (m)} = \frac{\text{Number of moles of solute}}{\text{Number of kilo-grams of the solvent}}$$

Let w_A grams of the solute of molecular mass m_A be present in w_B grams of the solvent, then

$$\text{Molality (m)} = \frac{w_A}{m_A \times w_B} \times 1000$$

Relation between mole fraction and Molality :

$$X_A = \frac{n}{N+n} \text{ and } X_B = \frac{N}{N+n}$$

$$\frac{X_A}{X_B} = \frac{n}{N} = \frac{\text{Moles of solute}}{\text{Moles of solvent}} = \frac{w_A \times m_B}{w_B \times m_A}$$

$$\frac{X_A \times 1000}{X_B \times m_B} = \frac{w_A \times 1000}{w_B \times m_A} = m \quad \text{or} \quad \frac{X_A \times 1000}{(1-X_A)m_B} = m$$

50. If the ratio of the mole fraction of a solute is changed from $\frac{1}{3}$ to $\frac{1}{2}$ in the 800 g of solvent then the ratio of molality will be :

(A) 1 : 3 (B) 3 : 1 (C) 4 : 3 (D) 1 : 2

51. The mole fraction of the solute in the 12 molal solution of Na_2CO_3 is :

(A) 0.822 (B) 0.177 (C) 1.77 (D) 0.0177

52. What is the quantity of water that should be added to 16 gm, methanol to make the mole fraction of methanol as 0.25 -

(A) 27gm. (B) 12gm. (C) 18gm. (D) 36 gm.

53. A 300 gm, 30% (w/w) NaOH solution is mixed with 500 gm 40% (w/w) NaOH solution. What is % (w/v) NaOH. if density of final solution is 2 gm/ml.

(A) 72.5 (B) 65 (C) 62.5 (D) None

54. What is the molality of final solution obtained in the above problem

(A) 1.422 (B) 14.22 (C) 15.22 (D) None

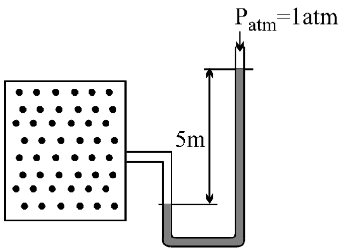
QUIZ # 1 GASEOUS STATE PHYSICAL CHEMISTRY

1. Automobile air bags are inflated with N_2 gas which is formed by the decomposition of solid sodium azide (NaN_3). The other product is Na - metal. Calculate the volume of N_2 gas at 27°C and 756 Torr formed by the decomposing of 125 gm of sod azide.

2. 3.6 gm of an ideal gas was injected into a bulb of internal volume of 8L at pressure P atmp and temp T-K. The bulb was then placed in a thermostat maintained at (T+ 15) K.0.6 gm of the gas was let off to keep the original pressure. Find P and T if mol weight of gas is 44.

3. Calculate the number of moles of gas present in the container of volume 10 lit at 300 K. If the manometer containing glycerin shows 5m difference in level as shown in diagram.

Given: $d_{glycerin}$ = 2.72 gm/ml, $d_{mercury}$ = 13.6 gm/ml.

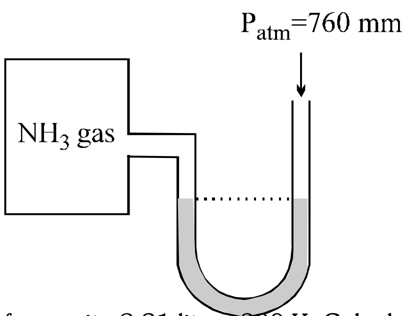


4. A toy balloon originally held 1.0 gm of He gas and had a radius 10 cm. During the night, 0.25 gm of the gas effused from the balloon. Assuming ideal gas behaivour, under these constant P and T conditions, what was the radius of the balloon the next morning.

5. While resting, the average human male use 0.2 dm^3 of O_2 per hour at 1 atm & 273 K for each kg of body mass. Assume that all this O_2 is used to produce energy by oxidising glucose in the body . What is the mass of glucose required per hour by a resting male having mass 60 kg . What volume, at 1 atm & 273 K of CO_2 would be produced.

6. An ideal gas is at a temperature of 200 K & at a pressure of 8.21 atm. It is subjected to change in volume by changing amount of the gas & a graph of n^2 vs V^2 ($litre^2$) is plotted. Is slope constant? If yes, calculate its value else justify why it is not constant.

7. A manometer attached to a flask contains NH_3 gas have no difference in mercury level initially as shown in diagram. After the sparking into the flask, it have difference of 19 cm in mercury level in two columns. Calculate % dissociation of ammonia.



8. 16 gm of O_2 was filled in a container of capacity 8.21 lit. at 300 K. Calculate

(i) Pressure exerted by O_2

(ii) Partial pressure of O_2 and O_3 if 50 % of oxygen is converted into ozone at same temperature.

(iii) Total pressure exerted by gases if 50% of oxygen is converted into ozone (O_3) at temperature 50 K.

9. 12 g N_2, 4 gm H_2 and 9 gm O_2 are put into a one litre container at 27°C. What is the total pressure.

10. 1.0×10^{-2} kg of hydrogen and 6.4×10^{-2} kg of oxygen are contained in a 10×10^{-3} m^3 flask at 473 K. Calculate the total pressure of the mixture. If a spark ignities the mixture. What will be the final pressure.

11. Calculate relative rate of effusion of SO_2 to CH_4 under given condition

(i) Under similar condition of pressure & temperature

(ii) Through a container containing SO_2 and CH_4 in 3:2 mass ratio

(iii) If the mixture obtained by effusing out a mixture ($n_{SO_2}/n_{CH_4} = 8/1$) for three effusing steps.

12. A gas mixture contains equal number of molecules of N_2 and SF_6, some of it is passed through a gaseous effusion apparatus . Calculate how many molecules of N_2 are present in the product gas for every 100 molecules of SF_6.

13. Two gases NO and O_2 were introduced at the two ends of a one metre long tube simultaneously (tube of uniform cross- section). At what distance from NO gas end , Brown fumes will be seen.

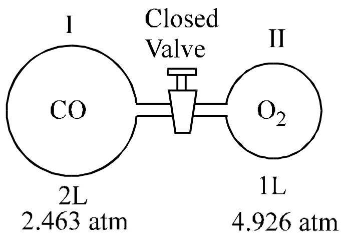
14. At 20°C two balloons of equal volume and porosity are filled to a pressure of 2 atm, one with 14 kg N_2 & other with 1 kg H_2 . The N_2 balloon leaks to a pressure of $\frac{1}{2}$ atm in one hour. How long will it take for H_2 balloon to leaks to a pressure of $\frac{1}{2}$ atm.

15. Pure O_2 diffuses through an aperture in 224 sec, whereas mixture of O_2 and another gas containing 80 % O_2 Takes 234 sec to effuse out same volume What is molecular weight of the gas?

16. Find the number of diffusion steps required to separate the isotopic mixture initially containing some amount of H_2 gas and 1 mol of D_2 gas in a container of 3 lit capacity maintained at 24.6 atm & 27 °C to the final mass ratio $\left(\frac{w_{D_2}}{w_{H_2}}\right)$ equal to $\frac{1}{4}$.

17. An iron cylinder contains helium at a pressure of 250 k pa and 27°C. The cylinder can withstand a pressure of 1×10^6 pa . The room in which cylinder is placed catches fire. Predict whether the cylinder will blow up before it metls or not. [melting point of cylinder = 1800 k]

18. Determine final pressure after the valve is left opened for a long time in the apparatus represented in figure. Assume that the temperature is fixed at 300 K. Under the given conditions assume no reaction of CO & O_2.



19. At what temperature in °C, the U_{rms} of SO_2 is equal to the average velocity of O_2 at 27°C.

20. Calculate U_{rms} of molecules of H_2 at 1 atmp if density of H_2 is 0.00009 g/cc.

21. A bulb of capacity 1 dm^3 contains 1.03×10^{23} H_2 molecules & pressure exerted by these molecules is 101.325 kPa. Calculate the average square molecular speed and the temperature.

22. The density of CO at 273 K and 1 atm is 1.2504 kg m^{-3}. Calculate (a) root mean square speed (b) the average speed and (c) most probable speed.

23. Calculate the temperature values at which the molecules of the first two members of the homologous series, C_nH_{2n+2} will have the same rms speed as CO_2 gas at 770 K. The normal b.p. of n-butane is 273 K. Assuming ideal gas behaviour of n-butane upto this temperature, calculate the mean velocity and the most probable velocity of its molecules at this temperature.

24. Calculate the temperatures at which the root mean square velocity, average velocity and most probable velocity of oxygen gas are all equal to 1500 ms^{-1}.

25. H_2 gas is kept inside a container A and container B each having volume 2 litre under different conditions which are described below. Determining the missing values with proper unit.

[R = 8 J mol^{-1} K^{-1} and $N_A = 6\times10^{23}$, N = No. of molecules]

Parameter	Container A	Container B
P	(i) ------	1 atm
T	300K	600 K
N	6×10^{20}	(ii) ------
Total Average KE	(iii) ------	(iv) ------
Ratio U_{mps}	(v) --------------	
Ratio Z_{11}	(vi) --------------	

26. Calculate the mean free path in CO_2 at 27°C and a pressure of 10^{-6} mm Hg. (molecular diameter = 460 pm)

27. The mean free path of the molecule of a certain gas at 300 K is 2.6×10^{-5} m. The collision diameter of the molecule is 0.26 nm. Calculate
(a) pressure of the gas, and
(b) number of molecules per unit volume of the gas.

ANSWERS (QUIZ # 1)

1. 71.4 L

2. P = 0.062 atm , T = 75 K

3. 0.94 mole

4. 9.08 cm

5. 16.07 gm ; 12 dm^3

6. $\frac{1}{4}$

7. 25 %

8. (i) 1.5 atm; (ii) O_2=0.75 atm, O_3 = 0.5 atm; (iii) 0.208 atm

9. 66.74 atm

10. $P_{total} = 27.54 \times 10^5$ N/m^2 , $P_{final} = 19.66 \times 10^5$N/m^2

11. (i) $\frac{1}{2}$; (ii) $\frac{3}{16}$; (iii) $\frac{1}{2}$

12. 228

13. 50.8 cm

14. 16 min

15. 46.6

16. 4

17. Yes

18. 3.284 atm

19. 236.3°C

20. 183,800 cm/sec

21. 8.88×10^5 (m /s)^2 ; 71.27 K

22. U_{RMS} = 493 m/s ,U_{mp} = 403m/s ,U_{av}=454.4 m/s

23. 280 K, 525 K , 315.7m/sec, 279.8m/sec

24. T_{RMS}= 2886 K, T_{av} = 3399 K, T_{mp}=4330K

25. (i) 0.012 atm; (ii) 2.5×10^{22}; (iii) 3.6 J; (iv) 300 J; (v) $\frac{1}{\sqrt{2}}$; (vi) 0.4×10^{-3} : 1

26. 3.3×10^3 cm

27. (a) 530.6 Pa, (b) 1281×10^{20} m^{-3}

QUESTION BANK

GASEOUS STATE

SINGLE CHOICE QUESTIONS

1. If density of vapours of a substance of molar mass 18 gm/mole at 1 atm pressure and 500 K is 0.36 kg m^{-3}, then value of Z for the vapours is : (Take R = 0.082 L atm $mole^{-1}$ K^{-1})

(A) $\frac{41}{50}$ (B) $\frac{50}{41}$ (C) 1.1 (D) 0.9

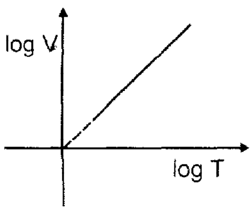
2. The density of gaseous HFat 1.0 atm and 300 Kis 3.17 g/L. Hence, HF in gaseous;

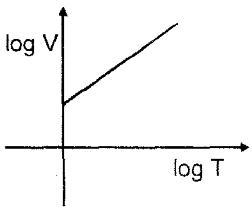
(A) Dimer (B) Monomer (C) Tetramer (D) Data

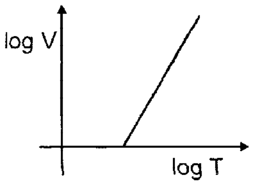
3. Equal amount (mass) of methane and ethane have their total translational kine 3 : 1 then their temperatures are in the ratio.

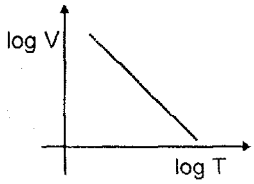
(A) 5:8 (B) 45:8 (C) 15:8 (D) 8: 5

4. Which of the following sketches is an isobar $\left(\frac{nR}{P} > 1\right)$

(A)


(B)


(C)


(D)


5. For a certain gas which deviates a little from ideal behaviour, the values of density, p were measured at different values of pressure, P. The plot of P/ρ on the Y-axis versus P on the X-axis was nonlinear and had an intercept on the Y-axis, which was equal to

(A) $\frac{RT}{M}$ (B) $\frac{M}{RT}$ (C) RT (D) $\frac{RT}{V}$

where M = molar mass.

6. A sample of water gas has a composition by volume of 50% H_2,45% CO and 5% CO_2. Calculate the volume in litre at S.T.P. of water gas which on treatment with excess of steam and total volume of H_2 will be 5 litre.The equation for the reaction is :

$$CO + H_2O \longrightarrow CO_2 + H_2$$

(A) 4.263 Litre (B) 5.263 Litre (C) 6.263 Litre (D) 7.263 Litre

7. If Pd v/s P(where P denotes pressure in atm and *d* denotes density in gm/L) is plotted for He gas (assume ideal) at a particular temperature. If $\left[\frac{d}{dP}(Pd)\right]_{P=8.21\,atm} = 5,$ then the temperature will be

(A) 160 K (B) 320 K (C) 80 K (D) none of these

8. 7 moles of a tetra-atomic non-linear gas 'A' at 10 atm and *T* K are mixed with 6 moles of another gas *B* at $\frac{T}{3}$K and 5 atm in a closed, rigid vessel without energy transfer with surroundings. If final temperature of mixture was $\frac{5T}{6}$ K, then gas *B* is? (Assuming all modes of energy are active)

(A) monoatomic (B) diatomic (C) triatomic (D) tetra atomic

9. Consider the reaction

$2X(g) + 3Y(g) \longrightarrow Z(g)$

Where gases X and Y are insoluble and inert to water and Z form a basic solution. In an experiment 3 mole each of X and Y are allowed to react in 15 lit flask at 500 K. When the reaction is complete, 5L of water is added to the flask and temperature is reduced to 300 K. The pressure in the flask is (neglect aqueous tension)

(A)1.64atm (B) 2.46atm (C) 4.92atm (D) 3.28atm

10. Three gases A, Band Care at same temperature if their r.m.s speed are in the ratio $1:\frac{1}{\sqrt{2}}:\frac{1}{\sqrt{3}}$ then their molar masses will be in the ratio :

(A) 1 : 2 : 3 (B) 3 : 2 : 1 (C) $1:\sqrt{2}:\sqrt{3}$ (D) $\sqrt{3}:\sqrt{2}:1$

11. There are n containers having volumes V, 2V,........nV. A fixed mass of a gas is filled in all containers connected with stopcock. All containers are at the same temperature if pressure of first container is P then final pressure when all stopcocks are opened is :

(A) $\frac{np}{n(n+1)}$ (B) $\frac{2np}{n(n+1)}$ (C) $\frac{3np}{n(n+1)}$ (D) $\frac{np}{2n(n+1)}$

12. A mixtureoftwogasesAand B in the mole ratio 2 : 3 is kept in a 2 litre vessel. A second 3 litre vessel has the same two gases in the mole ratio 3 : 5. Both gas mixtures have the same temperature and same total pressure. They are allowed to intermix and the final temperature and total pressure are the same as the initial values, the final volume being 5 litres. Given that the molar masses are M_A and M_B, what is the mean molar mass of the final mixture ?

(A) $\frac{77M_A+123M_B}{200}$ (B) $\frac{123M_A+77M_B}{200}$ (C) $\frac{77M_A+123M_B}{250}$ (D) $\frac{123M_A+77M_B}{250}$

13. At a certain temperature for which RT = 25 lit. atm. mol^{-1} , the density of a gas, in gm lit^{-1}, is **d = 2.00 P + 0.020 P²**, where P is the pressure in atmosphere. The molecular weight of the gas in gm mol^{-1} is

(A) 25 (B) 50 (C) 75 (D) 100

14. A mixture of carbon monoxide and carbon dioxide is found to have a density of 1.7 g/lit at S.T.P. The mole fraction of carbon monoxide is

(A) 0.37 (B) 0.40 (C) 0.30 (D) 0.50

15. If equal weights of oxygen and nitrogen are placed in separate containers of equal volume at the same temperature, which one of the following statements is true?

(mol wt: N_2 = 28, O_2 = 32)

(A) Both flasks contain the same number of molecules.

(B) The pressure in the nitrogen flask is greater than the one in the oxygen flask.

(C) More molecules are present in the oxygen flask.

(D) Molecules in the oxygen flask are moving faster on the average than the ones in the nitrogen flask.

16. The molecualr radius for a certain gas = 1.25 Å. What is a resonable estimate of the magnitude of the Vander Waals constant, b, for the gas ?

(A) 0.98×10^{-2} litre/mole (B) 1.43×10^{-2} litre/mole

(C) 1.97×10^{-2} litre/mole (D) 3.33×10^{-2} litre/mole

17. Given the value of their van der Waal's constant 'a' arange the following gases in the order of their expected liquification pressure at a temperature T. T is below the critical point of all the gases.

Gas	CH_4	Kr	N_2	Cl_2
'a' (atm L^2 . mol^{-1})	2.283	2.439	1.408	6.579

(A) $CH_4 < Kr < N_2 < Cl_2$ (B) $N_2 < CH_4 < Kr < Cl_2$

(C) $Cl_2 < Kr < CH_4 < N_2$ (D) $Cl_2 < N_2 < Kr < CH_4$

18. The van der Waal's constant 'b' for N_2 and H_2 has the values 0.039 lit^{-1} moM and 0.0266 lit mol^{-1} and 0.0266 lit mol^{-1}. The density of solid N_2 is 1 g cm^{-3}. Assuming the molecules in the solids to be close packed with the same percentage void, the density of solid H_2 would be (in g cm^{-3}).

(A) 0.105 (B) 0.682 (C) 1.466 (D) 0.071

19. Which of the following statements is not true?

(A) The ratio of the mean speed to the rms speed is independent of the temperature.

(B) The square of the mean speed of the molecules is equal to the mean squared speed at a certain temperature.

(C) Mean kinetic energy of the gas molecules at any given temperature is independent of the mean speed.

(D) The difference between rms speed and mean speed at any temperature for different gases diminishes as larger if larger molar masses are considered.

20. If the number of molecules of SO_2 (molecular weight = 64) effusing through an orifice of unit area of cross-section in unit time at 0°C and 1 atm pressure is n. The number of He molecules (atomic weight = 4) effusing under similar conditions at 273°C and 0.25 atm is

(A) $\frac{n}{\sqrt{2}}$ (B) $n\sqrt{2}$ (C) 2n (D) $\frac{n}{2}$

21. Two glass bulb A and B are connected by a very small tube (of negligible volume) having stop cock. Bulb A has a volume of 100 cm^3 and contains certain gas while bulb B is empty On opening the stop cock, the pressure in 'A' fell down by 60%. The volume of bulb B must be

(A) 200mL (B) 150 mL (C) 250 mL (D) 100 mL

22. A certain gas effuses out of two different vessels A and B. A has a circular orifice while B has a square orifice of length equal to the radius of the orifice of vessel A. The ratio of rate of diffusion of the gas from vessel A to that from vessel B is

(A) $\pi : 1$ (B) $1 : \pi$ (C) 1:1 (D) 3 : 2

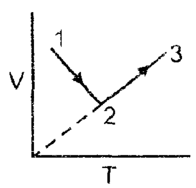
23. If 10^{23} gas molecules each having a mass of 10^{-25} kg, placed in a 1 L container moving with rms speed of 10^5 cm/sec. then the total KE of gasseous molecules and pressure exerted by molecules, respectively are :

(A) 10 kJ, 3.33×10^6Pa (B) 5 kJ, 3.33×10^6 Pa

(C) 10 kJ, 3.37×10^7 Pa (D) 5 kJ, 3.33×10^7 Pa

ONE OR MORE THAN ONE CORRECT OPTIONS

24. Following graph is constructed for the fixed amount of the gas.



(A) From 1 - 2 pressure will increase

(B) From 2 - 3 pressure remains constant

(C) Gas pressure at (3) is greater at state (1)

(D) From 1 - 2 pressure will decrease

25. Which of following statements are correct

(A) Average speed of molecules of a gas in a container is zero.

(B) All molecules in a gas are moving with the same speed.

(C) If an open container is heated from 300 K to 400 K the fraction of air which goes out with respect to originally present is 1/4 .

(D) If compressibility factor of a gas at STP is less than unity then its molar volume is less than 22.4 L at STP.

26. If the pressure of the gas contained in a closed vessel is increased by 20% when heated by 273°C then it's initial temperature must have been

(A) 1092°C (B)1029 K (C) 1365°C (D) 1365K

27. Two flask A and B have equal volumes. Flask A contains hydrogen at 600 K while flask B has same mass of CH_4 at 300 K. Then choose the correct options.

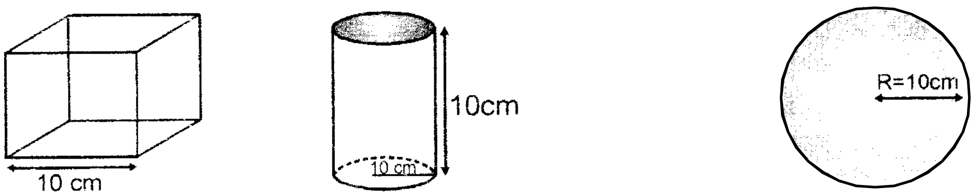
(A) In flask A the molecules move faster than B

(B) In flask B the molecules move faster than A

(C) Flask A contains greater number of molecules than B

(D) Flask B contains more molecules than A

28. There are three closed containers in which equal amount of the gas are filled.



(I) (II) (III)

If the containers are placed at the same temperatures then find the correct options :

(A) Pressure of the gas is minimum in (III) container

(B) Pressure of the gas is equal in (I) & (II) container

(C) Pressure of the gas is maximum in (I)

(D) The ratio of pressure in II and III container is 3 : 4

29. If the rms velocity of nitrogen and oxygen molecule are same at two different temperature and same pressure then

(A) Average speed of molecules is also same

(B) density (gm/lt.) of nitrogen and oxygen is also equal.

(C) Number of moles of each gas is also equal.

(D) most probable velocity of molecules is also equal.

COMPREHENSION BASED QUESTIONS

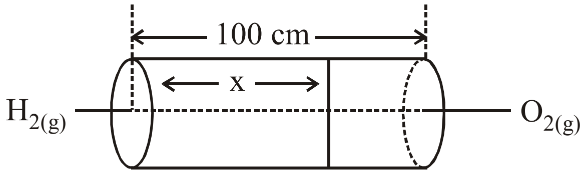
Comprehension # 1

When a gas is made to pass through a fine hole made in the wall of the container under a difference of pressure, it is known as effusion. The rate of effusion depends on the density, the temperature of the gas and the pressure gradient. Grahm's law states that a constant temperature and for constant pressure gradient the rates of effusion or diffusion of different gases are inversely proportional to the square root of their densities or molecular masses.

This law is very useful for calculating molecular weight, density, etc.... of gases. However, it should be noted that the law is true only for gases diffusing under low pressure gradient.

30. Rate of diffusion of hydrogen gas is $\sqrt{20}$ times rate of diffusion of a sample containing O_2 and O_3. Calculate the percentage (by moles) of O_2 in the sample.

(A) 75 % (B) 25 % (C) 50 % (D) 90 %

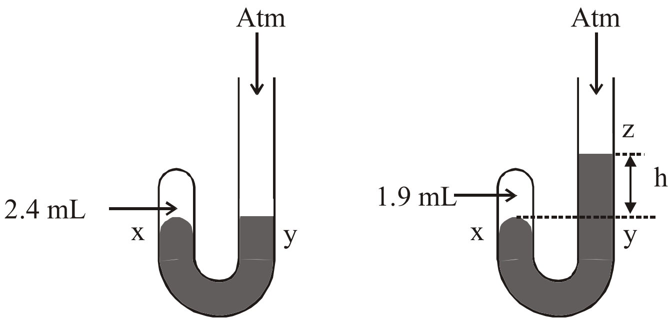
31. 

Value of x at which first drop of H_2O is formed :

(A) 20 cm (B) 80 cm (C) 50 cm (D) 60 cm

32. Given J-tube has 2.4 mL of air at a pressure of 1 atm. On adding mercury, volume of air is reduced to 1.9 mL as shown. Difference in the level of mercury in two columns is:

Gaseous



(A) 700 mm (B) 200 mm (C) 900 mm (D) 760 mm

Comprehension # 2

Ideal gas equation is represented as PV = nRT. Gases present in universe were found ideal in the Boyle's temperature range only and deviated more from ideal gas behaviour at high pressure and low temperature. The deviations are explained in terms of compressibility factor Z. For ideal behaviour $Z = \frac{PV}{nRT} = 1$. The main cause to show deviations were due to wrong assumptions made about forces of attractions (which becomes significant at high pressure) and volume occupied by molecules V in (PV = nRT). If volume occupied by gaseous molecule is not negligible, then the term V should be replaced by the ideal volume which is available for free motion of each molecules of gas i.e., for 1 mole gas.

$$V_{actual} = \text{volume of container} - \text{volume occupied by molecules}$$
$$= V - b$$

where b represents the effective volume of co-volume or excluded volume occupied by molecules present in one mole of gas. Similarly, for n mole gas, ($V_{actual} = V - nb$)

The excluded volume can be calculated by considering bimolecular collisions. The excluded is the volume occupied by the sphere of 2r for each pair of molecule.

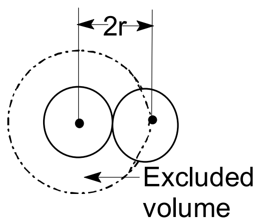
Thus, excluded volume for one pair of molecules $= \frac{4}{3}\pi(2r)^3 = \frac{4 \times 8\pi r^3}{3}$ ......(i)

$\therefore$ Excluded volume for 1 molecule $= \frac{2}{3} \times 8\pi r^3$ ....(ii)

$$= 4 \times \left(\frac{4}{3}\pi r^3\right)$$

=4 x volume of one molecule = 4V

$\therefore$ Excluded volume for N molecules = 4NV



33. Which of the following statements are correct?

(I) Larger is the value of $\frac{T_C}{P_C}$ for a gas larger would be its excluded volume

(II) Larger is the excluded volume of gas, more will be its critical volume

(III) The slope for an isochore obtained for a gas showing $P(V-b) = \text{RT}$ is $\left[\frac{R}{V-b}\right]$

(IV) The excluded volume for He is more than H_2.

(A) I, II, III (B) I, II, III,IV (C) II, III, IV (D) III, IV

34. Which of the following statements are correct?

(I) Rise in compressibility factor Z with increase in pressure is due to 'a'

(II) Rise in compressibility factor Z with increase in pressure is due to 'b'

(III) Ideal gas do not exist but is a useful concept

(IV) For 1 mole of a van der Waals's gas, $Z = 1 + \frac{\text{bP}}{\text{RT}} - \frac{a}{RTV} + \frac{ab}{RTV^2}$

(A) I, II, III, IV (B) II, III, IV (C) I, III, IV (D) I, IV

35. The ratio of coefficient of thermal expansion $\alpha = \frac{\left(\frac{\partial V}{\partial T}\right)_P}{V}$ and the isothermal compressibility $K = \frac{\left(\frac{\delta V}{\delta P}\right)_T}{V}$ for an ideal gas is:

(A) $-\frac{P}{T}$ (B) $\frac{P}{T}$ (C) $\frac{T}{P}$ (D) $-\frac{T}{P}$

Comprehension # 3

Average K.E per mole does not depend onthe nature of the gas but depends only on temperature

Average K.E per molecule $= \frac{\text{AverageK.E/mole}}{\text{N}} = \frac{3\text{RT}}{2\text{N}} = \frac{3}{2}\text{KT}$

Where K = Baltzmann constant

36. Which of the following expressions correctly represents the relationship between average molar kinetic energies of CO and N_2 molecules at the same temp?

(A) KE(CO) = K.E(N_2)

(B) KE(CO)>KE(N_2)

(C) KE(CO)<KE(N_2)

(D) cannot be predicted unless the volumes of the gases are given

37. The average K.e (in joule) of the molelcules in 8gm of methane at 27°C is

(A) 62.14×10^{-27} J (B) 72.68×10^{-21} J (C) 68.2×10^{-21} J (D) 62.14×10^{-29} J

38. Which of the following is valid at absolute zero?

(A) K.E of the gas becomes zero but the molecular motion does not become zero

(B) K.E and molecular motion becomes zero

(C) K.E of the gas decreases but does not become zero

(D) none

MATCH THE COLUMN

39. On the basis of the kinetic theory of gases (for an ideal gas)

	Column-I		**Column-II**
(A)	Vapour density of hydrogen gas (numerical value)	(P)	1
(B)	Numerical value of mass of hydrogen molecule in amu	(Q)	2
(C)	If V = constant & P is increased by 1% then T decreases by how much %	(R)	– 1
(D)	If P = 10 atm, V = .1 L, T = 24.08 K then n = ____	(S)	1/2

40.

	Column-I		**Column-II**
(A)	Total moles	(P)	$\sum_{i=1}^{n} \frac{m_i}{M_i}$
(B)	Total pressure	(Q)	$P \sum_{i=1}^{n} X_i$
(C)	Total Volume	(R)	$\sum_{i=1}^{n} P_i$
(D)	Total mass	(S)	None of these

Where : m_i : mass of i^{th} gas , P : total pressure, M_i : molecular mass of i^{th} gas, x_1 : mole fraction of i^{th} gas P_i : partial pressure of i^{th} gas

QUIZ # 1 THERMODYNAMICS PHYSICAL CHEMISTRY

True /false :

1. Mark as true or false with appropriate reason -

	Assertion	T/F	Reason
1.	ΔH is a state function		
2.	C_V is independent of T for a perfect (ideal) gas.		
3.	$\Delta U = q + w$ for every thermodynamic system at rest in the absence of external field.		
4.	A process in which final temperature equals initial temperature must be an isothermal process		
5.	U remain constant in every isothermal process in a closed system		
6.	$q = 0$ for every cyclic process.		
7.	$\Delta U = 0$ for every cyclic process.		
8.	A thermodynamic process is specified by specifying only initial and final state of system.		
9.	P-V work is usually negligible for solid and liquid.		
10.	If neither heat nor matter can enter or leave a system, that system must be isolated.		

Single Choice :

2. Which of the following statements is correct?

(A) The presence of reacting species in a covered beaker is an example of open system.

(B) There is an exchange of energy as well as matter between the system and the surroundings in a closed system.

(C) The presence of reactants in a closed vessel made up of copper is an example of a closed system.

(D) The presence of reactants in a thermos flask or any other closed insulated vessel is an example of a closed system.

3. Thermodynamics is **not** concerned about______.

(A) energy changes involved in a chemical reaction.

(B) the extent to which a chemical reaction proceeds.

(C) the rate at which a reaction proceeds.

(D) the feasibility of a chemical reaction.

4. The volume of gas is reduced to half from its original volume. The specific heat will be ______.

(A) reduce to half (B) be doubled (C) remain constant (D) increase four times

5. During an experiment an ideal gas is found to obey an additional law PV^3 = constant. The initial temperature of gas is 600 K what will be final temperature is gas expands to it's double volume.

(A) 1200 K (B) 2400 K (C) 300 K (D) 150 K

6. A gas expands slowly against a variable pressure given by $p = \frac{10}{V}$ bar, where V is volume of gas at each stage of expansion. During expansion from volume 10 L to 100 L the gas undergoes an increase in internal energy of 400 J. How much heat is absorbed by gas during expansion.

(A) 1900 J (B) 2300 J (C) 2700 J (D) 423 J

7. A system is provided 50 J of heat and work done on the system is 10 J. The change in internal energy during the process is :-

(A) 40 J (B) 60 J (C) 80 J (D) 50 J

8. The temperature of 1 mole of an ideal gas is increased by 1°C at constant pressure. The work done is

(A) –R (B) – 2R (C) – R/2 (D) – 3R

9. In an adiabatic process, no transfer of heat takes place between system and surroundings. Choose the correct option for free expansion of an ideal gas under adiabatic condition from the following.

(A) $q = 0, \Delta T \neq, 0, w = 0$ (B) $q \neq, 0, \Delta T = 0, w = 0$

(C) $q = 0, \Delta T = 0, w = 0$ (D) $q = 0, \Delta T < 0, w \neq 0$

10. The pressure-volume work for an ideal gas can be calculated by using the expression $w = -\int_{V_i}^{V_f} p_{ex}\, dV$. The work can also be calculated from the pV. plot by using the area under the curve within the specified limits. When an ideal gas is compressed (a) reversibly or (b) irreversibly from volume V_i to V_f . choose the correct option.

(A) w (reversible) = w (irreversible) (B) w (reversible) < w (irreversible)

(C) w (reversible) > w (irreversible) (D) w (reversible) = w (irreversible) + $p_{ex}.\Delta V$

11. For an ideal gas, the work of reversible expansion under isothermal condition can be calculated by using the expression $w = -nRT\, \ell n \frac{V_f}{V_i}$

A sample containing 1.0 mol of an ideal gas is expanded isothermally and reversibly to ten times of its original volume, in two separate experiments. The expansion is carried out at 300 K and at 600 K respectively. Choose the correct option.

(A) Work done at 600 K is 20 times the work done at 300 K.

(B) Work done at 300 K is twice the work done at 600 K.

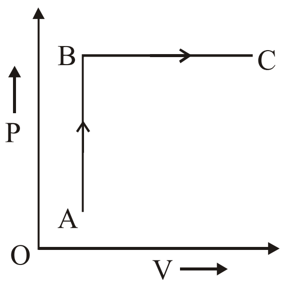
(C) Work done at 600 K is twice the work done at 300 K.

(D) $\Delta U = 0$ in both cases.

12. A thermodynamic process is shown in the following figure. In the process AB, 600 J of heat is added to the system and in BC, 200 J of heat is added to the system. The change in internal energy of the system in the process AC world be given

$P_A = 3 \times 10^4$ Pa, $V_A = 2 \times 10^{-3}$ m^3,

$P_B = 8 \times 10^4$ Pa, $V_C = 5 \times 10^{-3}$ m^3



(A) 560 J (B) 800 J

(C) 600 J (D) 640 J

13. Work done by a sample of an ideal gas in a process A is double the work done in anoher process B. The temperature rises through the same amount in the two processes. If C_A and C_B be the molar heat capacities for the two processes

(A) $C_A = C_B$ (B) $C_A > C_B$ (C) $C_A < C_B$ (D) None of these

14. One mole of a real gas is subjected to a process from (2 bar, 30 lit, 300 K) to (2 bar, 50 lit, 400 K)

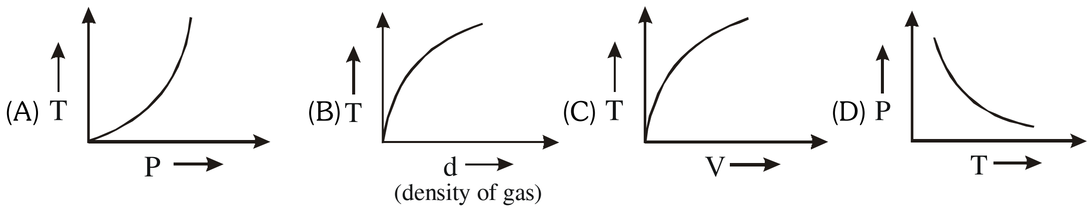
Given : $C_V = 40$ J/mol/K,

$C_p = 50$ J/mol/K

Calculate ΔU.

(A) 4000 J (B) 2000 J (C) 1000 J (D) 5000 J

15. Which of the following graph is correct for reversible adiabatic process for an ideal gas



16. An ideal gaseous sample at initial state i(P_0, V_0, T_0) is allowed to expand to volume $2V_0$ using two different process, in the first process the equation of process $2PV^2 = K_1$ and in second process the equation of the process is $PV = K_2$. Then

(A) Work done in first process will be greater that work in second process (magnitude wise)

(B) The order of value of work done cannot be compared unless we know the value of K_1 and K_2

(C) Value of work done (magnitude) in second process is greater in above expansion irrespective of the value of K_1 and K_2.

(D) I^{st} process is not possible.

17. A balloon is 1 m in diameter & contains air at 25°C & 1 bar pressure. If is filled with air isothermally & reversibly until the pressure reaches 5 bar. Assume pressure is proportional to diamerter of balloon. Caculate work done by air (atm m^3)

(A) 78 π (B) 156 π (C) 624 π (D) 625 π

18. During an adiabatic process, the pressure of a gas is found to be proportional to cube of its absolute temperature. The **poisson**'s ratio of gas is :-

(A) 3/2 (B) 7/2 (C) 5/3 (D) 9/7

Match the column:

19. Match the column : (Given process does not include chemical reaction and phase change)

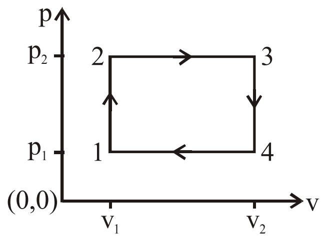
Column-I (Relation)	Column-II (Applicable to)
(A) $\Delta H = \Delta H + \Delta(PV)$	(P) Any matter undergoing any process.
(B) $\Delta H = n.C_P\,\Delta T$	(Q) Isochoric process involving any substance
(C) $q = \Delta U$	(R) Ideal gas, under any process
	(S) Ideal gas under isothermal process

Subjective:

20. One litre of N_2 and 2 litre of O_2 under identical condition of T & P are mixed. Find the volume of mixture if the pressure of the mixture is reduced to half of initial value.

21. Two glass bulbs A and B are connected by a very small tube having a stop cock. Bulb A has a volume of 100 ml and contained the gas, while bulb B was empty on opening the stop cock, the pressure fell down to 40% at constant temperature. Find out the volume of bulb B in mL.

22. A gas has been subjected to an isochoric and isobaric cycle. Plot the graph of this cycle in the pressure-density, V–T and P-T co-ordinates.



ANSWER-KEY (QUIZ # 1)

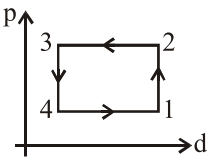
1. (1-F, 2-F, 3 -F, 4-F, 5-F, 6-F, 7-T, 8-F, 9-T, 10- F) **2.** (C) **3.** (C) **4.** (C)

5. (D) **6.** (C) **7.** (B) **8.** (A) **9.** (C) **10.** (B) **11.** (C,D)

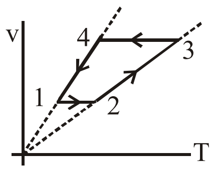
12. (A) **13.** (B) **14.** (C) **15.** (B) **16.** (C) **17.** (A) **18.** (A)

19. (A)→(P,Q,R,S) ; (B)→(R,S) ; (C)→(Q) **20.** 6 lt. **21.** (6)

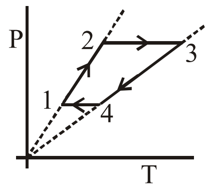
22. (i)



(ii)



(iii)



QUESTION BANK

THERMODYNAMICS & THERMOCHEMISTRY

SINGLE CHOICE QUESTIONS

1. A 10 g piece of iron (C = 0.45 J/g°C) at 100°C is dropped into 25 g of water (C = 4.2 J/g°C) at 27°C. Find temperature of the iron and water system at thermal equilibrium.

(A) 30°C (B) 33°C (C) 40°C (D) None of these

2. When freezing of a liquid takes place in a system :

(A) may have $q > 0$ or $q < 0$ depending on the liquid

(B) is represented by $q > 0$

(C) is represented by $q < 0$

(D) has $q = 0$

3. A sample of liquid in a thermally insulated container (a calorimeter) is stirred for 2 hr. by a mechanical linkage to a motor in the surrounding, for this process :

(A) $w < 0$; $q = 0$; $\Delta U = 0$ (B) $w > 0$; $q > 0$; $\Delta U > 0$

(C) $w < 0$; $q > 0$; $\Delta U = 0$ (D) $w > 0$, $q = 0$; $\Delta U > 0$

4. The work done by the gas in reversible adiabatic expansion process is :

(A) $\frac{P_2V_2 - P_1V_1}{\gamma - 1}$ (B) $\frac{nR(T_1 - T_2)}{\gamma - 1}$ (C) $\frac{P_2V_2 - P_1V_1}{\gamma}$ (D) None of these

5. For a reversible adiabatic ideal gas expansion $\frac{dP}{P}$ is equal to :

(A) $\gamma\frac{dV}{V}$ (B) $-\gamma\frac{dV}{V}$ (C) $\left(\frac{\gamma}{\gamma-1}\right)\frac{dV}{V}$ (D) $\frac{dV}{V}$

6. For a process to be spontaneous at constant *T* and *P* :

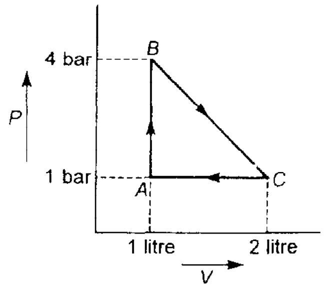
(A) $(\Delta G)_{system}$ must be negative (B) $(\Delta G)_{system}$ must be negative

(C) $(\Delta S)_{system}$ must be positive (D) $(\Delta S)_{system}$ must be negative

7. A rigid and insulated tank of 3 m^3 volume is divided into two compartments. One compartment of volume of 2 m^3 contains an ideal gas at 0.8314 MPa and 400 K and while the second compartment of volume 1 m^3 contains the same gas at 8.314 MPa and 500 K. If the partition between the two compartments is ruptured, the final temperature of the gas is :

(A) 420 K (B) 450 K (C) 480 K (D) None of these

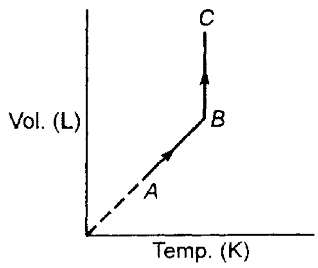
8. One mole of an ideal gas is carried through the reversible cyclic process as shown in figure. The max. temperature attained by the gas during the cycle :



(A) $\frac{7}{6R}$ (B) $\frac{12}{49R}$

(C) $\frac{49}{12R}$ (D) None of these

9. A triatomic linear (neglect vibration degree of freedom) gas of two mole is taken through a reversible process ideal starting from A as shown in figure.



The volume ratio $\frac{V_B}{V_A} = 4$. If the temperature at A is -73° C

(i) Work done by the gas in AB process is :

(A) 6.16 kJ (B) 308.3 kJ (C) 9.97 kJ (D) 0 J

(ii) Total enthalpy change in both steps is :

(A) 3000 R (B) 4200 R (C) 2100 R (D) 0

10. 10 mole of an ideal gas is heated at constant pressure of one atmosphere from 27°C to 127°C. If $C_{v,m} = 21.686 + 10^{-3}$ T, then ΔH for the process is :

(A) 3000 J (B) 3350 J (C) 3700 J (D) 30350

11. 2 mole of an ideal monoatomic gas undergoes a reversible process for which $PV^2 = C$. The gas is expanded from initial volume of 1 L to final volume of 3 L starting from initial temperature of 300 K. Find ΔH for the process :

(A) –600 R (B) –1000 R (C) –3000 R (D) None of these

12. One mole of an ideal monoatomic gas at 27°C is subjected to a reversible isoentroptic compression until final temperature reached to 327°C. If the initial pressure was 1.0 atm then find the value of In P_2: (Given : In 2 = 0.7)

(A) 1.75 atm (B) 0.176 atm (C) 1.0395 atm (D) 2.0 atm

13. Two mole of an ideal gas is expanded irreversibly and isothermally at 37°C until its volume in doubled and 3.41 kJ heat is absorbed from surrounding. ΔS_{total} (system + surrounding) is :

(A) –0.52 J/K (B) 0.52 J/K (C) 22.52 J/K (D) 0

14. For a perfectly crystalline solid $C_{p,m} = aT^3 + bT$, where a and b are constant. If $C_{p,m}$ is 0.40 J/K mol at 10 K and 0.92 J/K mol at 20 K, then molar entropy at 20 K is :

(A) 0.92 J/K mol (B) 8.66 J/K mol (C) 0.813 J/K mol (D) None of these

15. Fixed mass of an ideal gas contained in a 24.63 L sealed rigid vessel at 1 atm is heated from –73°C to 27°C. Calculate change in Gibb's energy if entropy of gas is a function of temperature as

$S = 2 + 10^{-2}$ T (J/K): (Use 1 atm L = 0.1 kJ)

(A) 1231.5 J (B) 1281.5 J (C) 781.5 J (D) 0

16. Calculate $\Delta_f G°$ for (NH_4Cl, s) at 310 K.

Given : $\Delta_f H°(NH_4Cl, s) = -314.5$ kJ/mol ; $\Delta_r C_p = 0$

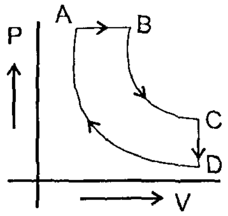
$S^o_{N_2(g)} = 192\,JK^{-1}\,mol^{-1}$; $S^o_{H_2(g)} = 130.5\,JK^{-1}\,mol^{-1}$;

$S^o_{Cl_2(g)} = 233\,JK^{-1}\,mol^{-1}$; $S^o_{NH_4Cl(s)} = 99.5\,JK^{-1}\,mol^{-1}$

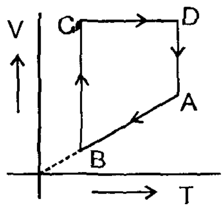
All given data at 300 K.

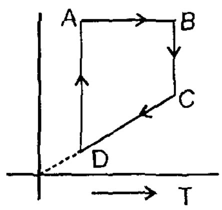
(A) –198.56 kJ/mol (B) – 426.7 kJ/mol (C) – 202.3 kJ/mol (D) None of these

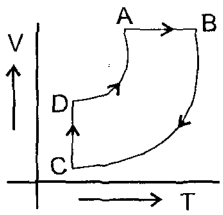
17. A cyclic proces ABCD is shown in PV diagram for an ideal gas.

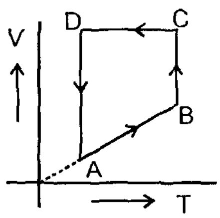


Which of the following diagram represents the same process

(A)


(B)


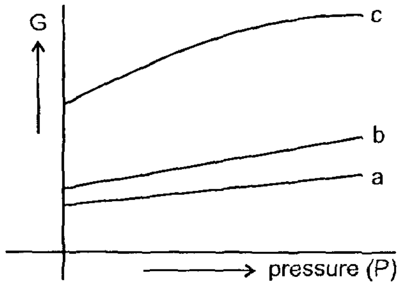
(C)


(D)


18. The change in free energy accompanied by the isothermal reversible expansion of 1 mol of an ideal gas when it doubles its volume is ΔG_1 The change in free energy accompanied by sudden isothermal irreversible doubling volume of 1 mole of the same gas is ΔG_2 Ratio of ΔG_1 and ΔG_2 is

(A) 1 (B) $\frac{1}{2}$ (C) -1 (D) $-\frac{1}{2}$

19. The following curve represents the variation of Gibbs function 'G' with pressure at constant temperature.

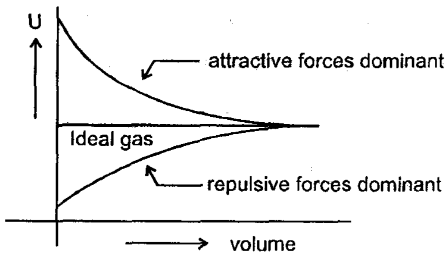


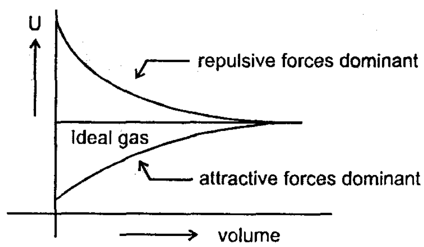
Correct match of given plots with the physical state of a substance is

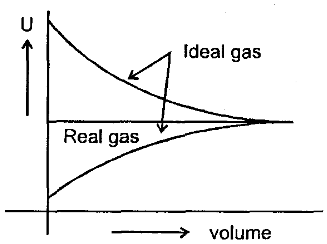
(A) c – solid , a – gas , b – liquid (B) c – gas, b – liquid, a – solid

(C) a – liquid, b – solid, c – gas (D) c – gas, b – solid, a – liquid

20. Which of the following diagram correctly represents the variation of internal energy (U) of gas under expansion at constant temperature.

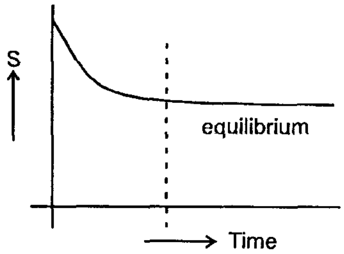
(A)


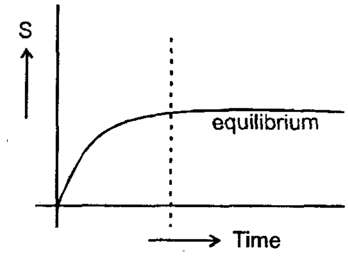
(B)


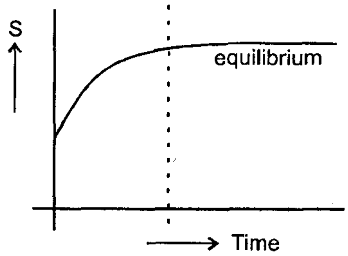
(C)


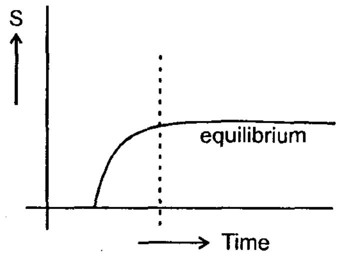
(D) None of the above

21. Which correctly represents the entropy (s) of an isolated system during a process.

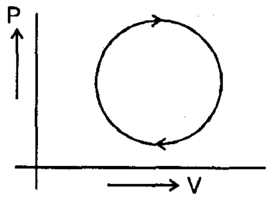
(A) 

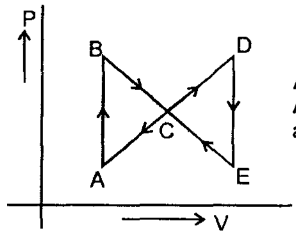
(B) 

(C) 

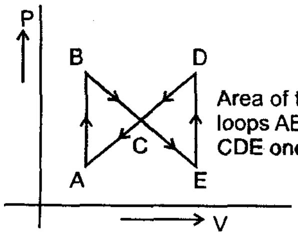
(D) 

22. Which of the following represents positive work i.e. work of compression

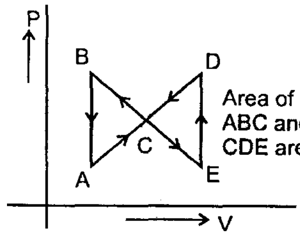
(A) 

(B) 

Area of loops ABC and CDE are equal

(C) 

Area of the loops ABC and CDE one equal

(D) 

Area of loops ABC and CDE are equal

23. For a perfectly crystalline solid $C_{m} = aT^3$, where *a* is constant. If $C_{p\ m}$ is 0.42 J/K-mol at 10 K, molar entropy

(A) 0.42 J/K-mol (B) 0.14 J/K-mol (C) 4.2 J/K-mol (D) Zero

24. One mole of an ideal monoatomic gas expands isothermally against constant external pressure of 1 atm from initial volume of 1L to a state where its final pressure becomes equal to external pressure. If initial temperature of gas is 300 K then total entropy change of system in the above process is : [R = 0.082 L atm mol^{-1} K^{-1} = 8.3 J mol^{-1} K^{-1}]

(A) 0 (B) $R\ell n\,(24.6)$ (C) $R\ell n\,(2490)$ (D) $\frac{3}{2}R\ell n\,(24.6)$

25. At 1000 K water vapour at 1atm. has been found to be dissociated into H_2 and O_2 to the extent of 3 × 10^{-5} %. Calculate the standard free energy decrease of the system, assuming ideal behaviour.

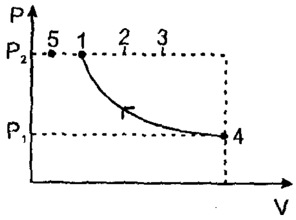
(A) $-\Delta G = 90{,}060$ cal (B) $-\Delta G = 20$ cal (C) $-\Delta G = 480$ cal (D) $-\Delta G = -45760$ cal

26. When 1 mole of an ideal gas at 20 atm pressure and 15 L volume expands such that the final pressure becomes 10 atm and the final volume become 60 L. Calculate entropy change for the process

($C_{p,\,m}$ = 30.96 J mole^{-1} K^{-1})

(A) 80.2J.k^{-1}mol^{-1} (B) 62.42 kJ.k^{-1}mo|$^{-1}$ (C) 120 × 10^{2} Jk^{-1}mol^{-1} (D)27.22 J.K^{-1}mol^{-1}

27. An ideal gas is taken from the same initial pressure P_1 to the same final pressure P_2 by three different processes. If it is known that point 1 corresponds to a reversible adiabatic and point 2 corresponds to a single stage adiabatic then



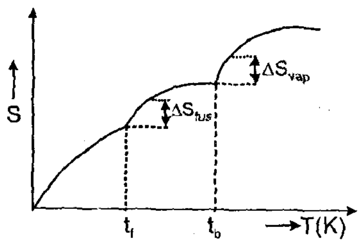
(A) Point 3 may be a two stage adiabatic.

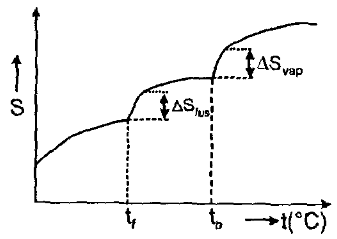
(B) the average K.E. of the gas is maximum at point 1

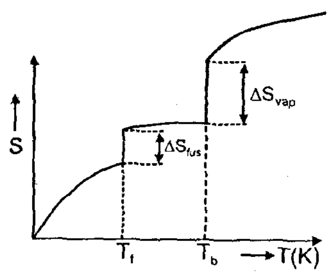
(C) Work done by surrounding in reaching point number '3' will be maximum

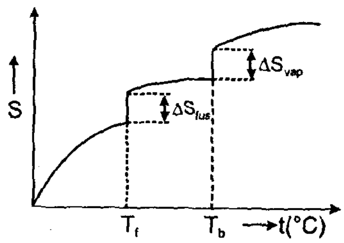
(D) If point 4 and point 5 lie along a reversible isotherm then $T_5 < T_1$.

28. Which of the following graphs best illustrates the variation of entropy of a substance with temp.

(A)


(B)


(C)


(D)


29. During winters, moisture condenses in the form of dew and can be seen on plant leaves and grass. The entropy of the system in such cases decreases as liquids possess lesser disorder as compared to gases. With reference to the second law, which statement is **correct,** for the above process ?

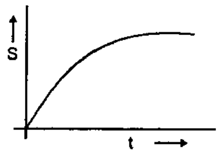
(A) The randomness of the universe decreases

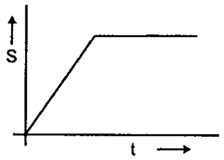
(B) The randomness of the surroundings decreases

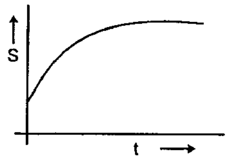
(C) Increase is randomness of surroundings equals the decrease in randomness of system

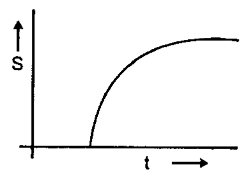
(D) The increase in randomness of the surroundings is greater as compared to the decrease in randomness of the system.

30. A copper block of mass 'm' at temperature 'T_1' is kept in the open atmosphere at temperature 'T_2' where $T_2 > T_1$. The variation of entropy of the copper block with time is best illustrated by

(A)


(B)


(C)


(D)


31. For which process will ΔH° and ΔG° be expected to be most similar.

(A) $2Al(s) + Fe_2O_3(s) \longrightarrow 2Fe(s) + Al_2O_3(s)$

(B) $2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$

(C) $2NO_2(g) \longrightarrow N_2O_4(g)$

(D) $2H_2(g) + O_2(g) \longrightarrow 2H_2O(g)$

ONE OR MORE THAN ONE CORRECT OPTIONS

32. In an isothermal irreversible expansion of an ideal gas as per IUPAC sign convention :

(A) $\Delta U = 0$ (B) $\Delta H = 0$ (C) $w = nRT \ln \frac{P_1}{P_2}$ (D) $w = -q$

33. In an adiabatic process is one in which :

(A) energy is transferred as heat

(B) no energy is transfer as heat

(C) $\Delta U = w$

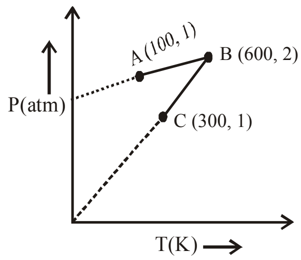
(D) the temp, of gas increases in a reversible adiabatic compression

34. In a adiabatic process, the work involved during expansion or compression of an ideal gas is given by :

(A) $nC_V\Delta T$ (B) $\frac{nR}{\gamma - 1}(T_2 - T_1)$

(C) $-nRP_{ext}\left[\frac{T_2P_1 - T_1P_2}{P_1P_2}\right]$ (D) $-2.303\,RT \log V_2/V_1$

35. One mole of an ideal gas is subjected to a two step reversible process ($A \rightarrow B$ and $B \rightarrow C$). The pressure at *A* and C is same. Mark the correct statement(s) :



(A) Work involved in the path *AB* is zero

(B) In the path *AB* work will be done on the gas by the surrounding

(C) Volume of gas at *C* = 3 × volume of gas at *A*

(D) Volume of gas at *B* is 16.42 litres

36. Assume ideal gas behaviour for all the gases considered and neglect vibrational degrees of freedom. Separate equimolar samples of Ne, O_2, CO_2 and SO_2 were subjected to a two process as mentioned. Initially all are at same state of temperature and pressure.

Step I $\longrightarrow$ All undergo reversible adiabatic expansion to attain same final volume, which is double the original volume thereby causing the decreases in their temperature.

Step II $\longrightarrow$ After step I all are given appropriate amount of heat isochorically to restore the original temperature.

Mark the correct option (s) :

(A) Due to step I only, the decrease in temperature will be maximum for Ne

(B) During step II, heat given will be minimum for SO_2

(C) There will be no change in internal energy for any of the gas after both the steps of process are competed

(D) The P–V graph of O_2 and CO_2 will be same

37. Which of the following is/are correct?

(A) $\Delta H = \Delta U + \Delta (PV)$ when P and V both changes

(B) $\Delta H = \Delta U + P\Delta V$ when pressure is constant

(C) $\Delta H = \Delta U + V\Delta P$ when volume is constant

(D) $\Delta H = \Delta U + P\Delta V + V\Delta P$ when P and V both changes

38. The value of $\Delta H_{transition}$ of C (graphite) $\longrightarrow$ C (diamond) is 1.9 kJ/mol at 25°C entropy of graphite is higher than entropy of diamond. This implies that :

(A) C (diamond) is more thermodynamically stable than C (graphite) at 25°C

(B) C (graphite) is more thermodynamically stable than C (diamond) at 25°C

(C) diamond will provide more heat on complete combustion at 25°C

(D) $\Delta G_{transition}$ of C (diamond) $\longrightarrow$ C (graphite) is –ve

39. Which of the following statement(s) is/are false?

(A) All adiabatic processes are isoentropic (or isentropic) processes

(B) When $(\Delta G_{system})_{T,p} < 0$; the reaction must be exothermic

(C) $dG = VdP - SdT$ is applicable for closed system, both PV and non-PV work

(D) The heat of vaporisation of water at 100°C is 40.6 kJ/mol. When 9 gm of water vapour condenses to liquid at 100°C of 1 atm, then $\Delta S_{system} = 54.42$ J/K

40. Which of the following statement(s) is/are true?

(A) $\Delta E = 0$ for combustion of $C_2H_6(g)$ in a sealed rigid adiabatic container

(B) $\Delta_f H°$ (S, monoclinic) $\neq 0$

(C) If dissociation energy of CH_4 (g) is 1656 kJ/mol and C_2H_6(g) is 2812 kJ/mol, then value of C—C bond energy will be 328 kJ/mol

(D) If $\Delta H_f (H_2O,g) = -242$kJ/mol; $\Delta H_{vap} (H_2O, l) = 44$ kJ/mol , then, $\Delta_f H° (OH^-, aq.)$ will be –142 kJ/mol

41. One mole of an ideal diatomic gas ($C_v = 5$ cal) was transformed from initial 25°C and 1 L to the state when temperature is 100°C and volume 10 L. Then for this process(R = 2 calories/mol/K) (take calories as unit of energy and Kelvin for temperature)

(A) $\Delta H = 525$

(B) $\Delta S = 5 \ln \dfrac{373}{298} + 2 \ln 10$

(C) $\Delta E = 525$

(D) ΔG of the process can not be calculated using given information.

42. Which of the following statement(s) is/are **false ?**

(A) $\Delta_f S$ for $\dfrac{1}{2} Cl_2(g) \rightarrow Cl(g)$ is positive.

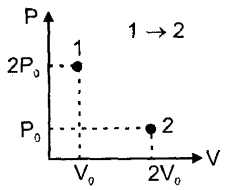
(B) $\Delta E < 0$ for combustion of CH_4(g) in a sealed container with rigid adiabatic system.

(C) ΔG is always zero for a reversible process in a closed system.

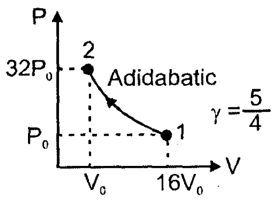
(D) $\Delta G°$ for an ideal gas reaction is a function of pressure.

MATCH THE COLUMN

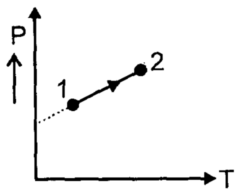
43. **Column -1** **(For a definite amount of an ideal gas)** | **Column -II** **[Enthalpy change / work done]**

(A) 

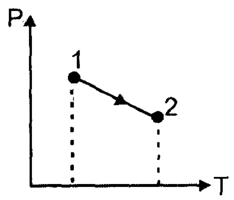
(p) $\Delta H > 0$

(B) 

(q) $\Delta H = 0$

(C) 

(r) $W > 0$

(D) 

(s) $W < 0$

44.

Column -1	**Column -II**
(A) For spontaneous reaction	(p) $\sum(\text{B.E.})_R - \sum(\text{B.E.})_P$
(B) For endothermic reaction	(q) $\Delta H = \Delta E$
(C) Bond dissociation energy	(r) $(\Delta G)_{T,P} = -\text{ ve}$
(D) For solids and liquids in a thermochemical reaction	(s) $\sum H_P > \sum H_R$

45.

Column -1	**Column -II**
(A) $\Delta H = q_p$	(p) $\Delta H_f^\circ = 0$
(B) Kirchhoff's equation	(q) A definite quantity
(C) H^+ (aq.)	(r) Path function
(D) Spontaneous process	(s) $(\Delta G)_{T,P} > 0$
	(t) $\Delta S^\circ_{\text{Total}} > 0$
	(u) $\Delta H_2 - \Delta H_1 = \Delta C_p (T_2 - T_1)$

46.

Column -1	**Column -II**
(A) Perfectly crystalline solid	(p) $\Delta H_{sys} = 0$
(B) Reversible cyclic process	(q) $\lim\limits_{T \to 0} S \to 0$
(C) Irreversible cyclic process	(r) $\Delta S_{\text{surrounding}} = 0$
(D) Any adiabatic process	(s) $\Delta U_{sys} = 0$
	(t) net heat change is zero

COMPREHENSION BASED QUESTIONS

Passage for Qu. No. 47 - 48

Chemical reactions are invariably associated with transfer of energy either in form of heat or light. In the laboratory, heat changes in physical and chemical processes are measured with an instrument called calorimeter. Heat change in the process is calculated as :

$q = ms\,\Delta T$ $\quad$ s = specific heat J/K-gm

$= C\Delta T$ $\quad$ C = heat capacity J/K

In Bomb calorimeter measurement are taking at constant volume.

So heat of reaction at constant volume : $q = \Delta U$ (Internal energy change)

Here heat is released by combustioin so sign of ΔU and heat should be negative. In water colarimeter measurement are taking at constant pressure.

So, $q_p = \Delta H$

$q_p = q_v + P\Delta V$

$\Delta H = \Delta U + \Delta n_g RT$

47. Match the list-I and list - II and select the answer :

List - I (Unbalanced reaction)	List-II
(a) $CH_4(g) + O_2(g) \longrightarrow CO_2(g) + H_2O\ (\ell)$	(i) $\Delta H = \Delta U$
(b) $C_6H_6(\ell) + O_2(g) \longrightarrow 6CO_2(g) + 3H_2O\ (\ell)$	(ii) $\Delta H = \Delta U - RT$
(c) $S(\ell) + O_2(g) \longrightarrow SO_2(g)$	(iii) $\Delta H = \Delta U - 2RT$
(d) $2CO(g) + O_2(g) \longrightarrow 2CO_2(g)$	(iv) $\Delta H = \Delta U - \frac{3}{2}RT$

(A) (a) - (iii) (b) - (iv) (c) - (i) (d) - (ii)
(B) (a) - (ii) (b) - (iii) (c) - (i) (d) - (iv)
(C) (a) - (iii) (b) - (iv) (c) - (ii) (d) - (i)
(D) (a) - (ii) (b) - (i) (c) - (iii) (d) - (iv)

48. The heat capacity of a bomb calorimeter is 100 J/K. When 1 gm C_7H_{16} (ℓ) was burnt in this calorimeter, the temperature increase by 5°C. The value of ΔH in KJ/mole at 300 K temperature will be -

(A) – 50 (B) – 40.03 (C) – 59.97 (D) none

Passage for Qu. No. 49 - 51

J.W. Gibbs and H.Von Helmoltz had given two equation and are known as Gibbs-Helmholtz equation. One equation can be expressed in terms of change in free energy (ΔG) and enthalpy (ΔH) while other can be expressed in terms of change in internal energy (ΔE) and work function (ΔW)

$$\Delta G = \Delta H + T.d\left(\frac{\Delta G}{dT}\right)_P \quad \text{............ (1)}$$

$$\Delta W = \Delta E + T.d\left(\frac{\Delta W}{dT}\right)_V \quad \text{............ (2)}$$

Where T is temperature equation (1) is obtained at constant pressure while equation (B) is obtained at constant volume system. It is observed that for the reaction.

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$$

Free energy change at 25°C is –33 kJ

while at 35°C is – 28 kJ which are at a constant pressure.

49. What woujd be the difference between enthalpy change at 25°C and 35°C for a given reaction :

(A) 4 kJ (B) 5 kJ (C) 3 kJ (D) zero

50. What would be the free energy change at 30°C -

(A) 30.5 kJ (B) 33 kJ (C) – 28 kJ (D) – 30.5 kJ

51. Internal energy change at 25°C is E_1 while at 35°C is E_2 then -

(A) $E_1 = E_2$ (B) $E_2 > E_1$ (C) $E_1 > E_2$ (D) None of these

Passage for Qu. No. 52 - 56

A gaseous sample is generally allowed to do only expansion/compression type of work against its surroundings. The work done in case of an irreversible expansion (in the intermediate stages of expansion/compression the states of gases are not defined). The work done can be calculated using $dw = -p_{ext}\, dV$

while in case of reversible process the work done can be calculated using

dw = – PdV where P is pressure of gas at some intermediate stages. Like for an isothermal reversible process.

Since $P = \frac{nRT}{V}$ so

$$w = \int dw = \int_{V_i}^{V_f} \frac{nRT}{V}\,.\,dV = -nRT\ \ln\left(\frac{V_f}{V_i}\right)$$

Since dw = - PdV so magnitude of work done can also be calculated by calculating the area under the PV curve of the reversible process in PV diagram.

52. An ideal gaseous sample at initial state i (P_0 V_0,T_0) is allowed to expand to volume $2V_0$ using two different process; in the first process the equation of process is $PV^2 = K_1$ and in second process the equation of the process is $PV = K_2$. Then

(A) work done in first process will be greater than work in second process (magnitude wise)

(B) The order of values of work done can not be compared unless we know the value of K_1 and K_2.

(C) value of work done (magnitude) in second process is greater in above expansion irrespective of the value

(D) I^{st} process is not possible

53. There are two samples of same gas initially under similar initial state. Gases of both the samples are expanded. I^{st} sample using reversible isothermal process and II^{nd} sample using reversible adiabatic process till final pressures of both the samples becomes half of initial pressure, then

(A) Final volume of I^{st} sample < final volume of II^{nd} sample

(B) Final volume of II^{nd} sample < final volume of I^{st} sample

(C) final volumes will be equal

(D) Information is insufficient

54. In the above problem

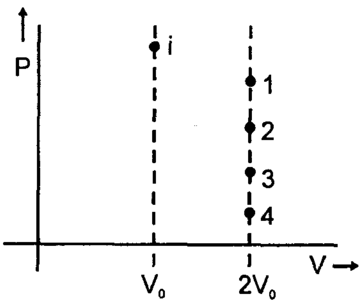
(A) work done by gas in I^{st} sample > work done by gas in II^{nd} sample

(B) work done by gas in II^{nd} sample > work done by gas in I^{st} sample

(C) work done by gas in I^{st} sample = work done by gas in II^{nd} sample

(D) none of these

55. If four identical samples of an ideal gas initially at similar state (P_0, V_0, T_0) are allowed to expand to double their volumes by four differen processes.



I : by isothermal irreversible process

II : by reversible process having equation $P^2 V$ = constant

III : by reversible adiabatic process

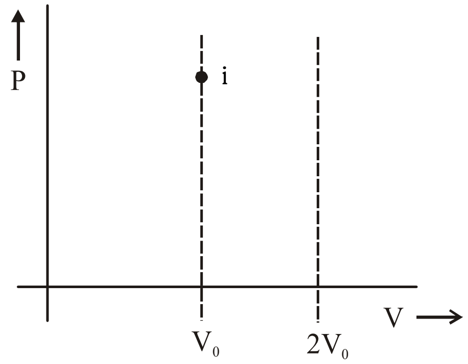
IV : by irreversible adiabatic expansion against constant external pressure.

Then in the graph shown the final state is represented by four different points then, the correct match can be :

(A) 1 - 1, 2 - II, 3 - III, 4 - IV (B) 1 - II, 2 -1, 3 - IV, 4 - III

(C) 2 - III, 3 - II, 4 - I, 1 - IV (D) 3 - II , 1 -1, 3 - IV, 4 - III

56. Two samples (initially under same states) of an ideal gas are first allowed to expand to double their volume using irreversible isothermal expansion against constant external pressure, then samples are returned back to their original volume first by reversible adiabatic process and second by reversible process having equation PV^2 = constant then :



(A) final temperature of both samples will be equal

(B) final temperature of first sample will be greater than of second sample

(C) final temperature of second sample will be greater than of first sample

(D) none of these

Passage for Qu. No. 57 - 59

Spontaneity of any process can be predicted with the help of ΔS_{total}. But this requires calculation of changes in system as will as surroundings. If some criteria (depending upon the system only) can be developed for checking spontaneity under specific conditions, then that would be a more useful parameter. The criteria can be derived from Clausius inequality.

$T\,dS \geq q$, > sign for rev. process

= sign for rev. process

or $TdS > dU - W$ for an irr. process

or $T\,dS_{sys} > dU_{sys} + p\,dV$ [consider no non-PV work]

If U and V are constant.

$T\,dS_{sys} > 0$

or $(dS)_{U,V} > 0$ for spontaneous process

if V and T are constant

$T\,dS > dU$

or $dU - T\,dS < 0$

As temperature is constant, $dU - d(TS) < 0$ or $d(U - TS) < 0$

Another state function A (Helmholtz's function) = U – TS

A decrease in Helmholtz function (A) under constant volume and temperature is the criteria of spontaneity of a process.

57. For a spontaneous process, if entropy and volume are constant, the internal energy of system must

(A) increase (B) decrease (C) remain constant (D) be zero.

58. If a state function G is defined such as G = H - TS, then dG must decrease for a spontaneous process (involving only PVwork) occuring at

(A) Constant volume and temperature (B) Constant pressure and temperature

(C) Constant volume and pressure (D) Constnat entropy and volume

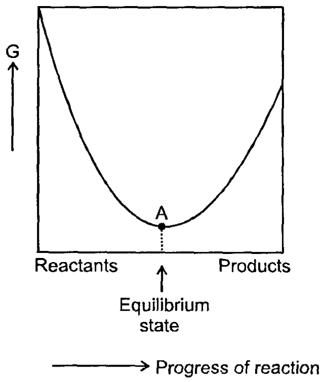
59. For an irreversible cyclic process $\oint \frac{dq}{T}$ is :

(A) equal to zero (B) greater than zero

(C) less than zero (D) equal to change in Gibb's energy.

Passage for Qu. No. 60 - 62

For a reversible reaction at constant temperature and at constant pressure the equilibrium composition of reaction mixture corresponds to the lowest point on **Gibbls energy Vs progress of reaction** diagrams as shown. At equilibrium Gibbs energy of reaction is equal to Zero.

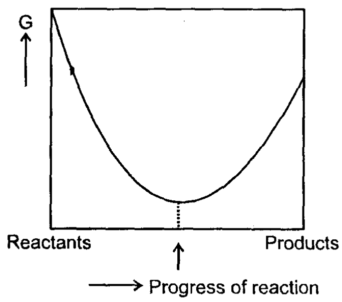


60. The value of $\log_{10} k_{eq}$ is equal to [k_{eq} is the equilibrium constant]

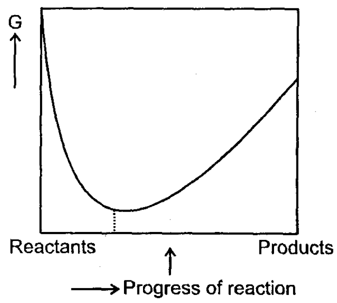
(A) $-\frac{\Delta G^{\circ}}{RT}$ (B) $-\frac{T\Delta S^{\circ} - \Delta H^{\circ}}{2.303RT}$ (C) $\frac{\Delta H^{\circ} - T\Delta S^{\circ}}{RT}$ (D) $\frac{RT}{T\Delta S^{\circ} - \Delta H^{\circ}}$

61. Which diagram represents the large value of equilibrium constant for the reversible reaction

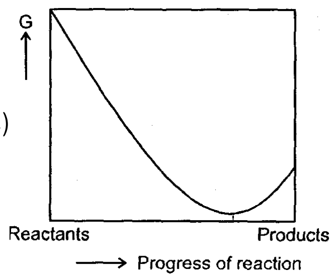
(A)



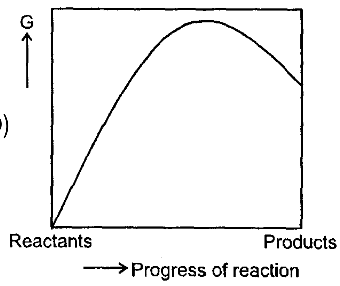
(B)



(C)



(D)



62. For a reaction $M_2O_{(s)} \longrightarrow 2M_{(s)} + \frac{1}{2}O_{2(g)}$, $\Delta H = 30$ KJ/mol and $\Delta S = 0.07$ KJ/mol/K at 1 atm. The reaction would not be spontaneous at temperatures

(A) > 428 K (B) < 428 K (C) < 100 K (D) > 100 K

Passage for Qu. No. 63 - 65

The contributions of both heat (enthalpy) and randomness (entropy) shall be considered to the overall spontaneity of a process. When deciding about the spontaneity of a chemical reaction or other process, we define a quantity called the Gibb's energy change (ΔG).

$\Delta G = \Delta H - T\Delta S$

where, ΔH = Enthalpy change ; ΔS = Entropy change ; T = Temperature in kelvin.

If $\Delta G < 0$, Process is spontaneous; $\Delta G = 0$, Process is at equilibrium, $\Delta G > 0$, Process is non-spontaneous.

63. For the change H_2O (s, 273 K, 2 atm) $\longrightarrow$ H_2O (ℓ, 273K, 2 atm), choose the correct option. ;

(A) $\Delta G = 0$ (B) $\Delta G < 0$ (C) $\Delta G > 0$ (D) None

64. 5 mol of liquid water is compressed from 1 bar to 10 bar at constant temperature. Change in Gibb's energy (ΔG) in Joule is : [Density of water = 1000 kg/m^3]. ;

(A) 18 (B) 225 (C) 450 (D) None of these

65. Quick lime, CaO is produced by heating limestone, $CaCO_3$ to drive off CO_2 gas.

$CaCO_3$ (s) $\longrightarrow$ CaO (s) + CO_2 (g), ($\Delta H°$) = 180 KJ, $\Delta S°$ = 150 J/K.

Assuming that variation of enthalpy change and entropy change with temperature to be negligible, choose the correct option:

(A) Decomposition of $CaCO_3$ (s) is never spontaneous.

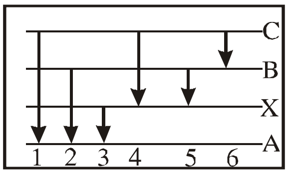
(B) Decomposition of $CaCO_3$ (s) becomes spontaneous when temperature is less than 900°C.

(C) Decomposition of $CaCO_3$ (s) becomes non-spontaneous when temperature is greater than 1000°C

(D) Decomposition of $CaCO_3$ (s) becomes spontaneous when temperature is greater than 927°C.

SINGLE CHOICE CORRECT

1. The figure indicates the energy level diagram for the origin of six spectral lines in emission spectrum(e.g. line no. 5 arises from the transition from level B to X) which of the following spectral lines will not occur in the absorption spectrum :–



(A) 1, 2, 3 (B) 3, 2 (C) 4, 5, 6 (D) 3, 2, 1

2. The energy of an electron in the first orbit of He^+ is -871.6×10^{-20} J. The energy of the electron in the first orbit of hydrogen would be :

(A) -871.6×10^{-20} J (B) -435.8×10^{-20} J (C) -217.9×10^{-20} J (D) -108.9×10^{-20} J

3. Ratio of total energy of an e^- in 1^{st} orbit and potential energy of e^- in 4^{th} orbit of a H-like specie is:

(A) 2 (B) 4 (C) 8 (D) 16

4. An e^- of He^+ makes a tranistion and emits 6^{th} line of Balmer series. Similar wavelength of radiation is absorbed by hydrogen like specie to give 9^{th} line of paschen series in its spectrum. The value of Z of the hydrogen like specie is

(A) 1 (B) 2 (C) 3 (D) 4

5. Potential energy of electron present in 2^{nd} orbit of Li^{2+} is : (r_0 = Radius of 1^{st} Bohr's orbit)

(A) $\frac{e^2}{4\pi\epsilon_0 r_0}$ (B) $-\frac{3e^2}{4\pi\epsilon_0 r_0}$ (C) $-\frac{3e^2}{16\pi\epsilon_0 r_0}$ (D) $-\frac{9e^2}{16\pi\epsilon_0 r_0}$

6. Suppose an electron is attracted towards the origin by a force $\frac{k}{r}$ where 'k' is a constant and 'r' is the distance of the electron from the origin. By applying Bohr model to this system, the radius of the n^{th} orbital of the electron is found to be 'r_n' and the kinetic energy of the electron to be 'T_n'. Then which of the following is true?

(A) $r_n \propto n^2$ (B) T_n independent of n

(C) $T_n \propto n^2$ (D) $T_n \propto \frac{1}{n}$

7. If electron is present in n_1 orbit of hydrogen atom and n_2 orbit of He^+ ions then which of the following relation is correct for the same value of potential energy of electron in both the chemcial specie

(A) $4n_2^2 = n_1^2$ (B) $n_1 = 2n_2$ (C) $2n_1 = n_2$ (D) $n_2 = 4n_1$

COMPREHENION BASED QUESTION

COMPREHENION QUESTION 8 to 10.

During one of its ambitions project "ISRO" has gathered information through its satellites for a planet in another galaxy. As far as structure of atoms is considered , it is very similar to the structure of an atom in our planet except energy of an orbit in which electron resides is $(E_n) = -\frac{45 \times Z}{(n+1)}$ eV/atom.

8. Calculate wavelength (in Å) of photon emitted when electron makes transition from 7th excited state to 1st excited state in H-atom of that planet :

(A) 1240 Å (B) 12400 Å (C) 310 Å (D) 3100 Å

9. If photon of wavelength 7 nm strikes on an atom of hydrogen on that planet ,wavelength of ejected electron will be approximately:

(A) 0.1 Å (B) 1 Å (C) 5 Å (D) 11 Å

10. If collection of H-atom and He^+ ion makes transition from 2^{nd} orbit and 5^{th} orbit respectively. Total different spectral lines emitted :

(A) 10 (B) 11 (C) 6 (D) 9

ONE OR MORE THAN ONE CHOICE CORRECT

11. For two Bohr orbits n_1 & n_2 which follow the relationships,

$(n_2^2 - n_1^2) = 21$ and $(n_2 - n_1) = 3$.

If an electron makes transition from n_2 to n_1 directly then :

(A) Third longest wavelength of Balmer series is emitted

(B) Longest frequency of Lyman series is emitted

(C) Longest frequency of Balmer series is emitted

(D) Only one spectral line is emitted

12. Select the correct expression(s) according to Bohr model -

(A) $TE = \dfrac{-2\pi^2k^2z^2e^4m}{n^2h^2}$ (B) $PE = \dfrac{-4\pi^2k^2z^2e^4m}{n^2h^2}$ (C) $v = \dfrac{2\pi kze^2}{nh}$ (D) $\dfrac{hc}{\lambda} = 13.6z^2\left(\dfrac{1}{n_1^2} - \dfrac{1}{n_2^2}\right)$

MATCH THE COLUMN

13. Match the column

	Column-I	Column-II
(A)	Electron moving in 2^{nd} orbit in He^+ ion	(P) Radius of orbit in which electron is moving is 0.529 Å
(B)	Electron moving in 3^{rd} orbit in H-atom	(Q) Total energy of electron is (–)13.6 × 9eV
(C)	Electron moving in 1^{st} orbit in Li^{+2} ion	(R) Velocity of electron is $\dfrac{2.188 \times 10^6}{3}$ m/sec
(D)	Electron moving in 2^{nd} orbit in Be^{+3} ion	(S) De-broglie wavelength of electron is $\sqrt{\dfrac{150}{13.6}}$ Å
		(T) Total energy of electron is –13.6 eV

14. Match the Column :

$r_{n,z}$ = Radius of n^{th} orbit of a single electron specie having atomic number 'z'.

$v_{n,z}$ = Velocity of electron in n^{th} orbit of a single electron specie having atomic number 'z'

$E_{n,z}$ = Magnitude of total energy of electron in n^{th} orbit of a single electron specie having atomic number 'z'

$K_{n,z}$ = Magnitude of kinetic energy of electron in n^{th} orbit of a single electron specie having atomic number 'z'

$P_{n,z}$ = Magnitude of potential energy of electron in n^{th} orbit of a single electron specie having atomic number 'z'

$f_{n,z}$ = Frequency of electron in n^{th} orbit of a single electron specie having atomic number 'z'

$T_{n,z}$ = Time period of electron in n^{th} orbit of a single electron specie having atomic number 'z'

Column-I		Column-II	
(A)	$\dfrac{f_{1,2}}{f_{1,1}}$	(P)	$\dfrac{r_{2,3}}{r_{1,3}}$
(B)	$\dfrac{\left\|E_{3,2}\right\|}{K_{3,1}}$	(Q)	$\dfrac{T_{1,1}}{T_{1,2}}$
(C)	$\left(\dfrac{1}{\sqrt{2}}\right)\left(\sqrt{\dfrac{P_{2,1}}{E_{3,1}}}\right)$	(R)	$2\left(\sqrt{\dfrac{K_{2,3}}{\left\|E_{1,2}\right\|}}\right)$
(D)	$\left(\dfrac{E_{1,4}}{E_{2,2}}\right)$	(S)	$\dfrac{v_{2,3}}{v_{1,1}}$
		(T)	$4\left(\dfrac{v_{2,5}-v_{2,1}}{v_{2,3}-v_{2,2}}\right)$

INTEGER TYPE

15. A hydrogen like atom with atomic number 'Z' is in higher excited state of quantum number 'n'.This xcited state atom can make a transiton to the first excited state by successively emitting two photons of energies 10 eV and 68.2 eV respectivley. Alternatively, the atom from the same excited state can make a transition to the 2nd excited state by emitting two photons of energies 4.25 eV and 5.95 eV respectively. Calculate the value of 'Z'.

16. A collection of H-atom in its 3rd excited state makes transition to ground state calculate minimum number of atoms required to emit all possible radiation.

ANSWERS (QUIZ#1)

1. (C) **2. (C)** **3. (C)** **4. (C)** **5. (D)** **6. (B)** **7. (C)** **8. (A)**
9. (B) **10. (A)** **11. (A, D)** **12. (A,B,C,D)**
13. (A) - S,T, (B) - R, (C) - Q, (D) -P 14. (A) → **(P,Q), (B) →(P,Q), (C) →(R,S), (D) →(T)**
15. (6) **16. (4)**

QUIZ # 2

1. What will be the ratio of de Broglie wavelengths for two electrons which are accelerated through 100 volts and 400 volts ?

(A) 2 : 1 (B) 1 : 4 (C) 1 : 2 (D) 4 : 1

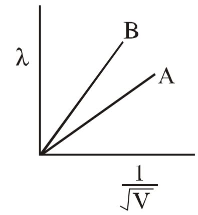
2. A photon of wavelength 12.4 nm is used to emit an electron from ground state of He^{+}, calculate the de-broglie wavelength of emitted electron -

(A) 1.2 Å (B) 2.2 Å (C) 1.8 Å (D) 2.7 Å

4. An atom has a mass of 0.02 kg and uncertainty in its velocity is 9.218×10^{-6} m/s then uncertainty in position is (h = 6.626×10^{-34} Js)

(A) 2.86×10^{-28} m (B) 2.86×10^{-32} cm (C) 1.5×10^{-27} m (D) 3.9×10^{-10} m

5. De-Broglie wavelength of two particles A & B are plotted against $\left(\frac{1}{\sqrt{V}}\right)$; where V is potential on the particles. Which of the following relation is correct about mass of particle



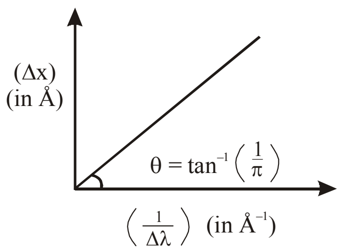
(A) $M_A = M_B$

(B) $M_A > M_B$

(C) $M_A < M_B$

(D) $M_A \le M_B$

6. A graph is plotted between uncertainity in position and inverse of uncerntanity in wavelength for an electron. We get a straight line passing through origin. Calculate voltage through which electron is accelerated with -



(A) 150 V (B) 75 V (C) 37.5 V (D) 300 V

7. If the radius of second Bohr orbit is x, then de Broglie wavelength of electron in 3rd orbit is nearly:

(A) $\pi x/3$ (B) $6\pi x$ (C) 9x (D) $\left(\frac{3}{2}\right)\pi x$

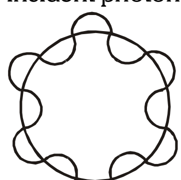
8. A particle initially at rest having charge q coulomb. & mass m kg is accelerated by a potential difference of V volts. What would be its K.E. & de broglie wavelength respectively after acceleration

(A) qV, $\frac{h}{\sqrt{2qVm}}$ (B) $\frac{h}{\sqrt{2qVm}}$, qV (C) qV, $\frac{h}{mV}$ (D) $\frac{h}{mV}$, qV

INTEGER TYPE

10. A light source of 32 watt emits light of $\lambda = 620$ nm if all the emitted photons are made to strike a metal plate of work function = 4.5 eV. Then the magnitude of photocurrent is $x \times 10^{18}$ [If 40% of the incident photon eject photoelectrons]. Then calculate the value of x.

11. The figure shows a sample of H-atoms having electron revolving in higher orbit 'n'.



If this electron makes transition from this orbit 'n' to ground state, No. of paschen lines emitted are.

12. deBroglie wavelength associated with an electron in 4^{th} orbit of hydrogen atom is a $\times$ (πr_0) where r_0 is radius of 1^{st} orbit of hydrogen atom, find value of 'a'.

13. Determine the de Broglie's wavelength of electron emitted by a metal whose thrashold frequency is 2.25×10^{14} Hz when exposed to visible radiation of wavelength 500 nm.

ANSWERS (QUIZ # 2)

1. (A) **2.** (C) **4.** (A) **5.** (B) **6.** (C) **7.** (D) **8.** (A) **10.** (0)

11. (3) **12.** (8) **13.** (9.84)

QUESTION BANK
ATOMIC STRUCTURE
SINGLE CHOICE QUESTIONS

1. An electron in an atom jumps in such a way that its kinetic energy changes from *x* to x/4. The change in potential energy will be :

(A) $+\frac{3}{2}x$ (B) $-\frac{3}{8}x$ (C) $+\frac{3}{4}x$ (D) $-\frac{3}{4}x$

2. The kinetic and potential energy (in eV) of electron present in third Bohr's orbit of hydrogen atom are respectively:

(A) –1.51,-3.02 (B) 1.51,–3.02 (C) –3.02,1.51 (D) 1.51,–1.51

3. According to Bohr's atomic theory, which of the following is correct?

(A) Potential energy of electron $\propto \frac{Z^2}{n^2}$

(B) The product of velocity of electron and principle quantum number $(n) \propto Z^2$

(C) Frequency of revolution of electron in an orbit $\propto \frac{Z^2}{n^3}$

(D) Coulombic force of attraction on the electron $\propto \frac{Z^2}{n^2}$

4. Find the value of wave number ($\bar{v}$) in terms of Rydberg's constant, when transition of electron takes place between two levels of He^+ ion whose sum is 4 and difference is 2.

(A) $\frac{8R}{9}$ (B) $\frac{32R}{9}$ (C) $\frac{3R}{4}$ (D) none of these

5. An excited state of H atom emits a photon of wavelength λ and returns in the ground state, the principal quantum number of excited state is given by :

(A) $\sqrt{\lambda R(\lambda R-1)}$ (B) $\sqrt{\frac{\lambda R}{(\lambda R-1)}}$ (C) $\sqrt{\lambda R(\lambda R-1)}$ (D) $\sqrt{\frac{(\lambda R-1)}{\lambda R}}$

6. How do the energy gaps between successive electron energy levels in an atom vary from low to high *n* values?

(A) All energy gaps are the same

(B) The energy gap decreases as *n* increases

(C) The energy gap increases as *n* increases

(D) The energy gap changes unpredictably as *n* increases

7. If radiation corresponding to second line of "Balmer series" of Li^{2+} ion, knocked out electron from first excited state of H-atom, then kinetic energy of ejected electron would be:

(A) 2.55 eV (B) 4.25 eV (C) 11.25 eV (D) 19.55 eV

8. Number of waves produced by an electron in one complete revolution in n^{th} orbit is :

(A) *n* (B) n^2 (C) (n + l) (D) (2n + l)

9. If in Bohr's model, for unielectronic atom, time period of revolution is represented as $T_{n,Z}$ where *n* represents shell no. and Z represents atomic number then the value of $T_{1,2} : T_{2,1}$ will be :

(A) 8 : 1 (B) 1 : 8 (C) 1 : 1 (D) 1 : 32

10. Be^{3+} and a proton are accelerated by the same potential, their de-Broglie wavelengths have the ratio (assume mass of proton = mass of neutron) :

(A) 1 : 2 (B) 1 : 4 (C) 1 : 1 (D) $1:3\sqrt{3}$

11. The mass of a particle is 10^{-10} g and its radius is 2×10^{-4} cm. If its velocity is 10^{-6} cm sec^{-1} with 0.0001% uncertainty in measurement, the uncertainty in its position is :

(A) 5.2×10^{-8} m (B) 5.2×10^{-7} m (C) 5.2×10^{-6} m (D) 5.2×10^{-9}

12. If an electron is travelling at 200 m/s within 1 m/s uncertainty, what is the theoretical uncertainty in its position in |im (micrometer)?

(A) 14.5 μm (B) 29 μm (C) 58 μm (D) 114 μm

13. Which of the following statement(s) is/are consistent with the Bohr theory of the atom (and no others)?

(1) An electron can remain in a particular orbit as long as it continuously absorbs radiation of a definite frequency.

(2) The lowest energy orbits are those closest to the nucleus.

(3) All electrons can jump from the *K* shell to the M shell by emitting radiation of a definite frequency.

(A) 1, 2, 3 (B) 2 only (C) 3 only (D) 1, 2

14. The mass of a proton is 1836 times more than the mass of an electron. If a sub-atomic particle of mass (m') 207 times the mass of electron is captured by the nucleus, then the first ionization potential of H :

(A) decreases (B) increases

(C) remains same (D) may be decrease or increase

15. What is the maximum wavelength line in the Lyman series of He^+ ion?

(A) 3R (B) $\frac{1}{3R}$ (C) $\frac{4}{4R}$ (D) $\frac{9R}{4}$

16. Electromagnetic radiation (photon) with highest wavelength results when an electron in the hydrogen atom falls from $n = 6$ to :

(A) n = l (B) n = 2 (C) n = 3 (D) n = 5

17. The ratio of magnetic moments of Fe (III) and Co (III) is :

a) $\sqrt{5}:\sqrt{7}$ (B) $\sqrt{35}:\sqrt{15}$ (C) 7 : 3 (D) none of these

18. Read the following statements and choose the correct option:

(I) If the radius of the first Bohr orbit of hydrogen atom is r, then radius 2nd orbit of Li^{2+} would be 4r

(II) For a s-orbital electron, the orbital angular momentum is zero

(A) only I is correct (B) only II is correct (C) both are correct (D) both are incorrect

19. What is the maximum number of electrons in an atom that can have the quantum numbers n = 3 and *l* = 2?

(A) 2 (B) 5 (C) 6 (D) 10

20. In a 3d subshell, all the five orbitals are degenerate. What does it mean?

(A) All the orbitals have the same orientation. (B) All the orbitals have the same shape.

(C) All the orbitals have the same energy. (D) All the orbitals are unoccupied.

21. Which of the following statements is incorrect?

(A) The concepts of "penetration" and "shielding" are important in deciding the energetic ordering of orbitals in multi-electron atoms

(B) A wave-function can have positive and negative values

(C) "Radial nodes" can appear in radial probability distribution functions

(D) The absolute size of an orbital is given by the principal quantum number.

22. For similar orbitals having different values of *n* :

(A) the most probable distance increases with increase in *n*

(B) the most probable distance decreases with increase in *n*

(C) the most probable distance remains constant with increase in *n*

(D) none of these

23. A sushell n = 5, *l* = 3 can accommodate :

(A) 10 electrons (B) 14 electrons (C) 18 electrons (D) None of these

24. The set of quantum numbers, n = 3, *l* = 2, $m_l = 0$

(A) describes an electron in a 2s orbital

(B) is not allowed

(C) describes an electron in a 3p orbital

(D) describes one of the five orbitals of a similar type

25. Which of the following sets of quantum numbers describes the electron which is removed most easily from a potassium atom in its ground state ?

(A) $n = 3, l = 1, m_1 = 1\ m_s = -\frac{1}{2}$ (B) $n = 2, l = 1, m_1 = 0, m_s = -\frac{1}{2}$

(C) $n = 4, l = 0, m_1 = 1, m_s = +\frac{1}{2}$ (D) $n = 4, l = 0, m_l = 0, m_s = +\frac{1}{2}$

26. A single electron in an ion has ionization energy equal to 217.6 eV What is the total number of neutrons present in one ion of it?

(A) 2 (B) 4 (C) 5 (D) 9

27. A small particle of mass *m* moves in.such a way that P. E. $= -\frac{1}{2}mkr^2$, where *k* is a constant and r is the distance of the particle from origin. Assuming Bohr's model of quantization of angular momentum and circular orbit, r is directly proportional to :

(A) n^2 (B) *n* (C) $\sqrt{n}$ (D) none of these

28. The ratio of the radius difference between 4th and 3rd orbit of H-atom and that of Li^{2+} ion is :

(A) 1 : 1 (B) 3 : 1 (C) 3 : 4 (D) 9 : 1

29. In a collection of H-atoms, all the electrons jump from *n* = 5 to ground level finally (directly or indirectly), without emitting any line in Balmer series. The number of possible different radiations is :

(A) 10 (B) 8 (C) 7 (D) 6

30. An element undergoes a reaction as shown :

$X + 2e^- \longrightarrow X^{2-}$, energy released = 30.87 eV/atom. If the energy released, is used to dissociate 4 gms of H_2 molecules, equally into H^+ and H , where H is excited state of H atoms where the electron travels in orbit whose circumference equal to four times its de Broglie's wavelength. Determine the least moles of *X* that would be required :

Given : I.E. of H = 13.6 eV/atom, bond energy of H_2 = 4.526 eV/molecule.

(A) 1 (B) 2 (C) 3 (D) 4

31. If the energy of H-atom in the ground state is -*E*, the velocity of photo-electron emitted when a photon having energy E_p strikes a stationary Li^{2+} ion in ground state, is given by :

(A) $v = \sqrt{\frac{2(E_p - E)}{m}}$ (B) $v = \sqrt{\frac{2(E_p + 9E)}{m}}$ (C) $v = \sqrt{\frac{2(E_p - 9E)}{m}}$ (D) $v = \sqrt{\frac{2(E_p - 3E)}{m}}$

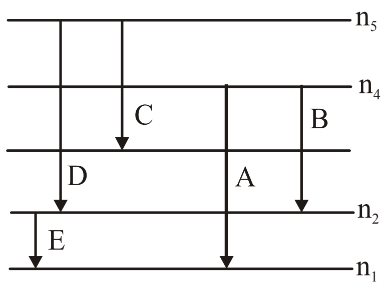
32. A hydrogen like species (atomic number Z) is presept in a higher excited state of quantum number *n*. This excited atom can make a transition to the first excited state by successive emission of two photons of energies 10.20 eV and 17.0 eV respectively. Alternatively, the atom from the same excited state can make a transition to the second excited state by successive emission of two photons of energy 4.25 eV and 5.95 eV respectively. Determine the value of Z.

(A) 1 (B) 2 (C) 3 (D) 4

33. Probability of finding the electron ψ^2 of 's' orbital doesn't depend upon -

(A) Distance from nucleus (r) (B) Energy of s-orbital
(C) Principal quantum number (D) Azimuthal quantum number

34. For a hypothetical H like atom which follows Bohr's model, some spectral lines were observed as shown. If it is known that line 'E' belongs to the visible region, then the lines possibly belonging to ultra violet region will be (n, is not necessarily ground state)



[Assume for this atom, no spectral series shows overlaps with other series in the emmission spectrum]

(A) B and D (B) D only (C) C only (D) A only

35. The number of photons emitted in 10 hours by a 60 W sodium lamp (X of photon = 6000 Å)

(A) 6.50×10^{24} (B) 6.40×10^{23} (C) 8.40×10^{23} (D) 3.40×10^{23}

[Take hc = 12400 eV Å, h = Planck's constant, c = speed of light]

36. Radius of 3rd orbit of Li^{2+} ion is '*x*' cm then de-broglie wavelength of electrons in the 1st orbit of H-atom is

(A) $\frac{2\pi x}{3}$ cm (B) $6\pi\, x$ cm (C) $3\pi\, x$ cm (D) $\frac{2\pi x}{6}$ cm

37. Ratio of frequency of revolution of electron in the 2nd excited state of He and 2nd state of hydrogen is:

(A) $\frac{32}{27}$ (B) $\frac{27}{32}$ (C) 1/54 (D) 27/2

38. In any subshell,the maximum number of electrons having same value of spin quantum number is

(A) $\sqrt{\ell(\ell+1)}$ (B) $\ell+2$ (C) $2\ell+1$ (D) $4\ell+2$

39. A proton accelerated from rest through a potential difference of V volts has a wavelength X associated with it. An alpha particle in order to have the same wavelength must be accelerated from rest through a potential difference of

(A) V volt (B) 4V volt (C) 2V volt (D) $\frac{V}{8}$ volt

40. Consider the ground state of Cr atom (Z = 24). The numbers of electrons with the azimuthal quantum numbers, $\ell = 1$ and 2 are, respectively :

(A) 16 and 5 (B) 12 and 5 (C) 16 and 4 (D) 12 and 4

41. Maximum number of electrons having $(l + m)$ value equal to zero in $_{26}Fe$ may be

(A) 13 (B) 14 (C) 7 (D) 12

42. 4000 Å photon is used to break the iodine molecule, then the % of energy converted to the K.E. of iodine atoms if bond dissociation energy of I_2 molecule is 246.5 kJ/mol.

(A) 8 % (B 12% (C) 17% (D) 25 %

ONE OR MORE THAN ONE CORRECT OPTIONS

43. If the wave number of 1st line of Balmer series of H-atom is 'x' then then :

(A) wave number of 1st line of lyman series of the He^+ ion will be $\frac{108x}{5}$

(B) wave number of 1st line of lyman series of the He^+ ion will be $\frac{36x}{5}$

(C) the wave length of 2nd line of lyman series of H-atom is $\frac{5}{32x}$

(D) the wave length of 2nd line of lyman series of H-atom is $\frac{32x}{5}$

44. The wave functions of 3s and $3p_z$ orbitals are given by

$$\Psi_{3s} = \frac{1}{9\sqrt{3}}\left(\frac{1}{4\pi}\right)^{1/2}\left(\frac{z}{a_0}\right)^{3/2}\left(6 - \frac{4zr}{a_0} + \frac{4}{9}\cdot\frac{z^2r^2}{a_0^2}\right)e^{-zr/3a_0}$$

$$\Psi_{3P_z} = \frac{1}{9\sqrt{6}}\left(\frac{3}{4\pi}\right)^{1/2}\left(\frac{z}{a_0}\right)^{3/2}\left(4 - \frac{2zr}{3a_0}\right)\left(\frac{2zr}{3a_0}\right)e^{-zr/3a_0}\cos\theta$$

from these we can conclude

(A) number of nodal surface for $3p_z$ & 3s orbitals are equal

(B) the angular nodal surface of $3p_z$ orbital has the equation $\theta = \pi/2$

(C) The radial nodal surfaces of 3s orbital and $3p_z$ orbitals are at equal distance from the nucles

(D) 3s electron have greater penetrating power into the nucleus in comparison to $3p_z$ electrons

45. In a hydrogen like sample electron is in 2nd excited state, the Binding energy of 4th state of this sample is 13.6eV, then

(A) A 25 eV photon can set free the electron from the second excited state of this sample.

(B) 3 different types of photon will be observed if electrons make transition up to ground state from I second excited state

(C) If 23 eV photon is used then K.E. of the ejected electron is 1 eV.

(D) 2nd line of Balmer series of this sample has same energy value as 1st excitation energy of H- atoms.

46. 1st excitation potential for the H-like (hypothetical) sample is 24 V. Then :

(A) Ionisation energy of the sample is 36 eV

(B) Ionisation energy of the sample is 32 eV

(C) Binding energy of 3rd excited state is 2 eV

(D) 2nd excitation potential of the sample is $\frac{32 \times 8}{9}$ V

47. A hydrogen like atom in ground state absorbs 'n' photons having the same energy and it emits exactly same number of photons when electronic transition takes place. Then the energy of the absorbed photon may be

(A) 91.8eV (B) 40.8eV (C) 48.4eV (D) 54.4eV

48. In a hydrogen like sample two different types of photons A and B are produced by electronic transitio Photon B has it's wavelength in infrared region if photon A has more energy than B, then the photon A belong to the region.

(A) ultraviolet (B) visible (C) infrared (D) None

49. Hydrogen atoms in a particular excited state 'n', when all returned to ground state, 6 different photons are emitted. Which of the following is/are incorrect.

(A) out of 6 different photons only 2 photons have speed equal to that of visible light.

(B) If highest energy photon emitted from the above sample is incedent on the metal plate having work function 8 eV, KE of liberated photo-electron may be equal to or less than 4.75 eV.

(C) Total number of radial nodes in all the orbitals of n^{th} shell is 14.

(D) Total number of angular nodes in all the orbitals in $(n-1)^{th}$ shell is 13.

50. In a H-like sample electrons make transition from 4th excited state to 2nd state then

(A) 10 different spectral lines are observed

(B) 6 different spectral lines are observed

(C) number of lines belonging to the balmer series is 3

(D) Number of lines belonging to paschen series is 2.

MATCH THE COLUMN

51. P_n = potential energy, E_n = total energy
f = frequency, Z = atomic number
V_n = velocity in n^{th} orbit
T_n = time period in n^{th} orbit

	Column -1		Column -II
(A)	$E_n \propto r^y$, y =?	(p)	1/2
(B)	E_n / P_n	(q)	1
(C)	$\frac{1}{f_n^{-x}} \propto z$, x = ?	(r)	2
(D)	$(v_n \times T_n)^t \propto r_n$, t=?	(s)	–1

52. B. E. – Binding energy
I.E. – Ionization energy :

	Column -1		Column -II
(A)	B.E. of He^+ atom in an excited state	(p)	Infrared region
(B)	$7 \rightarrow 3$ transition in H-atom	(q)	3.4 eV
(C)	$5 \rightarrow 1$ transition in H-atom	(r)	13.6 eV
(D)	Series limit of Balmer series in H-atom	(s)	10 Spectral lines observed

53.

	Column -1		Column -II
(A)	$n = 6 \rightarrow n = 3$ (In H-atom)	(p)	10 lines in the spectrum
(B)	$n = 7 \rightarrow n = 3$ (In H-atom)	(q)	Spectral lines in visible region
(C)	$n = 5 \rightarrow n = 2$ (In H-atom)	(r)	6 lines in the spectrum
(D)	$n = 6 \rightarrow n = 2$ (In H-atom)	(s)	Spectral lines in infrared region

COMPREHENSION BASED QUESTIONS

Passage for Qu. No. 54 - 56

The French physicist Louis de Broglie in 1924 postulated that matter, like radiation, should exhibit a dual behaviour. He proposed the following relationship between the wavelength λ of a material particle, its linear momentum p and planck constant h.

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

The de Broglie relation implies that the wavelength of a particle should decrease as its velocity increases. It also implies that for a given velocity heavier particles should have shorter wavelength than lighter particles. The waves associated with particles in motion are called matter waves or de Broglie waves. These waves differ from the electromagnetic waves as they

(i) have lower velocities

(ii) have no electrical and magnetic fields and

(iii) are not emitted by the particle under consideration.

The experimental confirmation of the de Broglies relation was obtained when Davisson and Germer, in 1927, observed that a beam of electrons is diffractedby a nickel crystal. As diffraction is a characteristic property of waves, hence the beam of electron behaves as a wave, as proposed by de broglie.

54. If proton, electron and α-particle are moving with same kinetic energy then the order of their de-Broglie's wavelength.

(A) $\lambda_p > \lambda_e > \lambda_\alpha$ (B) $\lambda_\alpha > \lambda_p > \lambda_e$ (C) $\lambda_e > \lambda_p > \lambda_\alpha$ (D) $\lambda_e = \lambda_p < \lambda_\alpha$

55. Using Bohr's theory, the transition, so that the electrons de-Broglie wavelength becomes 3 times of its original value in He^+ ion will be

(A) $2 \longrightarrow 6$ (B) $2 \longrightarrow 4$ (C) $1 \longrightarrow 4$ (D) $1 \longrightarrow 6$

56. De-Broglie wavelength of an electron travelling with speed equal to 1% of the speed of light

(A) 400 pm (B) 120 pm (C) 242 pm (D) 375 pm

Passage for Qu. No. 57 - 59

If an electron is in any higher state n = n and makes a transition to ground state, then total no. of different photons emitted is equal to $\frac{n \times (n-1)}{2}$.

If an electrons is in any higher state $n = n_2$ and makes a transition toanother excited state $n = n_1$, then total no. of different photons emitted is equal to $\frac{\Delta n \times (\Delta n + 1)}{2}$, where $\Delta n = n_2 - n_1$.

Note : In case of single isolated atom if electron make transition from n^{th} state to the ground state then max. number of spectral lines observed = (n – 1).

57. Two samples are taken, in 1^{st} sample mole ratio of H to He^+ is 1 : 2. In II^{nd} sample mole ratio of H to He^+ is 2 : 1. If n_1 is number of spectral lines produced by 1^{st} sample and number of spectral lines produced by II^{nd} sample is n_2 then $n_1 : n_2$ will be :

(There is sufficient number of both types of atom and both are in same excited state in both sample).

(A) 2 : 1 (B) 1 : 2 (C) 1 : 1 (D) None of these

58. Which of the following statements is/are true :

(I) If H atoms are in 3^{rd} state then number of maximum spectral line produced by 2 atoms will be equal to numberof maximum spectral lines produced by 3 H-atoms

(II) For third excited state, maximum number of spectral line produced by He^+ and maximum number of spectral line produced by 4H-atom are not equal.

(III) Energy evolved from 3 to 2 transition in He^+ cannot used for any transition in Li^{2+}.

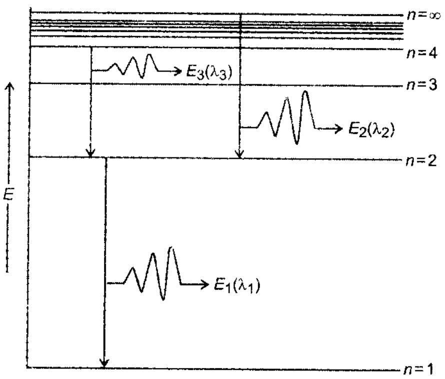
(A) I & III (B) II & III (C) All (D) I & II.

59. In a sample one H atom is in 1^{st} excited state, two He^+ ions is in II^{nd} excited state and three Li^{2+} ions is in III^{rd} excited state are present then maximum, number of spectral line which can be obtained when all possible transition terminated at n = 1.

(A) 11 (B) 12 (C) 13 (D) none of these

Passage for Qu. No. 60 - 64

Following is the emission spectrum of He^+ ion :



60. Which of the following relationship between the engergies E_1 and E_2 is correct :

(A) $E_1 = 4E_2$ (B) $E_2 = 3E_1$ (C) $E_1 = 3E_2$ (D) $E_2 = 3E_1$

61. When light of wavelength λ_2 is passed through a glass tube containing H-atoms in the ground state, they will be:

(A) excited to 5^{th} Bohr orbit (B) they will be excited to 10^{th} Bohr orbit

(C) No electronic transition would occur (D) All H-atoms will be ionised

62. When light of wavelength λ_3 is passed through a glass tube containing gaseous Li^{2+} ions in the excited state, it is absorbed leading to an electronic transition to the further higher excited state. In which Bohr orbit, the electron in Li^{2+} ion is present finally :

(A) 4 (B) 6 (C) 8 (D) 9

63. In the above shown energy diagram, the three wavelengths λ_1, λ_2 and λ_3 are in the ratio of .... respectively.

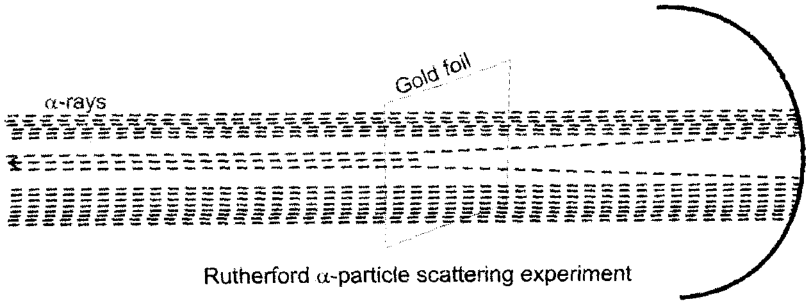
(A) 1 : 2 : 3 (B) 1 : 4 : 3 (C) 2 : 3 : 5 (D) 1 : 3 : 4

64. Line corresponding to which of the following wavelength(s) will not be observed in the emission spectrum of H-atom ?

(A) λ_1 (B) λ_2 (C) λ_3 (D) λ_1 and λ_2

Passage for Qu. No. 65 - 67

Rutherford carried out his experiments using α-particles. He bombarded α-particles on a thin gold foil and the rays coming out were observed on a screen as shown in the following diagram :



Rutherford α-particle scattering experiment

Observations :

I. Most of the α-particles passed underflected.

II. Very few of the α-particles deflected from their normal path.

III. One out of approximately 20,000 α-particles returned back on it original path.

Conclusions :

I. Most part of the atom is vacant.

II. There exists a positively-charged centre and most of the atomic mass is concentrated at this centre.

65. Had the Rutherford used aluminium foil in place of gold foil, which of his observations would have been affected the most ?

(A) Observaton - I (B) Observation - II (C) Observation - III (D) All I, II, III

66. Had the Rutherford used β-particles in place of α-particles, which of his observations would have been least affected ?

(A) Observation - I (B) Observation-II (C) Observation-III (D) None of these

67. Had the Rutherford used H^+ gas in place of α-particles, which of his observations would have been greatly affected ?

(A) Observation-I (B) Observation-II (C) Observation-III (D) Observations-II & III

QUIZ # 1 **CHEMICAL EQUILIBRIUM** **PHYSICAL CHEMISTRY**

Single Option Correct

1. Equilibrium constant for the given reaction is $K = 10^{20}$ at temperature 300 K

$A(s) + 2B\,(aq) \rightleftharpoons 2C\,(s) + D\,(aq) \qquad K = 10^{20}$

The equilibrium conc. of B starting with 1 mole of A and 1/3 mole/litre of B is

(A) $\sim 4 \times 10^{-11}$ M (B) $\sim 2 \times 10^{-10}$ (C) $\sim 2 \times 10^{-11}$ (D) none of these

2. When C & D are mixed together to give A & B according to the reaction

$A(g) + B(g) \rightleftharpoons C(g) + D(g)$, the reaction quotient Q, at the initial stages of the reaction :

(A) is zero (B) decreases with time

(C) is independent of time (D) increases with time

3. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ For the above homogeneous endothermic decomposition, $K_p = \dfrac{\alpha^2 P}{1-\alpha^2}$.

where P = Total pressure, α = degree of dissociation.

Then which of the following is false

(A) K_p increases with increase in total pressure at constant temperature.

(B) K_p increases with increase in temperature.

(C) α depends on P at constant temperature.

(D) K_p is independent of α at constant temperature.

4. For the reaction $PCl_5 \rightleftharpoons PCl_3(g) + Cl_2(g)$ Let α = degree of dissociation at temperature T and K_p = eq. const. Let P = Total pressure at equilibrium, then :

(A) $\alpha = \dfrac{K_P}{P+K_P}$ (B) $\alpha = \dfrac{P+K_P}{K_P}$ (C) $\alpha = \sqrt{\dfrac{K_P}{P+K_P}}$ (D) $\alpha = \sqrt{\dfrac{P+K_P}{K_P}}$

5. The equilibrium constant, K_p for $SO_{3(g)} + NO_{(g)} \rightleftharpoons SO_{2(g)} + NO_{2(g)}$ is 0.12 at 460°C Suppose you began with 6 moles SO_3, 10 moles NO, 3 moles SO_2 and 1 mole NO_2 in a closed fixed volume vessel. After some time it would be true that :

(A) There will be less than 6 moles SO_3 (B) There will be more than 6 moles SO_3

(C) The concentration of NO will increase. (D) Nothing will have changed.

6. For reaction $N_2O_4 \rightleftharpoons 2NO_2$ at given temperature if $K_p = \dfrac{8}{5}$ for 30% degree of dissociation at equilibrium then what will be new K_p for 50% dissociation of N_2O_4 at equilibrium at same temperature

(A) $\dfrac{5}{8}$ (B) $\dfrac{8}{5}$ (C) $\dfrac{2}{5}$ (D) $\dfrac{12}{5}$

7. Asample of pure NO_2 gas heated to 1000K decomposes : $2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$. The equilibreium constant K_p is 100 atm. Analysis shows that the partial pressure of O_2 is 0.25 atm. at equilibrium. The partial pressure of NO_2 at equilibrium is :

(A) 0.03 (B) 0.25 (C) 0.025 (D) 0.04

8. $\log \frac{K_P}{K_C} - \log RT = 0$ is a relationship for the reaction :

(A) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ (B) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

(C) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ (D) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

9. Consider two equilibrium $2Cl_2(g) + 2H_2O(g) \rightleftharpoons 4HCl(g) + O_2(g)$ and

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ simultaneously established in a closed rigid vessel. When some amount of HCl is added isothermally at equilibrium, Which of the following statements is correct.

(A) Amount of N_2 gas will decrease. (B) Amount of N_2 gas will increase

(C) Amount of N_2 gas will remain constant (D) Nothing can be said with certainty.

10. At a temperature T, a compound $AB_4(g)$ dissociates as $2AB_4(g) \rightleftharpoons A_2(g) + 4B_2(g)$ with a degree of dissociation x, which is small compared with unity. The expression of K_p in terms of x and total equilibrium pressure P is :

(A) $8P^3x^5$ (B) $256P^3x^5$ (C) $4Px^2$ (D) None of these

11. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$. The density of the equilibrium mixture for the above equilibrium is 0.23 gm/cc. The initial concentration of N_2O_4 is

(A) 2.5 M (B) 0.23 M (C) 5.2 M (D) can't be determined

12. HI dissociates according to the following equilibrium,

$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

In a particular experiment HI is found to dissociated to extent of 50% at equilibrium. The equilibrium mixture is found to require 100 ml of 1 M $Na_2S_2O_3$ solution for complete reduction of iodine (hypo completely reacts with iodine). the K_c for the reaction is

(A) 2.5×10^{-1} (B) 3.2 (C) 4.3×10^{-3} (D) data insufficient

13. $A(s) \rightleftharpoons B(g) + C(g)$; Total pressure at time of equilibrium is 40 atmosphere. If all the contents of this reaction has been shifted to another reactor of double capacity, then the total equilibrium pressure in new reactor will be (in atmosphere)

(A) 20 (B) 40 (C) 400 (D) 1600

14. 1 mole each of A & D is introduced in 1ℓ container simultaneous two equilibriums are established

$A \rightleftharpoons B + C$ $K_C = 10^6$ M

$B + D \rightleftharpoons A$ $K_C = 10^{-6}$ M^{-1}

Equilibrium concentration of A will be

(A) 10^{-6} M (B) 10^{-3} M (C) 10^{-12} M (D) 0.5×10^{-3} M

ANSWER KEY

1. (A)	2. (B)	3. (A)	4. (C)	5. (A)	6. (B)	7. (C)	8. (A)	9. (B)
10. (A)	11. (A)	12. (A)	13. (B)	14. (A)				

QUESTION BANK

CHEMICAL EQUILIBRIUM

SINGLE CHOICE QUESTIONS

1. The equilibrium constant K_c for the reaction $P_4(g) \rightleftharpoons 2P_2(g)$ is 1.4 at 400°C. Suppose that 3 moles of $P_4(g)$ and 2 moles of $P_2(g)$ are mixed in 2 litre container at 400°C. What is the value of reaction quotient (Q) ?

(A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) 1 (D) None of these

2. For a reversible gaseous reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$ at equilibrium, if some moles of H_2 are replaced by same number of moles of T_2 (T is tritium, isotope of H and assume isotopes do not have different chemical properties) without affecting other parameters, then :

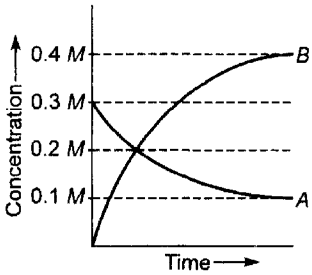
(A) The sample of ammonia obtained after sometime will be radioactive.

(B) Moles of N_2 after the change will be different as compared to moles of N_2 present before the change

(C) The value of K_p or K_c will change

(D) The average molecular mass of new equilibrium will be same as that of old equilibrium

3. The figure shows the change in concentration of species *A* and *B* as a function of time.



The equilibrium constant K_c for the reaction $A(g) \rightleftharpoons 2B(g)$ is :

(A) $K_c > 1$ (B) $K < 1$ (C) $K = 1$ (D) data insufficient

4. In the reaction $X(g) + Y(g) \rightleftharpoons 2Z(g)$, 2 mole of *X* , 1 mole of *Y* and 1 mole of *Z* are placed m a 10 litre vessel and allowed to reach equilibrium. If final concentration of Z is 0.2 M, then K_c for the given reaction is :

(A) 1.60 (B) $\frac{80}{3}$ (C) $\frac{16}{3}$ (D) None of these

5. In the presence of excess of anhydrous $SrCl_2$, the amount of water taken up is governed by $K_p = 10^{12}$ atm^{-4} for the following reaction at 273 K

$$SrCl_2 . 2H_2O(s) + 4H_2O(g) \rightleftharpoons SrCl_2 . 6H_2O(s)$$

What is equilibrium vapour pressure (in torr) of water in a closed vessel that contains $SrCl_2 . 2H_2O(s)$?

(A) 0.001 torr (B) 10^3 torr (C) 0.76 torr (D) 1.31 torr

6. $I_2(aq) + I^-(aq) \rightleftharpoons I_3^-(aq)$. We started with 1 mole of I_2 and 0.5 mole of I^- in one litre flask. After equilibrium is reached, excess of $AgNO_3$ gave 0.25 mole of yellow precipitate. Equilibrium constant is:

(A) 1.33 (B) 2.66 (C) 2.0 (D) 3.0

7. In the equilibrium $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, the partial pressure of SO_2, O_2 and SO_3 are 0.662, 0.10 and 0.331 atm respectively. What should be the partial pressure of oxygen so that the equilibrium concentrations of SO_2 and SO_3 are equal ?

(A) 0.4 atm (B) 1.0 atm (C) 0.8 atm (D) 0.25 atm

8. In a system $A(s) \rightleftharpoons 2B(g) + 3C(g)$, if the concentration of C at equilibrium is increased by a factor of 2, it will cause the equilibrium concentration of *B* to change to :

(A) two times the original value (B) one half of its original value

(C) $2\sqrt{2}$ times to the original value (D) $\frac{1}{2\sqrt{2}}$ times the original value

9. At 273 K and 1 atm, 10 litre of N_2O_4 decomposes to NO_2 according to equation

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

What is degree of dissociation (A) when the original volume is 25% less than that of existing volume ?

(A) 0.25 (B) 0.33 (C) 0.66 (D) 0.5

10. The equilibrium constant for the reaction $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$ is 5. How many moles of CO_2 must be added to 1 litre container already containing 3 moles each of CO and H_2O to make 2M equilibrium concentration of CO ?

(A) 15 (B) 19 (C) 5 (D) 20

11. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

For the reaction initially the mole ratio was 1 : 3 of $N_2 : H_2$. At equilibrium 50% of each has reacted. If the equilibrium pressure is *P*, the partial pressure of NH_3 at equilibrium is :

(A) $\frac{P}{3}$ (B) $\frac{P}{4}$ (C) $\frac{P}{6}$ (D) $\frac{P}{8}$

12. At certain temperature compound $AB_2(g)$ dissociates according to the reaction

$$2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$$

With degree of dissociation α, which is small compared with unity. The expression of K_p, in terms of a and initial pressure P is :

(A) $P\frac{\alpha^3}{2}$ (B) $\frac{P\alpha^2}{3}$ (C) $P\frac{\alpha^3}{3}$ (D) $\frac{P\alpha^2}{2}$

13. At 27°C and 1 atm pressure, N_2O_4 is 20% dissociation into NO_2. What is the density of equilibrium mixture of N_2O_4 and NO_2 at 27°C and 1 atm ?

(A) 3.11 g/litre (B) 2.11 g/litre (C) 4.5 g/litre (D) None of these

14. $COCl_2$ gas dissociates according to the equation, $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$. When heated to 700 K the density of the gas mixture at 1.16 atm and at equilibrium is 1.16 g/litre. The degree of dissociation of CO_2 at 700 K is :

(A) 0.28 (B) 0.50 (C) 0.72 (D) 0.42

15. Determine the volume of equilibrium constant *(K_c)* for the reaction

$$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$$

If 10 moles of A_2; 15 moles of B_2 and 5 moles of AB are placed in a 2 litre vessel and allowed to come to equilibrium. The final concentration of AB is 7.5 M :

(A) 4.5 (B) 1.5 (C) 0.6 (D) None of these

16. For the reaction $XCO_3(s) \rightleftharpoons XO(s) + CO_2(g)$, $K_p = 1.642$ atm at 727°C. If 4 moles of $XCO_3(s)$ was put into a 50 litre container and heated to 727°C.

What mole percent of the XCO_3 remains unreacted at equilibrium ?

(A) 20 (B) 25 (C) 50 (D) None of these

17. $AB_3(g)$ is dissociates as $AB_3(g) \rightleftharpoons AB_2(g) + \frac{1}{2} B_2(g)$.

When the initial pressure of AB_3 is 800 torr and the total pressure developed at equilibrium is 900 torr. What fraction of $AB_3(g)$ is dissociated ?

(A) 10% (B) 20% (C) 25% (D) 30%

18. At 200°C PCl_5 dissociates as follows :

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

It was found that the equilibrium vapours are 62 times as heavy as hydrogen. The degree of dissociation of $PC1_5$ at 200°C is nearly :

(A) 10% (B) 42% (C) 50% (D) 68%

19. For the dissociation reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, the degree of dissociation (A) in terms of K_p and total equilibrium pressure P is :

(A) $\alpha = \sqrt{\frac{4P + K_P}{K_P}}$ (B) $\alpha = \sqrt{\frac{K_P}{4P + K_P}}$ (C) $\alpha = \sqrt{\frac{K_P}{4P}}$ (D) None of these

20. For the reaction (1) and (2)

$A(g) \rightleftharpoons B(g) + C(g)$ .....(1)

$X(g) \rightleftharpoons 2Y(g)$ .....(2)

Given, $K_{P_1} : K_{P_2} = 9 : 1$

If the degree of dissociation of A(g) and *X(g)* be same then the total pressure at equilibrium (1) and (2) are in the ratio :

(A) 3 : 1 (B) 36 : 1 (C) 1 : 1 (D) 0.5 : 1

21. The volume of the reaction vessel containing an equilibrium mixture is increased in the following reaction

$SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$

When equilibrium is re-established :

(A) the amount of $Cl_2(g)$ remains unchanged (B) the amount of $Cl_2(g)$ increases

(C) the amount of $SO_2Cl_2(g)$ increases (D) the amount of $SO_2(g)$ decreases

22. Some inert gas is added at constant volume to the following reaction at equilibrium

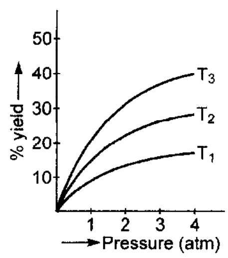
$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

Predict the effect of adding the inert gas :

(A) the equilibrium shifts in the forward direction (B) the equilibrium shifts in the backward direction

(C) the equilibrium remains unaffected (D) the value of K_p is increased

23. The preparation of $SO_3(g)$ by reaction $SO_2(g) + 1/2\,O_2(g) \rightleftharpoons SO_3(g)$ is an exothermic reaction. If the preparation follows the following temperature-pressure relationship for its % yield, then for temperatures T_2, T_2 and T_3. The correct option is :



(A) $T_3 > T_2 > T_1$ (B) $T_1 > T_2 > T_3$ (C) $T_1 = T_2 = T_3$

(D) Nothing could be predicted about temperature through given information

24. For which of the following reactions is product formation favoured by low pressure and high temperature?

(A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$; $\Delta H° = -9.4kJ$

(B) $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$; $\Delta H° = 172.5$ kJ

(C) $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH$; $\Delta H° = -21.7kJ$

(D) $3O_2(g) \rightleftharpoons 2O_3(g)$; $\Delta H° = 285kJ$

25. Consider the following reaction and determine which of the conditions will shift the equilibrium position to the right ?

$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g) + \text{Heat}$

(A) increasing the temperature
(B) increasing the pressure
(C) adding a catalyst
(D) none of the above is correct

26. A system at equilibrium is described by the equation of fixed temperature *T*.

$SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$

What effect will an increases in the total pressure caused by a decrease in volume have on the equilibrium?

(A) Concentration of SO_2C1_2 (g) increases
(B) Concentration of SO_2(g) increases
(C) Concentration of Cl_2(g) increases
(D) Concentration of all gases increases

27. The reaction $2NO_2(g) \rightleftharpoons N_2O_4(g)$ is an exothermic equilibrium. This means that :

(A) equilibration of this gas mixture will be slower at high temperature
(B) a mole of N_2O_4 will occupy twice the volume of a mole of NO_2 at the same
(C) the equilibrium will move to the right if an equilibrium mixture is cooled
(D) the position of equilibrium will move to the left with increasing gas pressure

28. Densities of diamond and graphite are 3.5 and 2.3 gm /mL

C (diamond) $\rightleftharpoons$ C (graphite); $\Delta_rH = -.9$ kJ/mol

favourable conditions for formation of graphite are :

(A) high pressure and low temperature
(B) low pressure and high temperature
(C) high pressure and high temperature
(D) low pressure and low temperature

29. The pressure on a sample of water at its triple point is reduced while the temperature is held constant. Which phases changes are favoured ?

(I) melting of ice (II) sublimation of ice (III) vaporization of liquid water

(A) I only (B) III only (C) II only (D) II and III

30. A plot of Gibbs energy of a reaction mixture against the extent of the reaction is :

(A) minimum at equilibrium
(B) zero at equilibrium
(C) maximum at equilibrium
(D) None of these

31. Solid $Ca(HCO_3)_2$ decomposes as

$Ca(HCO_3)_2(s) \rightleftharpoons CaCO_3(s) + CO_2(g) + H_2O(g)$

If the total pressure is 0.2 bar at 420 K, what is the standard free energy change for the given reaction (Δ_rG^0)?

(A) 840 kJ/mol (B) 3.86 kJ/mol (C) 6.98 kJ/mol (D) 16.083 kJ/mol

32. One mole of N_2(g) is mixed with 2 moles of H_2(g) in a 4 litre vessel. If 50% of N_2(g) is converted to NH_3(g) by the following reaction :

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

What will be the value of K_c for the following equilibrium ?

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

(A) 256 (B) 16 (C) $\frac{1}{16}$ (D) None of these

33. Assume that the decomposition of HNO_3 can be represented by the following equation

$4HNO_3(g) \rightleftharpoons 4NO_2(g) + 2H_2O(g) + O_2(g)$

and the reaction approaches equilibrium at 400 K temperature and 30 atm pressure. At equilibrium partial pressure of HNO_3 is 2 atm.

Calculate K_c in $(mol/L)^3$ at 400 K :

(Use : $R = 0.08$ atm-L/mol-K)

(A) 4 (B) 8 (C) 16 (D) 32

34. For the equilibrium

$LiCl . 3NH_3(s) \rightleftharpoons LiCl\text{-}NH_3(s) + 2NH_3(g);$

$K_p = 9$ atm^2 at 37°C. A 5 litre vessel contains 0.1 mole of $LiCl \cdot NH_3$. How many moles of NH_3 should be added to the flask at this temperature to derive the backward reaction for completion ?

Use : $R = 0.082$ atm-l/mol K

(A) 0.2 (B) 0.59 (C) 0.69 (D) 0.79

35. Ammonium carbamate dissociates as

$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

In a closed vessel containing ammonium carbamate in equilibrium, ammonia is added such that partial pressure of NH_3 now equals to the original total pressure. Calculate the ratio of partial pressure of CO_2 now to the original partial pressure of CO_2 :

(A) 4 (B) 9 (C) $\frac{4}{9}$ (D) $\frac{2}{9}$

36. At a certain temperature and 2 atm pressure equilibrium constant (K_p) is 25 for the reaction

$SO_2(g) + NO_2(g) \rightleftharpoons SO_3(g) + NO(g)$

Initially if we take 2 moles of each of the four gases and 2 moles of inert gas, what would be the equilibrium partial pressure of NO_2?

(A) 1.33 atm (B) 0.1665 atm (C) 0.133 atm (D) None of these

37. Determine the degree of association (polymerization) for the reaction in aqueous solution

$6HCHO \rightleftharpoons C_6H_{12}O_6$

If observed (mean) molar mass of HCHO and $C_6H_{12}O_6$ is 150 :

(A) 0.50 (B) 0.833 (C) 0.90 (D) 0.96

38. The equilibrium constant K_p for the reaction

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is 4.5

What would be the average molar mass (in g/mol) of an equilibrium mixture of N_2O_4 and NO_2 formed by the dissociation of pure N_2O_4 at a total pressure of 2 atm ?

(A) 69 (B) 57.5 (C) 80.5 (D) 85.5

39. Rate of diffusion of ozonized oxygen is $0.4\sqrt{5}$ times that of pure oxygen. What is the per cent degree of association of oxygen assuming pure O_2 in the sample initially ?

(A) 20 (B) 40 (C) 60 (D) None of these

40. One mole of SO_3 was placed in a two litre vessel at a certain temperature. The following equilibrium was established in the vessel

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

the equilibrium mixture reacted with 0.2 mole $KMnO_4$ in acidic medium. Hence, K_C is :

(A) 0.50 (B) 0.25 (C) 0.125 (D) None of these

41. The equilibrium constant for the ionization of $RNH_2(g)$ in water as

$RNH_2(g) + H_2O\ (l) \rightleftharpoons RNH^+_3\ (aq) + OH^-\ (aq)$

is 8×10^{-6} at 25°C. Find the pH of a solution at equilibrium when pressure of $RNH_2(g)$ is 0.5 bar :

(A) ≈ 12.3 (B) ≈ 11.3 (C) ≈ 11.45 (D) None

42. When N_2O_5 is heated at certain temperature, it dissociates as $N_2O_5(g) \rightleftharpoons N_2O_3(g) + O_2(g)$; $K_c = 2.5$. At the same time N_2O_3 also decomposes as : $N_2O_3(g) \rightleftharpoons N_2O(g) + O_2(g)$. If initially 4.0 moles of N_2O_5 are taken in 1.0 litre flask and allowed to dissociation, concentration of O_2 at equilibrium is 2.5 M. Equilibrium concentration of N_2O_5 is :

(A) 1.0 M (B) 1.5 M (C) 2.166 M (D) 1.846 M

43. Two solid compounds X and Y dissociates at a certain temperature as follows

$X(s) \rightleftharpoons A(g) + 2B\ (g);$ $K_{p_1} = 9 \times 10^{-3}\ atm^3$

$Y(s) \rightleftharpoons 2B(g) + C(g);$ $K_{p_2} = 4.5 \times 10^{-3}\ atm^3$

The total pressure of gases over a mixture of X and Y is :

(A) 4.5 atm (B) 0.45 atm (C) 0.6 atm (D) None of these

ONE OR MORE THAN ONE CORRECT OPTIONS

44. Which of the following is correct about the chemical equilibrium?

(A) $(\Delta G)_{T,p} = 0$

(B) Equilibrium constant is independent of initial concentration of reactants

(C) Catalyst has no effect on equilibrium state

(D) Reaction stops at equilibrium

45. Consider the reaction $2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g) + \text{Heat}$

Under what conditions shift is undeterminable?

(A) Addition of O_2 and decrease in volume

(B) Addition of CO and removal of CO_2 at constant volume

(C) Increase in temperature and decrease in volume

(D) Addition of CO and increase in temperature at constant volume

46. What will be the effect of addition of catalyst at constant temperature?

(A) The equilibrium constant will remain constant (B) ΔH of the reaction will remain constant

(C) k_f and k_b will increase upto same extent (D) equilibrium composition will change

47. For the reaction $PC1_5\ (g) \rightleftharpoons PC1_3\ (g) + Cl_2\ (g)$, the forward reaction at constant temperature is favoured by :

(A) introducing an inert gas at constant volume (B) introducing chlorine gas at constant volume

(C) introducing an inert gas at constant pressure (D) foaming the volmi of On comlm

48. Increase in the pressure for the following equilibrium : $H_2O(\ell) \rightleftharpoons H_2O(g)$, result in the :

(A) formation of more $H_2O(l)$ (B) formation of more $H_2O(g)$

(C) increase in b.p. of $H_2O(l)$ (D) decrease in b.p. of $H_2O(l)$

QUESTION BANK
IONIC EQUILIBRIUM

(SINGLE CHOICE QUESTION)

1. What concentration of $HCOO^-$ is present in a solution of 0.01 M HCOOH ($K_a = 1.8 \times 10^{-4}$) and 0.01 M HC1?

(A) 1.8×10^{-3} (B) 10^{-2} (C) 1.8×10^{-4} (D) 10^{-4}

2. How much water must be added to 300 mL of 0.2 M solution of CH_3COOH ($K_a = 1.8 \times 10^{-5}$) for the D.O.I, (A) of the acid to double?

(A) 600 mL (B) 900 mL (C) 1200 mL (D) 1500 mL

3. A hand book states that the solubility of RNH_2 (g) in water at 1 atm and 25°C is 22.41 volumes of RNH_2(g) per volume of water. (pK_b of RNH_2 = 4) Find the max. pOH that can be attained by dissolving RNH_2 in water :

(A) 1 (B) 2 (C) 4 (D) 6

4. 6.0 g weak acid HA (mol. wt. = 60 g/mol.) is dissolved in water and formed 10 m^3 solution.

If K_a(HA) = 10^{-9}, then pOH of solution is : [Given : log 4 = 0.6]

(A) 6.7 (B) greater than 6.7 and less than 7.0

(C) greater than 7.0 and less than 7.3 (D) greater than 7.3

5. If a salt of strong acid and weak base hydrolyses appreciably (α = 0.1), which of the following formula is to be used to calculate degree of hydrolysis 'α'?

(A) $\alpha = \sqrt{\frac{K_w}{K_a.a}}$ (B) $\alpha = \sqrt{\frac{K_w}{K_b.a}}$ (C) $\alpha = \sqrt{\frac{K_w}{K_a.K_b}}$ (D) None of these

6. Calculate the $[OH^-]$ in 0.01 M aqueous solution of NaOCN (K_b for $OCN^- = 10^{-10}$) :

(A) 10^{-6} M (B) 10^{-7} M (C) 10^{-8} M (D) None of these

7. Calculate $[H^+]$ at equivalent point between titration of 0.1 M, 25 mL of weak acid HA ($K_{a(HA)} = 10^{-5}$) with 0.05 *M* NaOH solution :

(A) 3×10^{-9} (B) 1.732×10^{-9} (C) 8 (D) 10

8. The degree of hydrolysis of a salt of weak acid (HA) and weak base (BOH) in its 0.1M solution is found to be 10%. If the molarity of the solution is 0.05 *M*, the percentage hydrolysis of the salt should be :

(A) 5% (B) 10% (C) 20% (D) None of these

9. H_3PO_4 is a weak triprotic acid; approximate pH of 0.1 M Na_2HPO_4 *(aq.)* is calculated by :

(A) $\frac{1}{2}[pK_{a_1} + pK_{a_2}]$ (B) $\frac{1}{2}[pK_{a_2} + pK_{a_3}]$ (C) $\frac{1}{2}[pK_{a_1} + pK_{a_3}]$ (D) $pK_{a_1} + pK_{a_2}$

10. 0.1 M formic acid solution is titrated against 0.1 M NaOH solution. What would be the difference in pH between 1/5 and 4/5 stages of neutralization of acid ?

(A) 2 log 3/4 (B) 2 log 1/5 (C) log 1/3 (D) 2 log 4

11. The pH of the resultant solution of 20 mL of 0.1 M H_3PO_4 and 20 mL of 0.1M Na_3PO_4 is :

(A) $pK_{a_1} + \log 2$ (B) pK_{a_1} (C) pK_{a_2} (D) $\frac{pK_{a_1} + pK_{a_2}}{2}$

12. During the titration of a weak diprotic acid (H_2A) against a strong base (NaOH), the pH of the solution half-way to the first equivalent point and that at the first equivalent point are given respectively by :

(A) pK_{a_1} and $pK_{a_1} + pK_{a_2}$ (B) $\sqrt{K_{a_1}C}$ and $\frac{pK_{a_1} + pK_{a_2}}{2}$

(C) pK_{a_1} and $\frac{pK_{a_1} + pK_{a_2}}{2}$ (D) pK_{a_1} and pK_{a_2}

13. 50 mL of a solution containing 10^{-3} mole of Ag^+ is mixed with 50 mL of a 0.1 M HC1 solution. How much $[Ag^+]$ remains in solution? (K_{sp} of AgCl = 1.0×10^{-10})

(A) 2.5×10^{-9} (B) 2.5×10^{-7} (C) 2.5×10^{-8} (D) 2.5×10^{-10}

14. The solubility product of AgCl is 10^{-10} M^2. The minimum volume (in m^3) of water required to dissolve 14.35 mg of AgCl is approximately :

(A) 0.01 (B) 0.1 (C) 100 (D) 10

15. Solubility of AgCl in 0.2 M NaCl is x and that in 0.1 M $AgNO_3$ is y then which of the following is correct?

(A) x = y (B) x > y (C) x < y (D) We cannot predict

16. Find moles of NH_4C1 required to prevent $Mg(OH)_2$ from precipitating in a litre of solution which contains 0.02 mole of NH_3 and 0.001 mole of Mg^{2+} ions.

Given : K_b $(NH_3) = 10^{-5}$; K_{sp} $[Mg(OH)_2] = 10^{-11}$

(A) 10^{-4} (B) 2×10^{-3} (C) 0.02 (D) 0.1

17. 20 mL of 0.1 M weak acid HA ($K_a = 10^{-5}$) is mixed with solution of 10 mL of 0.3 M HC1 and 10 mL of 0.1 M NaOH. Find the value of $[A^-]/([HA] + [A^-])$ in the resulting solution :

(A) 2×10^{-4} (B) 2×10^{-5} (C) 2×10^{-3} (D) 0.05

18. If first dissociation of $X(OH)_3$ is 100% where as second dissociation is 50% and third dissociation is negligible then the pH of 4×10^{-3} M $X(OH)_3$ is :

(A) 11.78 (B) 10.78 (C) 2.5 (D) 2.22

19. Calcium lactate is a salt of weak organic acid and strong base represented as $Ca(LaC)_2$. A saturated solution of $Ca(LaC)_2$ contains 0.6 mole in 2 litre solution. pOH of solution is 5.60. If 90% dissociation of the salt takes place then what is pK_a of lactic acid ?

(A) 2.8 – log (0.54) (B) 2.8 + log (0.54) (C) 2.8 + log (0.27) (D) None of these

20. What is the concentration of $CH_3COOH(aq.)$ in a solution prepared by dissolving 0.01 mole of

$NH_4^+CH_3COO^-$ in 1 L H_2O? $[K_{a(CH_3COOH)} = 1.8 \times 10^{-5}; K_{b(NH_4OH)} = 1.8 \times 10^{-5}]$

(A) 5.55×10^{-5} (B) 0.10 (C) 6.4×10^{-4} (D) 5.55×10^{-3}

21. How many gm of solid KOH must be added to 100 mL of a buffer solution ? Which is 0.1 M each w.r.t. acid HA and salt K A to make the pH of solution 6.0. [Given : pK_a (HA) = 5]

(A) 0.458 (B) 0.327 (C) 5.19 (D) None of these

22. A 1.025 g sample containing a weak acid HX (mol. wt. =82) is dissolved in 60 mL water and titrated with 0.25 M NaOH. When half of the acid was neutralised the pH was found to be 5.0 and at the equivalence point the pH is 9.0. Calculate weight percentage of HX in sample :

(A) 50% (B) 75% (C) 80% (D) None of these

23. Which of the following Expression for degree of ionization of a monoacidic base (BOH) in aqueous solution at appreciable concentration is not correct ?

(A) $\sqrt{\dfrac{K_b}{c}}$ (B) $\dfrac{1}{1+10^{(pK_b - pOH)}}$ (C) $\dfrac{K_w[H^+]}{K_b + K_w}$ (D) $\dfrac{K_b}{K_b + [OH^-]}$

24. A solution of weak acid HA was titrated with base NaOH. The equivalent point was reached when 40 mL of 0.1 M NaOH has been added. Now 20 mL of 0.1 M HCl were added to titrated solution, the pH was found to be 5.0. What will be the pH of the solution of obtained by mixing 20 mL of 0.2 M NaOH and 20 mL of 0.2 M HA ?

(A) 7 (B) 9 (C) 11 (D) None of these

25. A buffer solution 0.04 M in Na_2HPO_4 and 0.02 M in Na_3PO_4 is prepared. The electrolytic oxidation of 1.0 milli-mole of the organic compound RNHOH is carried out in 100 mL of the buffer. The reaction is

$$RNHOH + H_2O \longrightarrow RNO_2 + 4H^+ + 4e^-$$

The approximate pH of solution after the oxidation is complete is :

$[\text{Given: for } H_3PO_4, pK_{a_1} = 2.2; pK_{a_2} = 7.20; pK_{a_3} = 12]$

(A) 6.90 (B) 7.20 (C) 7.5 (D) None of these

26. When a 20 mL of 0.08 M weak base BOH is titrated with 0.08 HC1, the pH of the solution at the end point is 5. What will be the pOH if 10 mL of 0.04 M NaOH is added to the resulting solution ?

[Given : log 2 = 0.30 and log 3 = 0.48]

(A) 5.40 (B) 5.88 (C) 4.92 (D) None of these

27. In the titration of a solution of a weak acid HA and NaOH, the pH is 5.0 after 10 mL of NaOH solution has been added and 5.60 after 20 mL NaOH has been added.

What is the value of pK_a for HA ?

(A) 5.15 (B) 5.3 (C) 5.6 (D) None of these

28. 50 mL of 0.05 M Na_2CO_3 is titrated against 0.1 M HCl. On adding 40 mL of HCl, pH of the solution will be

$[\text{Given: For } H_2CO_3, pK_{a_1} = 6.35, pK_{a_2} = 10.33; \log 3 = 0.477, \log 2 = 0.30]$

(A) 6.35 (B) 6.526 (C) 8.34 (D) 6.173

29. A solution is 0.10 M $Ba(NO_3)_2$ and 0.10 M $Sr(NO_3)_2$. If solid Na_2CrO_4 is added to the solution, what is $[Ba^{2+}]$ when $SrCrO_4$ begins to precipitate?

$[K_{sp}(BaCrO_4) = 1.2 \times 10^{-10}; K_{sp}(SrCrO_4) = 3.5 \times 10^{-5}]$

(A) 7.4×10^{-7} (B) 2.0×10^{-7} (C) 6.1×10^{-7} (D) 3.4×10^{-7}

30. If 500 ml of 0.4 M $AgNO_3$ is mixed with 500 ml of 2M NH_3 solution then what is the concentration of $Ag(NH_3)^+$ in solution ?

Given: $K_{f_1}[Ag(NH_3)^+] = 10^3$; $K_{f_2}[Ag(NH_3)_2^+] = 10^4$

(A) 3.33×10^{-7}M (B) 3.33×10^{-5}M (C) 3×10^{-4}M (D) 10^{-7}M

31. The simultaneous solubility of AgCN ($K_{sp} = 2.5 \times 10^{-16}$) and AgCl ($K_{sp} = 1.6 \times 10^{-10}$) in 1.0 M NH_3(aq.) are respectively : $[\text{Given}: K_f[Ag(NH_3)_2^+] = 10^7]$

(A) $0.037, 5.78 \times 10^{-8}$ (B) $5.78 \times 10^{-8}, 0.037$

(C) $0.04, 6.25 \times 10^{-8}$ (D) None of these

32. What is the minimum pH required to prevent the precipitation of ZnS in a solution that is 0.01 M $ZnCl_2$ and saturated with 0.10 M H_2S ?

[Given : $K_{sp} = 10^{-21}$, $K_{a_1} \times K_{a_2} = 10^{-20}$]

(A) 0 (B) 1 (C) 2 (D) 4

33. The salt $Al(OH)_3$ is involved in the following two equilibria ,

$$Al(OH)_3(s) \rightleftharpoons Al^{3+}(aq.) + 3OH^-(aq.); K_{sp}$$

$$Al(OH)_3(s) + OH^-(aq.) \rightleftharpoons Al(OH)_4^-(aq.); K_c$$

Which of the following relationship is correct at which solubility is minimum ?

(A) $[OH^-] = \left(\frac{K_{sp}}{K_c}\right)^{1/3}$ (B) $[OH^-] = \left(\frac{K_c}{K_{sp}}\right)^{1/4}$ (C) $[OH^-] = \sqrt{\left(\frac{K_{sp}}{K_c}\right)^{1/4}}$ (D) None of these

34. The degree of dissociation of water in a 0.1 M aqueous solution of HCl at a certain temperature t°C is 3.6×10^{-15}. The temperature t must be :

(A) < 25°C (B) = 25°C

(C) > 25°C (D) insufficient data to predict

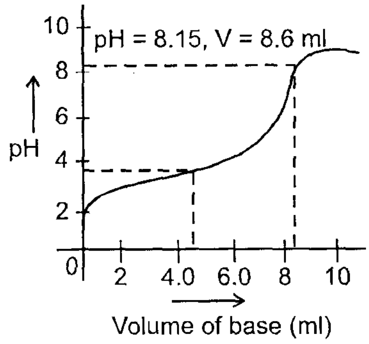
35. Which one is the correct expression below for the solution containing 'n' number of weak acids?

(A) $[H^+] = \sqrt{\sum_{i=1}^{n} \frac{K_i}{C_i}}$ (B) $[H^+] = \sqrt{\sum_{i=1}^{n} K_i C_i}$ (C) $[H^+] = \sum_{i=1}^{n} K_i C_i$ (D) none of these

36. The pH of glycine at the first half equivalence point is 2.34 and that at second half equivalence paint is 9.60. At the equivalence point (The first inflection point) The pH is :

(A) 3.63 (B) 2.34 (C) 5.97 (D) 11.94

37. The pK of a weak acid if titration progress is monitored as follows :



(A) Data insufficient (B) 8.15 (C) 3.88 (D) 4.28

38. A 1.458 g of Mg reacts with 80.0 ml of a HCl solution whose pH is -0.477. The change in pH after all Mg has reacted. (Assume constant volume. Mg = 24.3 g/mol.)(log 3 = 0.477)

(A) – 0.176 (B) + 0.477 (C) – 0.2385 (D) 0.3

39. Equal volume of two solution having pH = 2 and pH = 10 are mixed together at 90°C. Then pH of resulting solution is: (Take Kw at 90°C = 10^{-12})

(A) 2 +log 2 (B) 10 – log 2 (C) 7 (D) 6

40. Find the ΔpH(initial pH -final pH) when 100 ml 0.01 M HCl is added in a solution containig 0.1 m moles of $NaHCO_3$ solution of negligible volume (Ka_1 = 1Q-^{7}, Ka_1 =10^{-11} for H_2CO_3):

(A) 6 + 2 log3 (B) 6 – log 3 (C) 6 + 2 log 2 (D) 6 – 2 log 3

41. The ionization constant of benzoic acid is 6.46×10^{-5} and K_{SP} for silver benzoate is 2.5×10^{-13}. How many times silver benzoate more soluble in a buffer of pH = 3.19 as compared to its solubility in pure water?

(A) 3.317 (B) 9.5 (C) 1000 (D) 7.5

42. 30 ml of 0.06 M solution of the protonated form of an anion acid methonine (H_2A^+) is treated with 0.09 M NaOH. Calculate pH after addition of 20 ml of base. pKa_1 = 2.28 and pKa_2 = 9.2.

(A) 5.5 (B) 5.74 (C) 9.5 (D) None

43. A certain acid-base indicator is red in acid solution and blue in basic solution 75% of the indicator is present in the solution in its blue form at pH = 5. Calculate the pH at which the indicator shows 90% red form?

(A) 3.56 (B) 5.47 (C) 2.5 (D) 7.4

44. Ionisation constant of HA (weak acid) and BOH (weak base) are 3.0×10^{-7} each at 298K. The percent degree of hydrolysis of BA at the dilution of 10L is :

(A) 25 (B) 50 (C) 75 (D) 40

45. Which of the following concentrations of NH_4^+ will be sufficient to prevent the precipitation of $Mg(OH)_2$ from a solution which is 0.01 M $MgCl_2$and 0.1 M NH_3(aq). Given that: K_{sp} of

$Mg(OH)_2 = 2.5 \times 10^{-11}$ and $K_b$forNH$_3$(aq) = 2×10^{-5}.

(A) 0.01M (B) 0.02M (C) 0.001 M (D) 0.04M

46. Calculate the molar solubility of zinc tetrathiocyanato-N-mercurate (II) if its $K_{SP} = 2.2 \times 10^{-7}$.

(A) 0.00380 (B) 0.000469 (C) 0.0095 (D) 0.0183

47. An acid-base indicator which is a weak acid has a pK_a value = 5.45. At what cocentration ratio of sodium acetate to acetic acid would the indicator show a colour half-way between those of its acid and conjugate base forms? pK_a of acetic acid = 4.75. [log 2 = 0.3]

(A) 4 : 1 (B) 7 : 1 (C) 5 : 1 (D) 2 : 1

48. The indicator constant of phenolphthalein is approximately 10^{-10}. Asolutionis prepared by adding 100.1 c.c. of 0.01 N sodium hydroxide to 100.00c.c. of 0.01 N hydrochloric acid. If a few drops of phenolphthalein are now added, what fraction of the indicator is converted to its coloured form?

(A) $\frac{2}{3}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) none of these

49. A certain mixture of HCl and CH_3-COOH is 0.1 M in each of the acids. 20 ml of this solution is titrated against 0.1M NaOH. By how many units does the pH change from the start to the stage when the HCl is almost completely neutralised? K_a for acetic acid $= 1.8 \times 10^{-5}$.

(A) 2.03 (B) 0.775 (C) 2.325 (D) 3.172

50. A buffer solution is made by mixing a weak acid HA ($K_a = 10^{-6}$) with its salt NaA in equal amounts. What should be the amount of acid or salt that should be added to make 90 ml of buffer solution of buffer capacity 0.1 ? (change in pH on adding strong base or strong acid should be taken as one).

(A) 10 mmoles (B) 22 mmoles (C) 9 mmoles (D) 11 mmoles

(MULTIPLE CHOICE QUESTION)

51. Factor influencing the degree of ionization of a weak electrolyte is :

(A) dilution (B) temperature

(C) presence of other ions (D) nature of solvent

52. Which of the following statement (s) is/are correct about the ionic product of water ?

(A) K_i (ionization constant of water) < K_w (ionic product of water)

(B) $pK_i > pK_w$

(C) At 25°C $K_i = 1.8 \times 10^{-14}$

(D) Ionic product of water at 10°C is 10^{-14}

53. Which among the following statement is/are correct?

(A) $pH = -\log_{10} [H_3O^+]$ for dilute solution

(B) pH of H_2O decreases with increase of temperature

(C) pH can not more than 14

(D) If a solution is diluted ten times, its pH always increases by 1

54. If K_{a_1}, K_{a_2} and K_{a_3} be the first, second and third ionization constant of H_3PO_4 and $K_{a_1} >> K_{a_2} >> K_{a_3}$ which is/are correct :

(A) $[H^+] = \sqrt{K_{a_1}[H_3PO_4]}$ (B) $[H^+] = [HPO_4^{2-}]$ (C) $K_{a_2} = [HPO_4^{2-}]$ (D) $[HPO_4^{2-}] = [PO_4^{3-}]$

55. Which of the following mixtures can act as a buffer ?

(A) NaOH + HCOONa (1 : 1 molar ratio) (B) HCOOH + NaOH (2 : 1 molar ratio)

(C) NH_4Cl + NaOH (2 : 1 molar ratio) (D) HCOOH + NaOH (1 : 1 molar ratio)

56. Degree of hydrolysis (A) for a salt of strong acid and weak base is :

(A) independent of dilution (B) increases with dilution

(C) increases with decrease in K_b (D) increases with increase in temperature

57. Formic acid is a weak acid and hydrochloric acid is a strong acid. It follows that the :

(A) $[OH^-]$ of 0.01 M HC1 (aq.) will be less than that of 0.01 *M* HCOOH (aq.)

(B) solution containing 0.1 M NaOH (aq.) and 0.1 M HCOONa (aq.) is a buffer solution

(C) pH of 10^{-9} M HC1 (aq.) will be approximately 7 at 25°C

(D) pH of a solution formed by mixing equimolar quantities of HCOOH and HC1 will be less than that of a similar solution formed from HCOOH and HCOONa

58. A 1 litre solution of pH = 1 diluted upto 10 times. What volume of a solution with pH = 2 is to be added in diluted solution so that pH does not change :

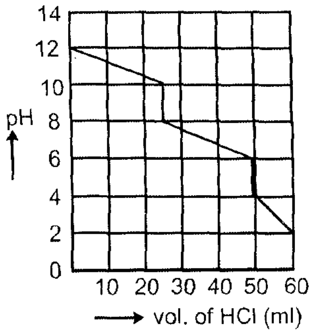
(A) 1 litre (B) 10 litre (C) 100 litre (D) 25 litre

59. The solubility of a sparingly soluble salt $A_x B_y$ in water at 25°C 1.4×10^{-4} M. The solubility product is 1.1×10^{-11}. The possibilities are :

(A) x = 1, y = 2 (B) x = 2, y = 1 (C) x = 1, y = 3 (D) x = 3, y = 1

60. The variation of pH during the titration of 0.5 N Na_2CO_3 with 0.5 N HCl is shown in the given graph. The following table indicates the colour and pH ranges of different indicators :

Indicator	Range of colour change	Colour in acid	Colour in base
Thymol blue	1.2 to 2.8	Red	Yellow
Bromocresol red	4.2 to 6.3	Red	Yellow
Bromothymol blue	6.0 to 7.6	Yellow	Blue
Cresolphethalein	8.2 to 9.8	Colourless	Red



Based on the graph and the table, which of the following statements are true ?

(A) The first equivalence point can be detected by cresolphthalein.

(B) The complete neutralisation can be detected by bromothymol blue.

(C) The second equivalence point can be detected by bromocresol red.

(D) The volume of HCl required for the first equivalence point is half the volume of HCl required for the second equivalence point.

61. Which of the following solutions when added to 1L of a 0.1 M CH_3COOH solution will cause no change in either the degree of dissociation of CH_3COOH or the pH of the solution.

$K_a = 1.8 \times 10^{-5}$ for CH_3COOH ?

(A) 3 mM HCOOH ($K_a = 6 \times 10^{-4}$) (B) 0.1 M CH_3COONa

(C) 1.34 mM HCl (D) 0.1 MCH_3COOH

62. Buffer solution A of a weak monoprotic acid and its sodium salt in the concentration ratio x: y has pH = $(pH)_1$ Buffer solution B of the same acid and its sodium salt in the concentration ratio y : x has pH = $(pH)_2$. If $(pH)_2 - (pH)_1 = 1$ unit and $(pH)_1 + (pH)_2 = 9.5$ units, then

(A) $pK_a = 4.75$ (B) $\frac{x}{y} = 2.36$ (C) $\frac{x}{y} = 3.162$ (D) $pK_a = 5.25$

ANSWER KEY

QUESTION BANK

MOLE & EQUIVALENT CONCEPT

SINGLE CHOICE QUESTIONS

1. (C) **2.** (D) **3.** (A) **4.** (C) **5.** (B) **6.** (A) **7.** (D) **8.** (D) **9.** (A) **10.** (B)
11. (B) **12.** (A) **13.** (D) **14.** (D) **15.** (C) **16.** (C) **17.** (C) **18.** (B) **19.** (B) **20.** (A)
21. (A) **22.** (A) **23.** (B)

MULTIPLE CHOICE QUESTIONS

24. (A,C,D) **25.** (All) **26.** (A,B,C) **27.** (A, B, D) **28.** (B, C) **29.** (A, B, C) **30.** (A, B, D)
31. (ALL) **32.** (A,C)

MATRIX MATCH QUESTIONS

33. A → (q, r); B → (p, s); C → (q, r); D → (r)

34. A → (p, s); B → (q, r, t); C → (p, q, s, t); D → (r, t)

35. A → (r); B → (s); C → (p); D → (q, r); E → (t)

COMPREHENSION BASED QUESTION

36. (B) **37.** (C) **38.** (A) **39.** (B) **40.** (B) **41.** (A) **42.** (B) **43.** (C) **44.** (B) **45.** (C)
46. (A) **47.** (B) **48.** (C) **49.** (B) **50.** (D) **51.** (B) **52.** (A) **53.** (A) **54.** (B)

GASEOUS STATE

1. (B) **2.** (C) **3.** (D) **4.** (B) **5.** (A) **6.** (B) **7.** (A) **8.** (B) **9.** (B) **10.** (A)
11. (B) **12.** (A) **13.** (B) **14.** (A) **15.** (B) **16.** (C) **17.** (C) **18.** (A) **19.**(B) **20.** (A)
21. (B) **22.** (A) **23.**(B) **24.** (A,B,C) **25.** (C,D) **26.** (A,D) **27.** (A,C)
28. (A, C) **29.** (A,B,D) **30.** (C) **31.** (B) **32.** (B) **33.**(A) **34.** (B) **35.** (A)
36.(A) **37.** (A) **38.** (B)

39. A → (P), B → (Q), C → (R), D → (S)

40. A → (P), B → (Q, R), C → (S), D → (S)

THERMODYNAMICS & THERMOCHEMISTRY

1. (A) **2.** (C) **3.** (D) **4.** (A) **5.** (B) **6.** (A) **7.** (C) **8.** (C) **9.** (i) (c) (ii) (b)
10. (D) **11.** (B) **12.** (A) **13.** (B) **14.** (C) **15.**(C) **16.** (A) **17.** (D) **18.** (A) **19.** (B)
20. (B) **21.** (C) **22.** (D) **23.** (B) **24.** (B) **25.** (D) **26.** (D) **27.** (D) **28.** (C) **29.** (D)
30. (C) **31.** (A) **32.**(A,B,D) **33.** (B,C,D) **34.** (A,B,C) **35.**(C)
36. (A,C,D) **37.** (A,B,C) **38.** (B,C,D) **39.** (A,B,C,D) **40.**(A,B,C)
41. (A,B,D) **42.** (B,C,D)

43. A → (s, q); B → (r, p); C → (s, p); D → (s, p)

44. A → (r); B → (s); C → (p); D → (q)

45. A → (q); B → (u); C → (p); D → (t)

46. A → (q); B → (p, r, s, t); C → (p, s); D → (r, t)

47. (A) **48.** (C) **49.** (D) **50.** (D) **51.** (B) **52.** (C) **53.** (B) **54.**(A) **55.** (B)
56.(C) **57.** (B) **58.**(B) **59.** (C) **60.** (B) **61.** (C) **62.** (B) **63.** (B) **64.** (D)
65.(D)

ATOMIC STRUCTURE

1. (A) 2. (B) 3. (C) 4. (B) 5. (B) 6. (B) 7. (D) 8.(A) 9.(D) 10.(D)
11.(A) 12.(C) 13.(B) 14.(B) 15.(B) 16.(D) 17.(D) 18.(B) 19.(D) 20.(C)
21.(D) 22.(A) 23.(B) 24.(D) 25.(D) 26.(C) 27.(C) 28.(B) 29.(D) 30.(B)
31.(C) 32.(C) 33. (D) 34. (D) 35. (A) 36. (A) 37. (A) 38. (C) 39. (D) 40.(B)
41. (B) 42. (C) 43. (A,C) 44. (A,B,D) 45. (A,B) 46.(B,C,D) 47. (A,B)
48. (A,B,C) 49. (A,C,D) 50. (B,C,D)

51. A→(s); B→(p); C→(p); D→(q)

52. A → (q, r); B→ (p, s); C → (s); D→ (q)

53. A → (r, s); B → (p, s); C→ (q, r); D→ (p, q)

54.(C) 55.(A) 56.(C) 57.(C) 58.(A) 59.(D) 60.(C) 61.(D) 62.(B) 63.(D)
64.(D) 65.(C) 66.(A) 67.(D)

CHEMICAL EQUILIBRIUM

SINGLE CHOICE QUESTIONS

1.(B) 2.(A) 3.(A) 4.(C) 5. (C) 6. (A) 7.(A) 8.(D) 9. (B) 10.(B)
11.(A) 12.(A) 13.(A) 14.(C) 15.(A) 16.(D) 17.(C) 18.(D) 19.(B) 20.(B)
21.(B) 22.(C) 23.(B) 24.(B) 25.(D) 26.(D) 27.(C) 28.(D) 29.(C,D) 30.(A)
31.(D) 32.(C) 33.(D) 34.(D) 35.(C) 36.(C) 37.(D) 38.(B) 39.(C) 40.(C)
41.(B) 42.(D) 43.(B)

MULTIPLE CHOICE QUESTIONS

44.(A,B,C) 45.(C,D) 46.(ABC) 47.(C,D) 48.(A,C)

IONIC EQUILIBRIUM

SINGLE CHOICE QUESTIONS

1. (C)	**2. (B)**	**3. (B)**	**4. (C)**	**5. (D)**	**6. (A)**	**7. (B)**	**8. (B)**	**9. (B)**	**10. (D)**
11. (C)	**12. (C)**	**13. (A)**	**14. (A)**	**15. (C)**	**16. (B)**	**17.(A)**	**18. (A)**	**19. (A)**	**20. (A)**
21. (A)	**22. (C)**	**23. (C)**	**24. (B)**	**25. (C)**	**26. (B)**	**27. (B)**	**28. (D)**	**29. (D)**	**30. (B)**
31. (D)	**32. (B)**	**33. (D)**	**34. (C)**	**35. (B)**	**36. (C)**	**37. (C)**	**38. (D)**	**39. (D)**	**40. (A)**
41. (A)	**42. (B)**	**43. (A)**	**44. (A)**	**45. (D)**	**46. (B)**	**47. (C)**	**48. (D)**	**49. (A)**	**50. (D)**

MULTIPLE CHOICE QUESTIONS

51. (A,B,C,D)	**52. (A,B)**	**53. (A,B)**	**54. (A,C)**	**55. (B,C)**	**56. (B,C,D)**
57. (A,C,D)	**58.(A,B,C,D)**	**59. (A,B)**	**60. (A,C,D)**	**61. (A,D)**	**62. (A,C)**